



Customer-Driven **Energy Solutions**

EEG- Energy Efficiency Group International Ltd

T : +357 22572020 | F : +357 22610222

E : hq@eegroup.info

16 Tefkrou Anthia, Dali Industrial Zone

2540 Dali, Cyprus

www.eegroup.info



FOUR SEASONS

HOTEL

SAN FRANCISCO AT EMBARCADERO



ENERGY AUDIT REPORT

FOUR SEASONS HOTEL

SAN FRANCISCO AT

EMBARCADERO

October 2022

EEG			
EEG – ENERGY EFFICIENCY INTERNATIONAL LTD		Document Type:	ASHRAE Level II Energy Audit Report
Energy Audit Report for the Four Seasons Hotel in San Francisco at Embarcadero - USA		Contract Ref:	
		Revision no.:	A
Document Control			
Revision / Date	Issued by:	Reviewed by:	Approved by:
A-for Client Review	Myriam Osman	Joe El Khoury Ronald Diab	Ronald Diab
B-for Client Approval			
Disclaimer			
<i>This report has been prepared by EEG with all reasonable skill, care and diligence within the terms of the contract with the client, incorporating our General Terms and Conditions of Business and taking account of the resources devoted to it by agreement with the client. The information contained in this report is, to the best of our knowledge, correct at the time of printing. The interpretations and recommendations are based on our experience, using reasonable professional skill and judgment, and based upon the information that was available to us. This report is confidential to the client and we accept no responsibility whatsoever to third parties to whom this report, or any part thereof, is made known. Any such party relies on the report at their own risk.</i>			

Table of Contents

Table of Contents	3
List of Tables	6
List of Figures	7
1. Introduction.....	12
1.1. Energy Efficiency Rationale	12
1.1. Energy Audit Goals and Tasks	13
1.1.1. End Goals.....	13
1.1.2. Methodology.....	13
2. Facility General Description.....	14
2.1. Introduction.....	14
2.1. Building Envelope	16
2.2. Occupancy	18
2.3. Automation and Metering	19
2.4. San Francisco Climatic Conditions	20
2.5. Utilities: Energy Sources and Water	22
3. Utilities Accounting	24
3.1. Electricity	24
3.1.1. Electricity Cost Breakdown	24
3.1.2. Electricity Overall Consumption and Cost	27
3.1.3. Load Profile.....	29
3.1.4. Regression Analysis.....	30
3.2. Natural Gas	32
3.2.1. Natural Gas Overall Consumption and Cost	32
3.2.2. Regression Analysis.....	33
3.3. Water	35
3.3.1. Water Overall Consumption and Cost.....	35
3.3.2. Water Consumption Breakdown.....	36
3.4. District Heating	38
3.4.1. District Heating Overall Consumption and Cost	38
3.4.2. Regression Analysis.....	39
3.5. Utilities Cost Breakdown	41
3.6. Energy & Water Use Intensity, Carbon Footprint	42
3.6.1. Energy Use Intensity (EUI)	42
3.6.2. Water Use Intensity (WUI)	43
3.6.3. Carbon Emissions Footprint	44

3.6.4.	CHSB – Cornell Sustainability Hotel Benchmark – Real Consumption	45
4.	Electrical Energy Load Analysis & Energy Balance	46
4.1.	Electrical Load Categories.....	46
4.1.1.	Lighting.....	47
4.1.2.	HVAC Fans	49
4.1.3.	Cooling	52
4.1.4.	Motors	57
4.1.5.	Kitchen and Refrigeration	61
4.1.6.	Others.....	62
4.2.	Energy Balance	63
5.	Thermal Energy	65
6.	Indoor Air Quality data logging	70
7.	Energy Efficiency Measures.....	78
7.1.	EEM-01 [CAPEX]: KW Demand Management 	80
7.1.1.	Rationale	80
7.1.2.	Efficiency Action and Energy Savings	80
7.1.3.	FSSFE Case.....	81
7.2.	EEM – 02: [CAPEX] Lighting Retrofit 	84
7.2.1.	Rationale	84
7.2.2.	Efficiency Action and Energy Savings	84
7.3.	EEM – 03: [LCNC] Lighting Control and Management 	86
7.3.1.	Rationale	86
7.3.2.	Efficiency Action and Energy Savings	87
7.4.	EEM-04: [LCNC] HVAC-Fans Control 	88
7.4.1.	Rationale	88
7.4.2.	Aggressive Scheduling	88
7.4.3.	Setpoint Changes	89
7.5.	EEM – 05: [LCNC] Chilled Water: Low ΔT Mitigation 	90
7.5.1.	Rationale	90
7.5.2.	Low ΔT Syndrome.....	90
7.5.3.	Efficiency Action and Energy Savings	91
7.6.	EEM – 06: [LCNC] Manage Guest Rooms During Unrented Mode 	93
7.6.1.	Rationale	93
7.6.2.	Efficiency Actions and Energy Savings	93
7.7.	EEM – 07: [CAPEX] Improve Hot Water/Steam network Thermal Insulation 	95
7.7.2.	Efficiency Action and Energy Savings.....	96
7.8.	EEM – 07: [LCNC] Install Vinyl Curtains on all Cold Rooms (positive and negative temperature) 	98

7.8.1.	Rationale	98
7.8.2.	Efficiency Action and Energy Savings	98
7.9.	EEM – 09: [CAPEX] Install VFDs on Fans 	99
7.9.1.	Rationale	99
7.9.2.	Efficiency Actions and Energy savings	99
7.10.	EEM – 10: [CAPEX] Retrofit Legacy FCUs by EC Motors FCU Type 	101
7.10.1.	Rationale	101
7.10.2.	Efficiency Action and Energy Savings	101
7.11.	EEM – 11: [CAPEX] Retrofit Electrical HW Tank in 36 th floor 	103
7.11.1.	Rationale	103
7.11.2.	Efficiency Action and Energy Savings	103
7.12.	EEM – 12: [CAPEX] Retrofit Chillers 3 and 4 serving guest rooms 	105
7.12.1.	Rationale	105
7.12.2.	Efficiency Action and Energy Savings	105
7.13.	EEM – 13: [CAPEX] Retrofit Heat Pump/DX Servicing Sky Bridges 	108
7.13.1.	Rationale	109
7.13.2.	Efficiency Action and Energy Savings	109
7.14.	EEM – 13: [CAPEX] BMS Upgrade and review possible pneumatic to DDC conversion 	110
7.14.1.	Rationale	110
7.14.2.	BMS Upgrade	110
7.14.3.	Convert Pneumatic Control to DDC	112
7.14.4.	Action Plan and Savings	113
8.	EEMs Summary Figures & Implementation Notes	115
8.1.	EEMs Summary – CAPEX, Cost Savings & Payback Period	115
8.2.	EEMs Summary – Electricity (KWh) and District Heating (Mlbs) Savings	116
8.3.	Implementation Notes and Carbon Reductions	117

List of Tables

Table 1: Monthly Occupancy Rate in FSSFE through the Baseline Period	18
Table 2: San Francisco Climatic Data	20
Table 3: Annual Utilities Costs	23
Table 4: PG&E Cost Breakdown	25
Table 5: Clean Energy Cost Breakdown	27
Table 6: FSSFE Monthly Electricity Consumption and Cost	27
Table 7: Regression Analysis Variables.....	30
Table 8: Electricity Projected Consumption, Residuals, and variation	31
Table 9: FSSFE Monthly Natural Gas Consumption and Cost	32
Table 10: Regression Analysis Variables.....	33
Table 11: Natural Gas Projected Consumption, Residuals, and variation	34
Table 12: FSSFE Monthly Water Consumption and Cost.....	35
Table 13: Water Breakdown During the Baseline Period	36
Table 14: California Title 20 – Water Efficiency Standards.....	37
Table 15: FSSFE Monthly District Heating Consumption and Cost.....	38
Table 16: Regression Analysis Variables.....	39
Table 17: District Heating Projected Consumption, Residuals, and variation	40
Table 18: Utilities Cost Breakdown	41
Table 19: Total Utilities Energy Breakdown.....	42
Table 20: FSSFE Carbon Emissions Monthly Breakdown.....	44
Table 21: CHSB benchmark – using real consumption	45
Table 22: HVAC Fans Sub-Categories Loads and Consumptions.....	51
Table 23: Electrical Load & Energy Consumption Breakdown by Category.....	63
Table 24: FSSFE Thermal Energy Breakdown.....	69
Table 25: Energy Efficiency Measures Color Codes.....	78
Table 26: Energy Efficiency Measures	78
Table 27: Electrical Load & Energy Consumption Breakdown by Category.....	82
Table 28: Cooling Capacity Variation vs. Delta-T Variation (Left) ‘ Cooling Capacity Variation vs. Chilled Water Flow Variation (Right).....	91
Table 29: Insulation thickness.....	97
Table 30: List of Fans and annual KWh consumption.....	100
Table 31: EEMs summary – CAPEX, Cost savings and Payback Period	115
Table 32: EEM Summary – Electricity (KWh) and District Heating (Mlbs) Savings.....	116

List of Figures

Figure 1: Four Seasons San Francisco at Embarcadero upper floors	14
Figure 2: Lobby & Bar F&B outlet (Left) , function room in the 2 nd floor (right).....	14
Figure 3: Gym (Left) , administration offices (right)	15
Figure 4: FSSFE Façade.....	16
Figure 5: FSSFE Façade.....	16
Figure 6: FSSFE Glazing Thermal Images.....	17
Figure 7: FSSFE Entrance.....	17
Figure 8: Monthly Occupancy Rate Variations.....	18
Figure 9: FSSFE- METASYS BMS snapshot	19
Figure 10: Telkonet Ecocentral GRMS	19
Figure 11: Temperature Variations in San Francisco	20
Figure 12: CDD and HDD Variations in San Francisco	21
Figure 13: Electrical Single Line Diagram	22
Figure 14: Utilities cost breakdown.....	23
Figure 15: PG&E – Delivery Services charges.....	24
Figure 16: PG&E – Maximum Demand	25
Figure 17: Clean Power – Maximum Demand	26
Figure 18: Clean Power – Electric Generation charges.....	26
Figure 19: FSSFE Electricity Monthly Variations along with the occupancy	28
Figure 20: FSSFE Electricity Monthly Variations along with the CDD	28
Figure 21: FSSFE KWh and KW recording for August 2022.....	29
Figure 22: FSSFE KW Demand for August 2-4 2022.....	29
Figure 23: Electrical Regression and Actual Energy Consumption	31
Figure 24: FSSFE Natural Gas Monthly Variations vs. occupancy	32
Figure 25: FSSFE Natural Gas consumption versus HDD	33
Figure 26: Natural Gas Regression and Actual Energy Consumption	34
Figure 27: FSSFE Water Consumption Monthly Variations.....	35
Figure 28: FSSFE Water Breakdown Variations.....	36
Figure 29: Up: Guest rooms’ aerator (Left), Shower (Right) Down: public toilet’s faucet (Left), Urinal (Right)	37
Figure 30: FSSFE District Heating Monthly Variations vs. HDD	38
Figure 31: District Heating Regression and Actual Energy Consumption	40
Figure 32: Utilities Cost Breakdown	41
Figure 33: Total Utilities Energy Breakdown.....	42
Figure 34: FSSFE Monthly WUI Variations	43
Figure 35: FSSFE Monthly Carbon Emissions	44
Figure 36: FSSFE Lighting	47
Figure 37: Lighting monthly energy consumption	48
Figure 38: AHU#4 serving 1 st floor – connected to Tower’s electricity feeder	49
Figure 39: Guest rooms’ corridor FCU specifications	49
Figure 40: Telkonet Thermostat (Left) connected to the GRMS platform (Right)	50
Figure 41: Supply Fan SF19 (Left), Kitchen Extractor Fan EF 60 (Right).....	50
Figure 42: BMS Fans Control (Below); Fans’ scheduling (Next Page).....	50
Figure 43: HVAC Fans monthly energy consumption.....	51
Figure 44: Analog measurements of readings in the pneumatic control system	52
Figure 45: Chillers 3 and 4 with the 4 compressors operation (Up) , ARI ratings (Down)	52

Figure 46: Chiller 5 – Multistack with 6 compressors (Left) ; Nameplate of one unit of 2 compressors (Right)	53
Figure 47: BMS Chilled Water Plant Controls – Chiller 3&4 (Below); Chiller 5 (Next Page)	53
Figure 48: Cooling Towers for Chiller 3,4 (Left) and 5 (Right).....	54
Figure 49: VFD on CT#3 (Left), VFD on CT#5 (Right).....	55
Figure 50: FSSFE Heat Pump	55
Figure 51: Cooling Sub-Categories Loads and Consumption	55
Figure 52: Cooling Monthly Energy Consumption	56
Figure 53: Primary Chilled water and condenser pumps for Chillers 3 and 4 (UP), Respective VFDs operation (Down)	57
Figure 54: Chiller 5 Primary Chilled Water and Condenser Pumps	58
Figure 55: Condenser and Chilled Water Pumps temperature during operations (4 th floor)	58
Figure 56: Hot Water Circulation Pumps	59
Figure 57: Hot Water Pumps.....	59
Figure 58: Air Compressor for the pneumatic controls	59
Figure 59: Guest (Left) and Service (Right) Lifts	60
Figure 60: Motors Monthly Energy Consumption.....	60
Figure 61: Kitchen Equipment (Left) and Cold Rooms (Right)	61
Figure 62: Kitchen and Refrigeration Energy Consumption.....	61
Figure 63: IT Equipment (Left) and Laundry Equipment (Right)	62
Figure 64: Others Monthly Energy Consumption.....	62
Figure 65: Electrical Load & Energy Breakdown percentages.....	63
Figure 66: Annual energy Consumption per end use	64
Figure 67: Monthly Electricity Consumption per end use	64
Figure 68: Hot Water Boilers (Left) and Nameplate (Right).....	65
Figure 69: HX 3 & 5 in the 48 th Floor (Left) and HX 7&8 in B1 (Right)	66
Figure 70: HX 4 in the 36 th floor with steam station (Up), 2xHot Water Tanks (Down)	66
Figure 71: Hot Water Pipes Thermal Images (Left) ; Clear related images (Right)	67
Figure 72: Steam network Thermal Images (Left); Clear related images (Right).....	68
Figure 73: Temp/Rh at the Bar	70
Figure 74: Temp/Rh at the Embassy Meeting Room.....	71
Figure 75: Temp/Rh at the Front Desk	71
Figure 76: Temp/Rh at the 46 th floor Guest rooms corridor.....	71
Figure 77: IAQ Recordings – Occupied Room (a to e)	72
Figure 78: IAQ Recordings – All locations (a to e)	74
Figure 79: KW Demand charges components within PG&E Transmission section in the utility bill	80
Figure 80: Using Battery Energy Storage for Peak Demand Management.....	81
Figure 81: FSSFE – Energy Storage Operation.....	81
Figure 82: Max Demand - Before and After the Management program	83
Figure 83: Areas with high potential of lighting control	86
Figure 84: E61 schedule of operation.....	88
Figure 85: Chiller 5 Supply/Return Chilled Water – Chiller 5 a & b, Chiller ¾ c	90
Figure 86: Guest rooms with windows and curtains	93
Figure 87: Thermal images (Left) clear image (right) of glazing and solar radiation in guest room...	93
Figure 88: FSSFE Steam and Hot Water Pipes – Thermal Losses	95
Figure 89: Jacket insulation on valves and heat exchangers	97
Figure 90: Cold rooms in FSSFE	98
Figure 91: Power and Speed Variations.....	99
Figure 92: ECM savings v/s PSC motors.....	102

Figure 93: Current 18KW water heater	103
Figure 94: Example of a Heat Pump installation for water heating	103
Figure 95: Current Carrier Water Cooled Chillers (CH3/CH4) serving upper floors	105
Figure 96: Water-Cooled MagLev® Modular Heat Recovery Chiller with Shell and Tube (Left) ; VME II®	106
Figure 97: Chiller Retrofit KWh – Before and After the retrofit.....	106
Figure 98: Chiller Retrofit- Heat Recovery	107
Figure 99: Example of Dry Coolers for condenser water cooling.....	108
Figure 100: Current Heat Pump CW setpoint	109
Figure 101: FSSFE BMS Chiller management (Below) 345 California Center building Chiller Plant BMS (Next Page).....	110
Figure 102: Annual Carbon Emissions under different EEMs implementation scenarios	117

Abbreviations

AC	Air Conditioning
BMS	Building Management System
CHW	Chilled Water System
CO ₂	Carbon Dioxide
COP	Coefficient of Performance ($KW_{cooling}/KW_{electrical}$)
EEM	Energy Efficiency Measures
ESCO	Energy Services Company
HVAC	Heating Ventilation Air Conditioning
IRR	Internal Rate of Return
KVA	kilo Volt Ampere
KW	Kilowatt
KWh	Kilowatt-hour
LED	Light Emitting Diode
m	Meter
m ²	Square meter
M3	Cubic meter
MDB	Main Distribution Board
°C	Degree Celsius
PF	Power Factor
RT	Refrigerated Tons
TR	Transformer
UPS	Uninterruptible Power Supply
VAV	Variable Air Volume
VFD	Variable Frequency Drive
W	Watt
MJ	Mega Joules
AED	United Arabs Emirates Dirham
JSIR	Jumeirah Saadiyat Island Resort

Conversion Factors

from	to	multiply by
meter	Feet	3.2808399
meter	Inches	39.370079
kg	Pounds	2.2046226
tons	Kg	1000
liter	Gallons (US beer)	0.264179
liter	Cubic feet	0.035315
kWh	MJ	3.6
kWh	BTU	3412
W/m ²	BTU/ft ²	0.317
kJ/kg K	BTU/lb °F	0.2388
W (heat flow rate)	BTU/h	3.412
W/m ² k	BTU/ft ² h °F	0.1761
kJ/kg	BTU/lb	0.4299
MJ/m ³	BTU/ft ³	26.84
Mm Hg (mercury)	Pa	133.332
Ft of water	kPa	2.98898
m ³ /kg	ft ³ /lb	16.02
m/s	ft/s	3.281

1. Introduction

Following an agreement with the Hotel Management, Energy Efficiency Group **EEG** has undertaken an ASHRAE Level II energy audit of the Four Seasons Hotel San Francisco at Embarcadero, located in San Francisco, USA (*abbreviated by FSSFE throughout this report*).

The energy audit was carried out by taking thorough field surveys and extensive energy analysis in the office. A close cooperation was made with the facility department throughout the audit who showed great support and provided quality assistance.

The present report is divided in two main sections. The first section includes a general description of the utilities costs (energy baseline) and the operation of all existing equipment (energy balance). The second section of the report comprises a list of suggested energy efficiency measure with

1.1. Energy Efficiency Rationale

Simply put, Energy Efficiency is using less energy to accomplish the same amount of work needed. Ever since energy emerged as a top priority for policymakers, it has become a critical economic issue. In other words, unsustainable energy supply has real implications. Efficient buildings offer considerable opportunities to save money while reducing greenhouse gas emissions and taking leadership in sustainable development.

Within all sectors, energy costs may be a small percentage of turnovers, but reducing those costs can directly impact the profitability as they go straight to the bottom line, making businesses more competitive, especially with the rising of energy prices. The implementation of energy efficiency and sustainability measures increases levels of occupants' comfort as well as improving staff's general morale.

Energy audits include a review of all energy systems such as cooling, lighting, heating, ventilation, and building envelope. An analysis then serves the purpose of identifying where and how a facility uses energy and developing the set of Energy Efficiency Measures [EEMs] that could optimize the overall energy costs and which can range from small energy projects to major capital investments on operations and maintenance.

The audit focuses on a first phase on Actions with little or no cost which are improvements in current operations and practices that involve little or no cost of execution [LCNC]. Then it develops a road map for additional Energy efficiency measures that would cover all loads and infrastructure and will have different levels of CAPEX (Capital Expenditures) and Prioritization.

1.1. Energy Audit Goals and Tasks

EEG's energy audit is considered an Investment Grade Energy audit which exceeds the ASHRAE Level II Procedure for Commercial Building Energy Auditing,

1.1.1. End Goals

The goals of an energy audit are to:

- ✓ Identify the types and costs of energy/utilities use.
- ✓ Understand how the energy is being used – and possibly wasted.
- ✓ Determine and analyze various EEM alternatives such as improved operational techniques and/or new equipment that could substantially reduce energy costs.
- ✓ Perform an economic analysis on those alternatives and determine which ones are cost effective for the business or industry involved.

1.1.2. Methodology

The energy audit's methodology includes the following set of tasks:

- ✓ Gathering all historical utilities billing database.
- ✓ Gathering all facility layout, description, load data, operational hours.
- ✓ Undertake an in-depth field survey and auditing covering all sections and utilities end uses of the facility, completing the load inventory, analyzing equipment status & operational performances...
- ✓ Perform all needed real time measurements and data logging.
- ✓ Build an Energy Simulation of the facility using an Energy Analysis software. This modeling will allow us to have a detailed load and cost breakdown (Energy balance) along an in depth study on the consumption of the facility.
- ✓ Investigate potential Energy Efficiency Measures (EEMs) from the low cost/no cost ones to those with low CAPEX and quick returns and then those with higher CAPEX and longer returns.
- ✓ Develop the chosen EEMs with their impact on the energy consumption and related financials.
- ✓ Prepare a comprehensive report that would be divided into 2 main parts and the appendixes:
- ✓ In depth analysis of the present situation including utilities analysis, energy baseline and energy balance, detailed systems description...
- ✓ EEMs: Technical analysis, savings development, financial analysis.
- ✓ Appendixes: Full load inventory list and any supporting technical documentation.

2. Facility General Description

2.1. Introduction

The Four Seasons San Francisco at Embarcadero is located in the 345 California Center building, at 222 Sansome Street. The hotel takes up the top 11 floors of the 48-story building, making it the highest hotel in San Francisco. Originally, the hotel was built and started operation back in 1986 with a different operator and then in 2019, it closed for renovations and was rebranded as the Four Seasons San Francisco at Embarcadero. The property opened in October 2020, but then closed again in December 2020 due to the Covid situation. and then finally reopened in May 2021. It consists of 155 guest rooms (127 standard guest rooms and 28 suites) over a total built up area of about 15,000 m²/161,500 ft².

Figure 1: Four Seasons San Francisco at Embarcadero upper floors



In addition to the guest rooms' floors (top 11 floors), the property occupies also three floors; the first floor includes the lobby and the only existing F&B Outlet 'Lounge & Bar' while the second floor includes the main kitchen as well as function rooms that are currently used only for internal gatherings and for storage.

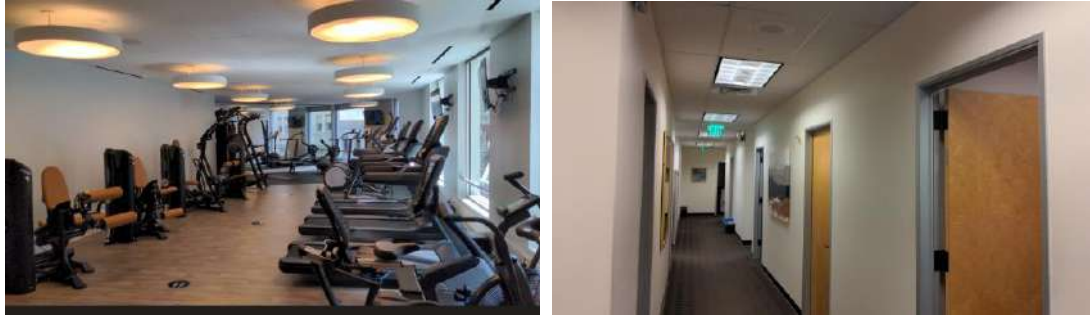
Figure 2: Lobby & Bar F&B outlet (Left) , function room in the 2nd floor (right)



The third floor is partially occupied by the hotel and is leased from the tower . It includes administration offices, the Gym and the SPA which still hasn't opened.

It is to be noted that the third floor's electrical loads are all connected to the tower's feeders and therefore the related electricity consumption is not part of the hotel's utility billing and is considered part of the leasing fees.

Figure 3: Gym (Left) , administration offices (right)



In addition to the previous floors, the hotel occupies part of the basement where the employee canteen, HR offices, lockers and the laundry are located.

2.1. Building Envelope

A building envelope is a major component of buildings energy performance and the physical separator between conditioned and unconditioned environments mainly including the resistance to heat and noise transfer. The building envelope includes all elements of the outer shell that maintain a dry, heated, or cooled indoor environment and facilitate its climate control. It is considered inefficient when it allows high rates of heat transfer between interior spaces and the outdoor environment. Eliminating heat gain and heat loss completely is not possible, but they can be minimized to improve efficiency.

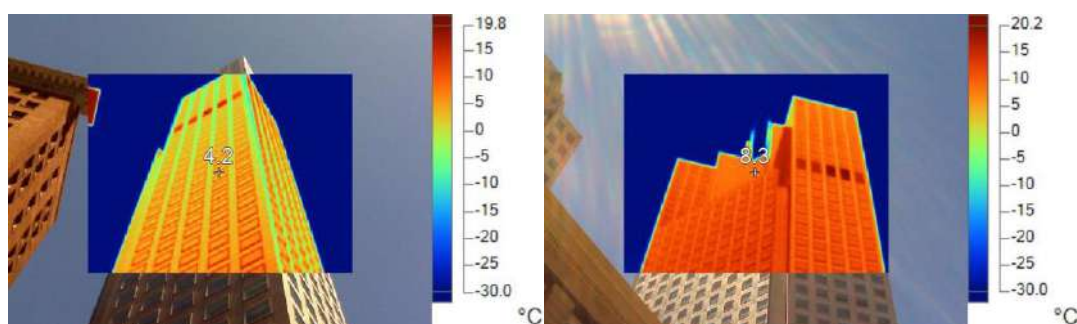
- **Architecture and Glazing:** California Tower where the FSSFE is located was originally built in 1986, and its façade never changed since . The tower consists of a steel frame with exterior granite panels features 88" x 88" Skidmore Signature Single Glazed Window as can be seen below. The windows to wall ratio is about 60%.

Figure 4: FSSFE Façade



The thermal images of the façade show a uniform heat gains which is both an indication of a low reflective glazing and the mild weather of San Francisco with the minimal difference between the outdoor and indoor temperatures.

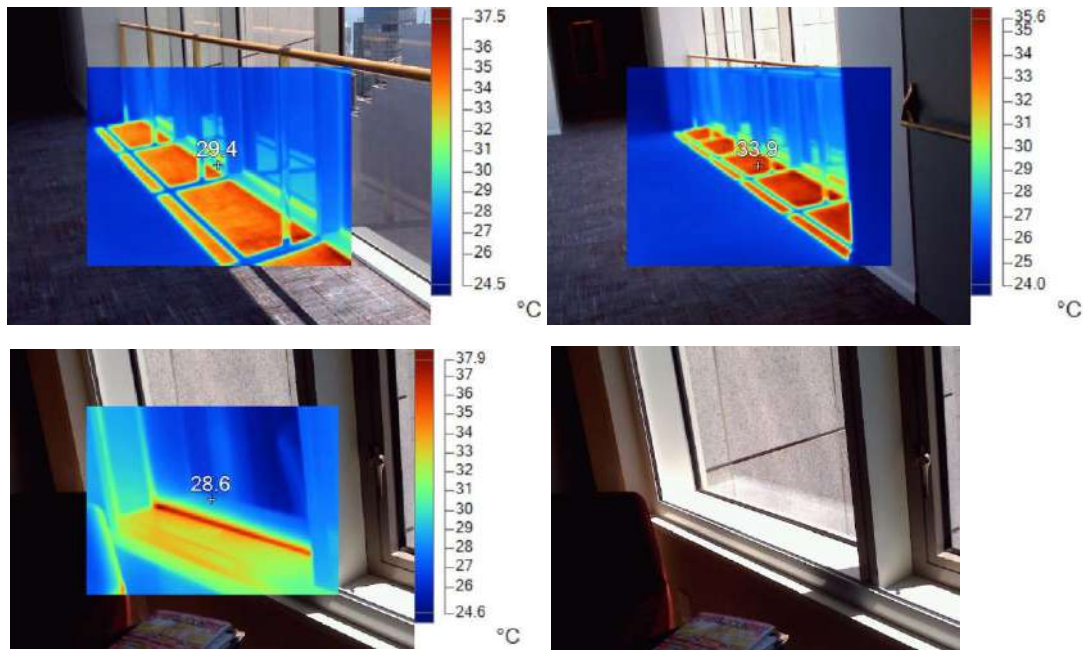
Figure 5: FSSFE Façade



As mentioned, all windows are of the single glazing type Class A, steel frame with exterior granite panels; features 88" x 88" Skidmore Signature Window. Completed in 1986, which obviously have high heat gains generally but the mild weather of San Francisco reduces their impact.

The below thermal images show the heat gains in the glazed corridor in between the two towers as well as in one of the guest room's window.

Figure 6: FSSFE Glazing Thermal Images



- **Entrance:** The entrance in FSSFE is done through mechanical doors which usually is a source of untreated air infiltration but is acceptable in the climate of San Francisco .

Figure 7: FSSFE Entrance



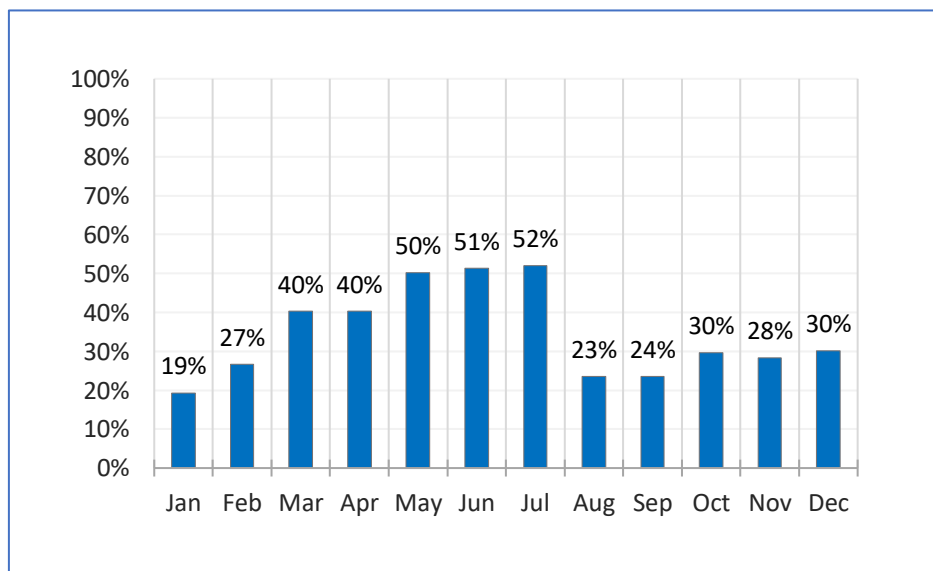
2.2. Occupancy

The following table depicts the occupancy rate variation of the resort during the baseline year which was chosen to be from July 2021 until June 2022. As can be seen, the occupancy was fluctuating during the baseline period between a minimum of 19% in January 2022 and a maximum of 52% in July 2021. The average occupancy rate throughout the baseline year is 35%.

Table 1: Monthly Occupancy Rate in FSSFE through the Baseline Period

Month	Occupancy	Availability	Occupancy Rate
Unit	Room Nights	Room Nights	%
Jan	928	4,805	19%
Feb	1,159	4,340	27%
Mar	1,936	4,805	40%
Apr	1,873	4,650	40%
May	2,410	4,805	50%
Jun	2,387	4,650	51%
Jul	2,497	4,805	52%
Aug	1,129	4,805	23%
Sep	1,093	4,650	24%
Oct	1,420	4,805	30%
Nov	1,319	4,650	28%
Dec	1,449	4,805	30%
TOTAL	19,600	56,575	35%
Maximum	2,497 [Jul]	4,805 [Jan]	52% [Jul]
Average	1,633	4,715	35%
Minimum	928 [Jan]	4,340 [Feb]	19% [Jan]

Figure 8: Monthly Occupancy Rate Variations



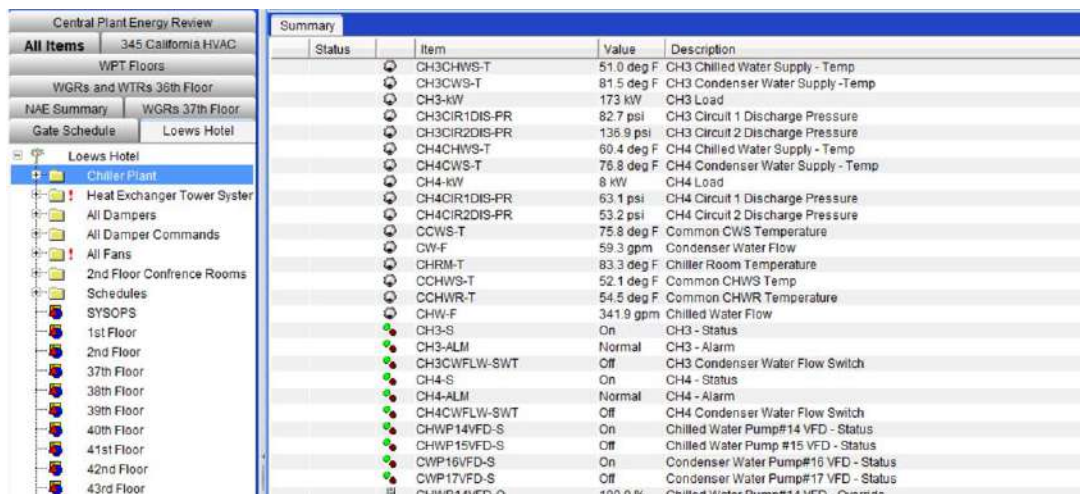
2.3. Automation and Metering

BMS

FSSFE has a Johnson Control Metasys BMS which is part of the 345 California Center one and which basis has been upgraded in 2009. The hotel's part was seen to virtually not having any graphical user interface on the key operations such as the chilled water plants and the district heating ones. In addition, automation implementation is rather limited. The below snapshot showcases a typical info and control page in the BMS.

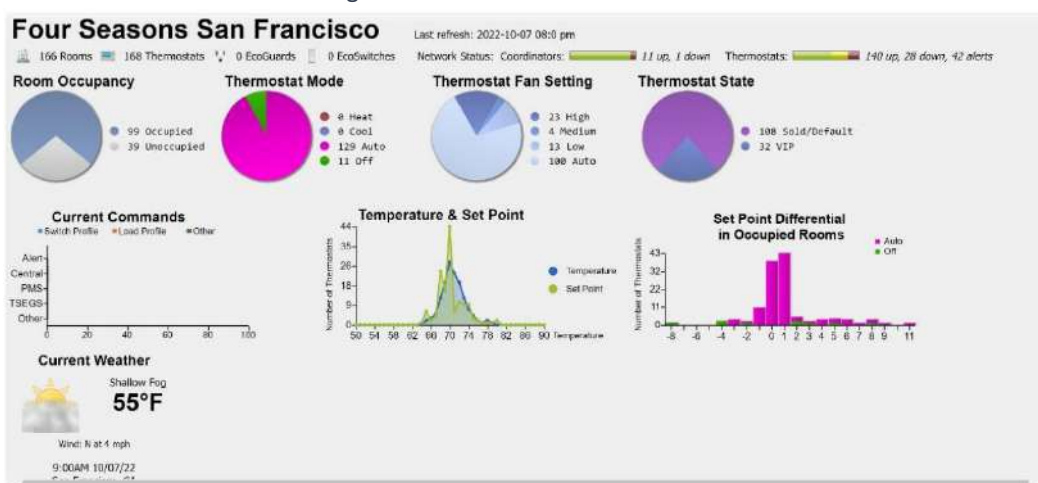
It is also important to note that the whole HVAC infrastructure in the hotel is of the pneumatic type with the absence of DDCs.

Figure 9: FSSFE- METASYS BMS snapshot



In addition to the BMS FSSFE has a Telkonet EcoCentral GRMS system that controls the thermostats and lighting systems of all guest rooms through a wireless infrastructure and a web based platform.

Figure 10: Telkonet Ecocentral GRMS



2.4. San Francisco Climatic Conditions

San Francisco has a warm-summer Mediterranean climate characteristic of California's coast, with moist, mild winters and dry summers. San Francisco's weather is strongly influenced by the cool currents of the Pacific Ocean on the west side of the city, and the water of San Francisco Bay to the north and east. This moderates temperature swings and produces a remarkably mild year-round climate with little seasonal temperature variation. On average, there are 73 rainy days a year, and annual precipitation averages 23.65 inches (601 mm).

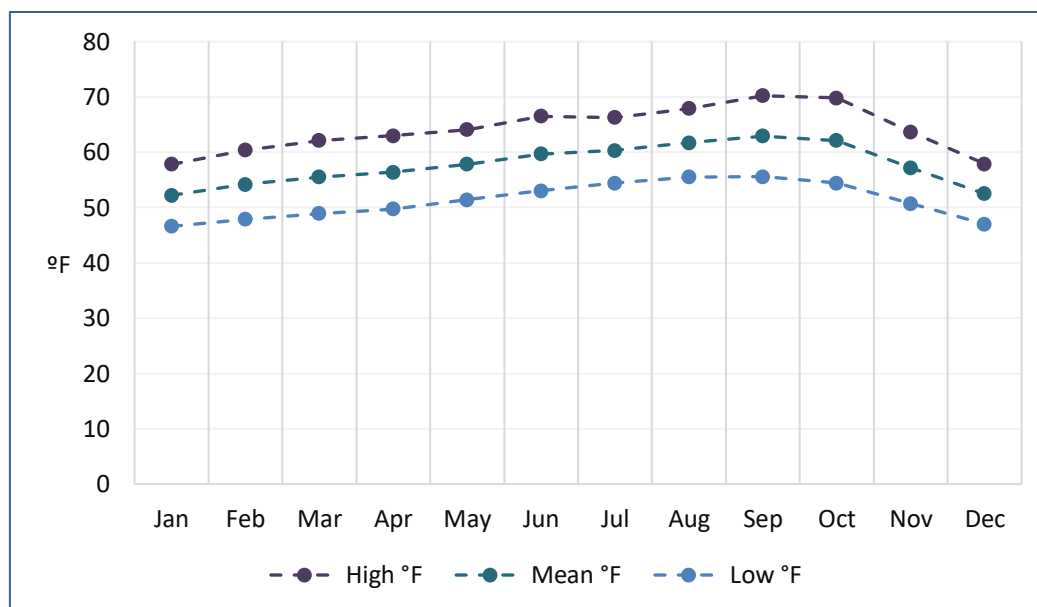
The following table depicts the weather data for San Francisco, which will help in many of the energy analysis present in the following chapters.

Table 2: San Francisco Climatic Data

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
High °F	58	60	62	63	64	67	66	68	70	70	64	58
Mean °F	52	54	56	56	58	60	60	62	63	62	57	53
Low °F	47	48	49	50	51	53	54	56	56	54	51	47
Prec. Inches	8	10	11	3	0	0	1	0	0	0	3	6
RH %	63%	60%	55%	50%	49%	55%	54%	52%	57%	58%	59%	64%
Sunshine H.	185.9	207.7	269.1	309.3	325.1	311.4	313.3	287.4	271.4	247.1	173.4	160.6

Source: Climate-Data.org

Figure 11: Temperature Variations in San Francisco



Degree Days:

Degree days are essentially a simplified representation of outside air-temperature data. They are widely used in the energy industry for calculations relating to the effect of outside air temperature on building energy consumption.

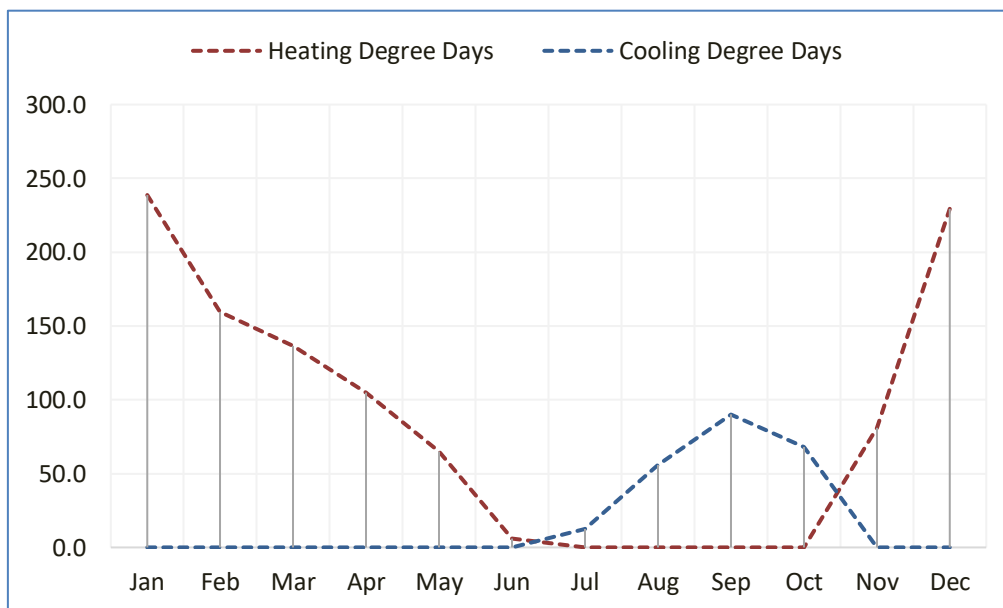
"Heating degree days", or "HDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was lower than a specific "base temperature" (or "balance point"). They are used for calculations relating to the energy consumption required to heat buildings.

"Cooling degree days", or "CDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was higher than a specific base temperature. They are used for calculations relating to the energy consumption required to cool buildings.

In the present report, CDD are calculated for San Francisco with a base temperature of 59.9°F (15.5°C). It was found that the maximum cooling degree day was 90 recorded in September, while the maximum heating degree day is 238.7 in January.

The below chart provides the CDD and HDD Variations in San Francisco across the baseline year.

Figure 12: CDD and HDD Variations in San Francisco



2.5. Utilities: Energy Sources and Water

FSSFE's utility sources are electricity, natural gas, water and district heating.

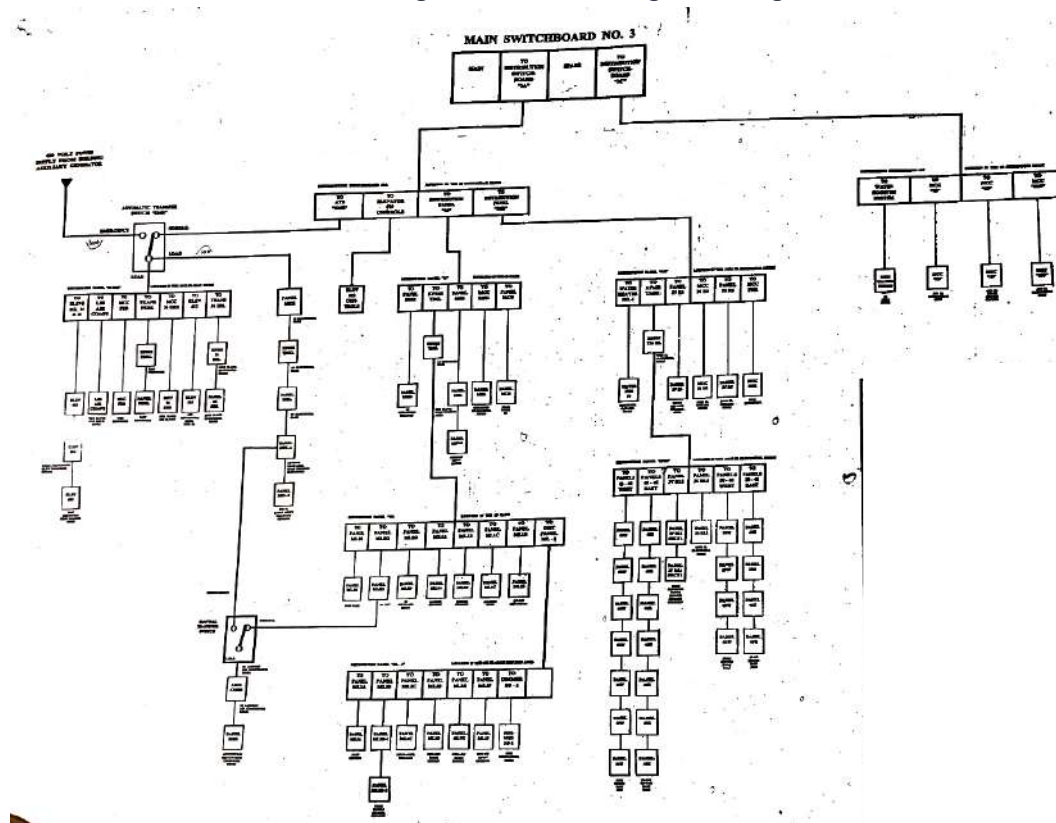
Electricity

Electricity in FSSFE is provided the local electric utility PG&E and distributed to all the hotel's panels through two main feeding lines from the Main Switchboard #3 A as can be seen in the below single line diagram.



It is to be noted that there are various parts of the Hotel property that are electrically serviced from the tower's feeders and are not therefore part of the property's utility's billing. One such example is the leased 3rd floor where the administration, Gym and SPA are located.

Figure 13: Electrical Single Line Diagram



Electricity average rate is at **\$0.23/KWh** for the the baseline period and the total charges are at **\$480,903** representing **68%** of the total energy & water related utilities expenditures.

Natural Gas

Natural Gas is provided to FSSFE and used for all Kitchens' operations as well for the gas driven boilers located in the 4th floor and used to produce domestic hot water for the lower floors.

Natural Gas rate is at **\$1.76/Therm** and the total charges are at **\$6,377** representing **1%** of the total utilities expenditures.

Water

Water in FSSFE is supplied by the local water supplier and distributed across all the hotel

The total water charges are at **\$115,755** per year representing **16%** of the total utilities cost.

District Heating

The district heating line is of the steam supply type with heat exchangers located in the 48th and B1 floors . They are responsible for providing space heating to the guest rooms and lower floor respectively as well as producing hot water in the upper floors.

District Heating rate is at **\$40.99/Mlb** and the total charges are at **\$99,034** representing **14%** of the total utilities expenditures.

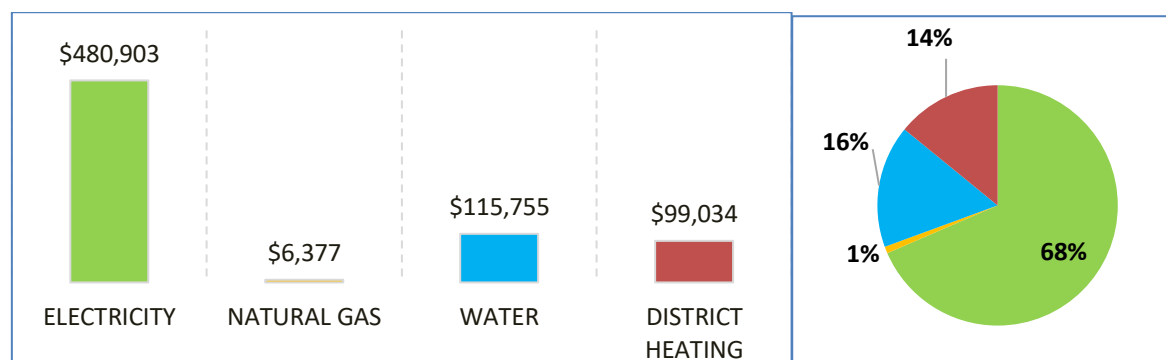
Total Utilities Cost

The below table provides the cost of each utility supply as compiled for the baseline year.

Table 3: Annual Utilities Costs

UTILITY	ELECTRICITY	NATURAL GAS	WATER	DISTRICT HEATING	ALL UTILITIES
TOTAL COST	\$480,903	\$6,377	\$115,755	\$99,034	\$702,069
MAXIMUM	61,720 (Jun)	830 (Jan)	10,991 (Jun)	13,351 (Jan)	81,286 (Jun)
AVERAGE	\$40,075	\$531	\$9,646	\$8,253	\$58,506
MINIMUM	31,117 (Dec)	343 (May)	8,675 (Jan)	5,937 (Sep)	51,865 (Nov)
PERCENTAGE	68%	1%	16%	14%	100%

Figure 14: Utilities cost breakdown



3. Utilities Accounting

The following sections detail the accounting numbers for the four utilities over the **baseline period from July 2021 until June 2022**. The annual consumption and related cost baseline are the aggregated consumption and associated costs for FSSFE throughout the baseline period. This energy baseline serves as a benchmark consumption to assess the current energy performance along with the calculation of all the proposed Energy Efficiency Measures. The present energy audit and the related energy simulation cover the full facility, excluding the leased part of the third floor.

3.1. Electricity

3.1.1. Electricity Cost Breakdown

The electricity bill of FSSFE has two components: the PG&E Delivery services and the Clean Energy Electricity Generation Services . Each source has demand charges, energy charges, as well as other charges, .

i. PG&E – Delivery Services

As can be seen in the below snapshot of the utility bill. The PG&E delivery services charges include multiple tariff on the electricity consumed as well as demand charges (three time slots), power factor adjustment, generation credit (which has been rising as of Q2 of 2022) and other charges.

It is to be noted that the Power Factor (PF) of FSSFE is on the average below 0.85 due to the absence of an adequate Automatic PF correction at the main distribution panels and the ageing mechanical infrastructure. However, the implications on the utility bills are not large as the penalty is small (below \$25 per month) and the maximum demand is measured in KW rather than KVA (which would have been affected by the PF)

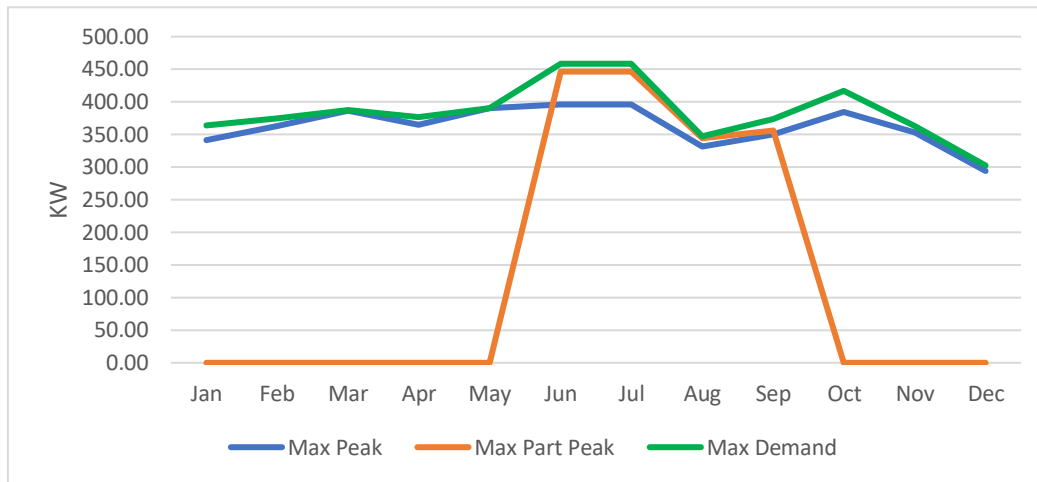
Figure 15: PG&E – Delivery Services charges

Customer Charge	30 days @ \$31.54779	\$946.43
Demand Charge ¹		
Max Peak	396.000000 kW @ \$32.90000	13,028.40
Max Part Peak	446.400000 kW @ \$6.81000	3,039.98
Max Demand	458.400000 kW @ \$26.07000	11,950.49
Energy Charges		
Peak	43,097.700000 kWh @ \$0.19598	8,446.29
Part Peak	32,186.700000 kWh @ \$0.15878	5,110.60
Off Peak	112,170.750000 kWh @ \$0.13246	14,858.14
Power Factor Adjustment (@ 83.00% Power Factor)		18.75
Generation Credit		-32,794.19
Power Charge Indifference Adjustment		3,505.41
Franchise Fee Surcharge		170.58
San Francisco Utility Users' Tax (7.500%)		2,108.27
SF Prop C Tax Surcharge		277.17
Total PG&E Electric Delivery Charges		\$30,666.32

A key component of of the PG&E charges is the Generation credit which topped the 50% credit from the total bill in June 2022.

The below chart shows the three Maximum Demand logs over 12 months. It is seen that the demand varies from 350KW to about 450KW which is topped in June and July , eventually due to higher chillers operation.

Figure 16: PG&E – Maximum Demand



The below table provides the charges of PG&E over the baseline year with the ‘Other charges’ having an annual credit of \$119,000 due to the Generation credit.

Table 4: PG&E Cost Breakdown

Month	Demand Charges	Energy Charges	Other Charges	Total Cost
-	\$	\$	\$	\$
Jan-22	10,051	22,842	-6,548	26,345
Feb-22	10,369	5,811	7,107	23,287
Mar-22	10,784	21,844	-13,234	19,394
Apr-22	10,459	22,418	-13,551	19,327
May-22	10,854	17,923	-7,952	20,825
Jun-22	28,019	28,415	-25,767	30,666
Jul-21	28,019	28,415	-25,967	30,466
Aug-21	18,748	19,816	-11,418	27,147
Sep-21	19,935	15,168	-8,106	26,997
Oct-21	10,229	21,516	-6,660	25,086
Nov-21	8,935	15,356	-1,647	22,644
Dec-21	7,455	18,093	-5,271	20,277
Total	173,857	237,617	-119,014	292,459
Maximum	28,019 (Jun)	28,415 (Jun)	7,107 (Feb)	30,666 (Jun)
Average	14,488	19,801	-9,918	24,372
Minimum	7,455 (Dec)	5,811 (Feb)	-25,967 (Jul)	19,327 (Apr)

ii. Clean Power – Electric Energy Charges

Similar to the PG&E charges, the Clean Power ones include the mix of energy (KWh) , Demand (KW) and other charges.

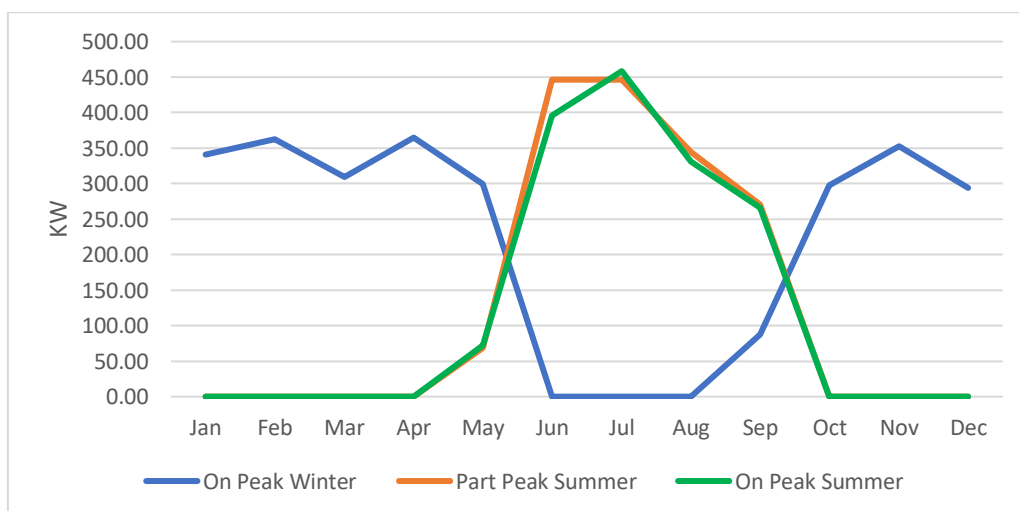
It is to be noted that FSSFE is currently **purchasing 33% of its total electricity from clean power** (Renewable Energy). The below figure provides a snapshot of the Clean Power charges in a typical utility bill.

Figure 17: Clean Power – Maximum Demand

Rate Schedule: B-19-S				
Demand - Part Peak - Summer	342.240000	KW	@ \$2.74000	\$937.74
Demand - Part Peak - Summer	104.160000	KW	@ \$3.08000	320.81
Demand - On Peak - Summer	303.600000	KW	@ \$18.85000	5,722.86
Demand - On Peak - Summer	92.400000	KW	@ \$21.22000	1,960.73
Generation - Off Peak - Summer	26,173.175000	kWh	@ \$0.07540	1,973.46
Generation - Off Peak - Summer	85,997.575000	kWh	@ \$0.09127	7,849.00
Generation - Part Peak - Summer	7,510.230000	kWh	@ \$0.09738	731.35
Generation - Part Peak - Summer	24,676.470000	kWh	@ \$0.11787	2,908.62
Generation - On Peak - Summer	10,056.130000	kWh	@ \$0.12847	1,291.91
Generation - On Peak - Summer	33,041.570000	kWh	@ \$0.15550	5,137.96
Net Charges				28,834.44
Local Utility Users Tax				2,162.58
Energy Commission Surcharge				56.24
CleanPowerSF has new rates! Visit cleanpowersf.org/rates to learn more.				
Total CleanPowerSF Electric Generation Charges				\$31,053.26

The below chart provides the three KW Demands logs for the 12 months as compiled from the Clean Power bills. Winter months have an average of 320KW as maximum demand while summer month tops the 450KW .

Figure 18: Clean Power – Electric Generation charges



The below table provides the Demand, Electricity and other charges of Clean Power .

Table 5: Clean Energy Cost Breakdown

Month	Demand Charges	Energy Charges	Other Charges	Total Cost
-	\$	\$	\$	\$
Jan-22	675	11,591	977	13,243
Feb-22	718	2,712	8,232	11,661
Mar-22	845	13,488	1,128	15,461
Apr-22	817	11,641	4,761	17,220
May-22	2,213	15,655	2,258	20,127
Jun-22	8,941	19,892	2,220	31,053
Jul-21	8,941	19,892	1,820	30,653
Aug-21	5,799	10,028	851	16,678
Sep-21	4,800	9,950	824	15,573
Oct-21	710	10,178	766	11,654
Nov-21	699	8,148	3,282	12,128
Dec-21	582	9,459	800	10,841
Total	35,740	142,633	27,920	206,293
Maximum	8,941 (Jun)	19,892 (Jun)	8,232 (Feb)	31,053 (Jun)
Average	2,978	11,886	2,327	17,191
Minimum	582 (Dec)	2,712 (Feb)	766 (Oct)	10,841 (Dec)

3.1.2. Electricity Overall Consumption and Cost

The total electricity charges are the aggregate of both the PG&E and Clean Power components. Electricity monthly consumptions and costs during the baseline period are highlighted in the table below. The annual electricity consumption of the facility is **2,129,180 KWh**.

Table 6: FSSFE Monthly Electricity Consumption and Cost

Month	Consumption		Rate	Deviation
Unit	KWh	\$	\$/KWh	-
Jan-22	218,370	\$39,588	\$0.181	19%
Feb-22	174,338	\$34,948	\$0.200	-2%
Mar-22	168,517	\$34,855	\$0.207	-5%
Apr-22	173,136	\$36,547	\$0.211	-2%
May-22	177,536	\$40,952	\$0.231	0%
Jun-22	187,455	\$61,720	\$0.329	5%
Jul-21	180,293	\$43,270	\$0.240	2%
Aug-21	162,693	\$43,824	\$0.269	-9%
Sep-21	168,063	\$42,570	\$0.253	-6%
Oct-21	188,505	\$36,740	\$0.195	6%
Nov-21	174,041	\$34,772	\$0.200	-2%
Dec-21	156,233	\$31,117	\$0.199	-14%
Total	2,129,180	\$480,903	\$0.23	-
Maximum	218,370 (Jan)	61,720 (Jun)	0.33 (Jun)	19% (Jan)
Average	177,432	\$40,075	\$0.23	-
Minimum	156,233 (Dec)	31,117 (Dec)	0.18 (Jan)	-14% (Dec)

The below figures show the electricity monthly consumption variation along with the occupancy and CDD respectively during the baseline year. Electricity consumption seems to be not highly correlated to both occupancy and CDD, since the weather in San Francisco is mild,

Figure 19: FSSFEE Electricity Monthly Variations along with the occupancy

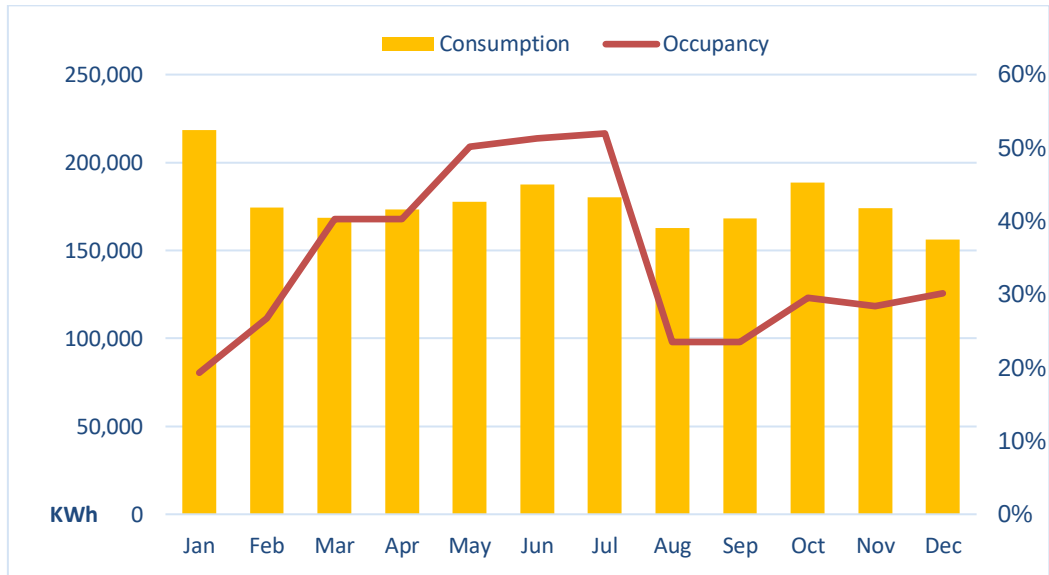
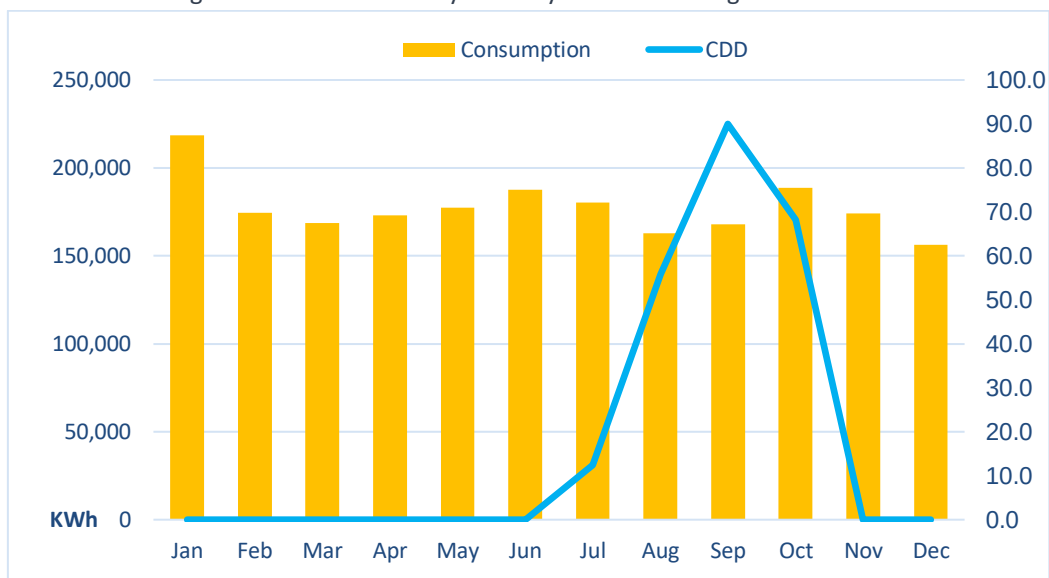


Figure 20: FSSFEE Electricity Monthly Variations along with the CDD

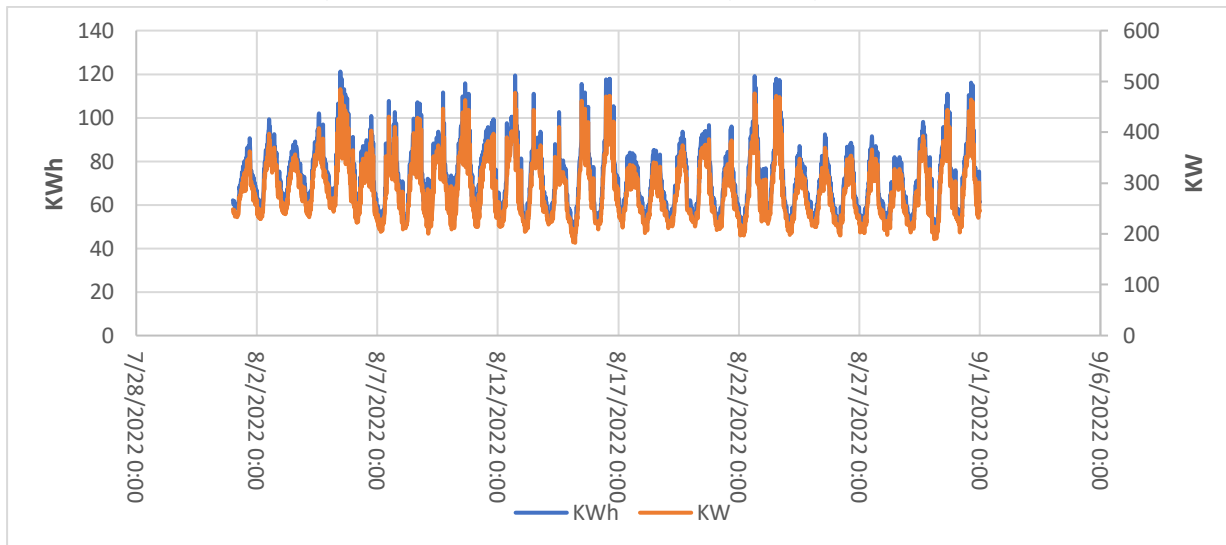


3.1.3. Load Profile

The below two charts provide a clear electrical load profile in August for FSSFE.

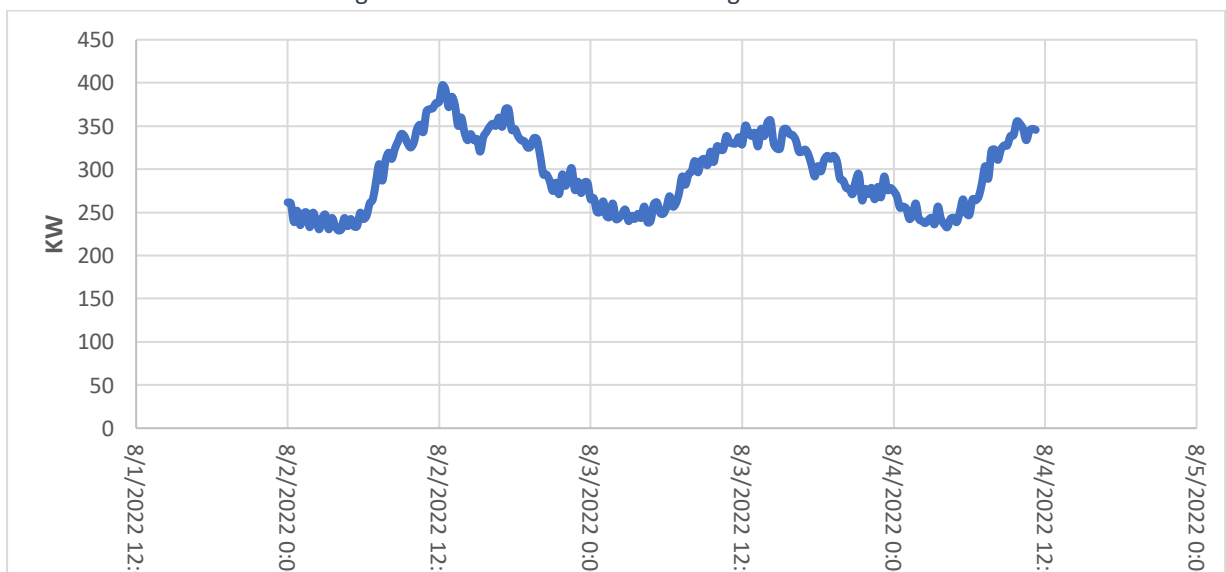
The first chart combines both the KWh and KW Demand (recorded every 15 minutes) for the whole month of August and showing a clear pattern between days and nights with average night lows of 55KWh per 15 minutes or 220 KWh for the full hour while the daily peaks topped the 121KWh per 15 minutes in some days but averaged around 100 KWh per 15 minutes or 400 KWh per hour.

Figure 21: FSSFE KWh and KW recording for August 2022



The below chart shows the KW Demand for 3 days clearly depicting the day/night variations with the night loads averaging 240KW and start increasing around 6hAM to reach peaks between 12hPM and 4hPM and then start going down as of 6hPM

Figure 22: FSSFE KW Demand for August 2-4 2022



3.1.4. Regression Analysis

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationship among variables. It focuses on the relationship between a dependent variable and one or more independent variables.

In the case of FSSFE, for a given month, a relationship exists between the energy consumption (the main parameter) and the following variables: the occupancy expressed in Room Nights and the weather conditions (mathematically represented by the heating degree days). Using the regression analysis, an empirical equation is modeled to generate projected energy consumption as a function of the above-mentioned variables. The model is linear, and the equation will be of the form:

$$\text{KWh} = A \times \text{OCC} + B \times \text{HDD} + C$$

OCC is the occupancy rate expressed in Room Nights and HDD is the Heating Degree Day

A, B and C are the coefficients to be calculated in the process of the regression analysis

In the present regression analysis, the historical figures used for the modeling are found in the below table.

Table 7: Regression Analysis Variables

Month	Occupancy	HDD
Unit	Room Nights	°C.Days
Jan-22	928	239
Feb-22	1,159	160
Mar-22	1,936	136
Apr-22	1,873	105
May-22	2,410	65
Jun-22	2,387	6
Jul-21	2,497	0
Aug-21	1,129	0
Sep-21	1,093	0
Oct-21	1,420	0
Nov-21	1,319	81
Dec-21	1,449	229
Total	19,600	1,021

Using these variables, and the energy consumption for the baseline period of each month, the coefficients of the regression model are obtained, leading to the below:

$$\text{KWh(per month)} = (-0.2) \times \text{OCC} + (28.3) \times \text{HDD} + (175,266.2)$$

The projected consumption would be then calculated based on the above equation and the following table provides both the real baseline electricity consumption as well as the projected one if the regression equation is used.

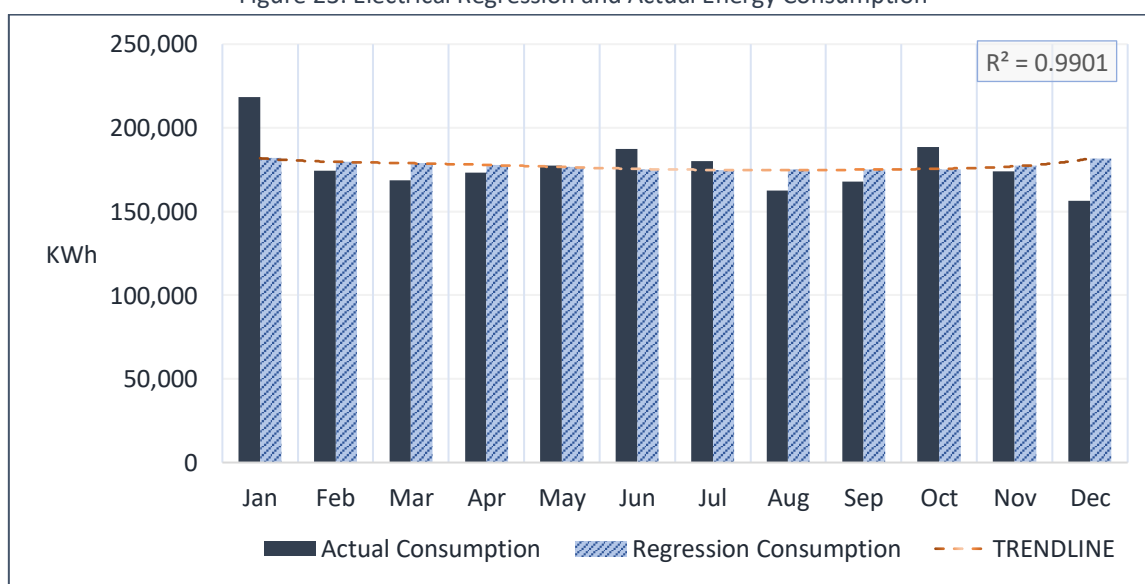
Table 8: Electricity Projected Consumption, Residuals, and variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	KWh	KWh	KWh	%
Jan-22	218,370	181,891	36,479	17%
Feb-22	174,338	179,614	-5,276	-3%
Mar-22	168,517	178,840	-10,323	-6%
Apr-22	173,136	177,959	-4,823	-3%
May-22	177,536	176,748	788	0%
Jun-22	187,455	175,076	12,379	7%
Jul-21	180,293	174,890	5,403	3%
Aug-21	162,693	175,096	-12,403	-8%
Sep-21	168,063	175,101	-7,039	-4%
Oct-21	188,505	175,052	13,453	7%
Nov-21	174,041	177,363	-3,322	-2%
Dec-21	156,233	181,549	-25,316	-16%
Total	2,129,180	2,129,180	0	0%

The residuals are the difference between the actual and the projected energy consumption for the same values of independent variables. The variation is the percentage of difference between the actual and the projected consumption. As can be seen, the annual regression provides an exact match (difference of 0); the following chart gives a clear visibility on the regression model and the actual consumption for the baseline period.

R-squared is a statistical measure linked to the regression model. It is also known as the coefficient of determination, which is the percentage of the response variable variation that is explained by a linear model (explained variation / total variation). R-squared is always between 0 and 100%: the higher the value, the better the model fits the data. In the case of FSSFE, R^2 was seen to be 0.9901, which means that the model can be considered as an acceptable model.

Figure 23: Electrical Regression and Actual Energy Consumption



3.2. Natural Gas

3.2.1. Natural Gas Overall Consumption and Cost

The total Natural Gas consumption for the baseline year is **3,615 Therms** annually at an average rate of **1.76 \$/Therm**. The following table depicts the monthly Gas consumption over the baseline year.

Table 9: FSSFE Monthly Natural Gas Consumption and Cost

Month	Consumption		Rate	Energy Value	Deviation
Unit	Therms	\$	\$/Therms	KWh	-
Jan-22	401	\$830	\$2.07	10,927	25%
Feb-22	313	\$639	\$2.04	8,529	4%
Mar-22	335	\$643	\$1.92	9,128	10%
Apr-22	285	\$468	\$1.64	7,777	-6%
May-22	191	\$343	\$1.80	5,192	-58%
Jun-22	282	\$517	\$1.83	7,697	-7%
Jul-21	292	\$345	\$1.18	7,957	-3%
Aug-21	283	\$345	\$1.22	7,711	-6%
Sep-21	264	\$386	\$1.46	7,202	-14%
Oct-21	292	\$499	\$1.71	7,954	-3%
Nov-21	270	\$546	\$2.02	7,368	-11%
Dec-21	406	\$816	\$2.01	11,063	26%
Total	3,615	\$6,377	\$1.76	98,506	-
Maximum	406 (Dec)	830 (Jan)	2.0700 (Jan)	11,063 (Dec)	26% (Dec)
Average	301	\$531	\$1.74	8,209	-
Minimum	191 (May)	343 (May)	1.1800 (Jul)	5,192 (May)	-58% (May)

The graphs below shows the monthly gas consumption versus occupancy and heating degree days (HDD). Gas consumption is highly correlated to both occupancy and HDD.

Figure 24: FSSFE Natural Gas Monthly Variations vs. occupancy

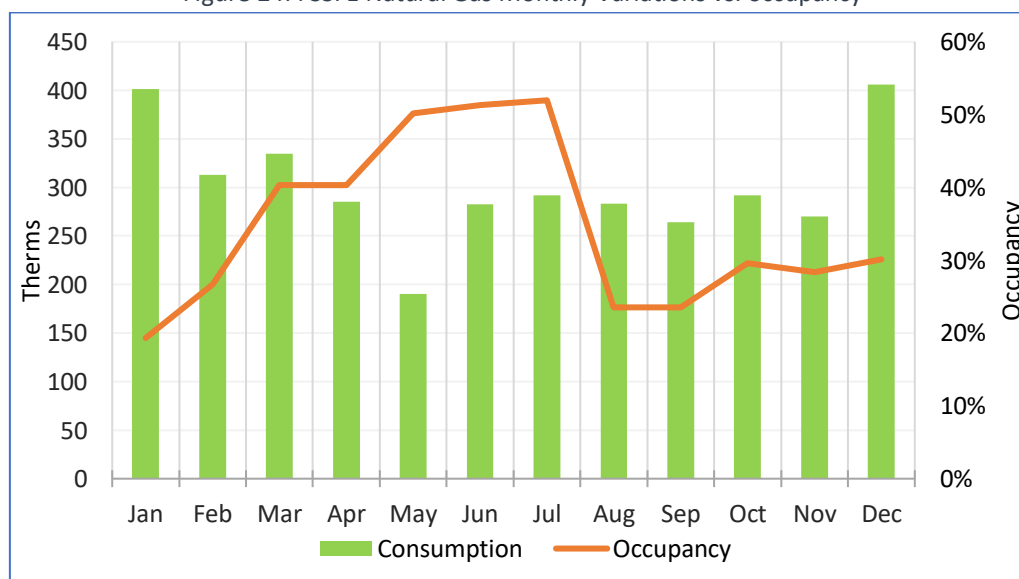
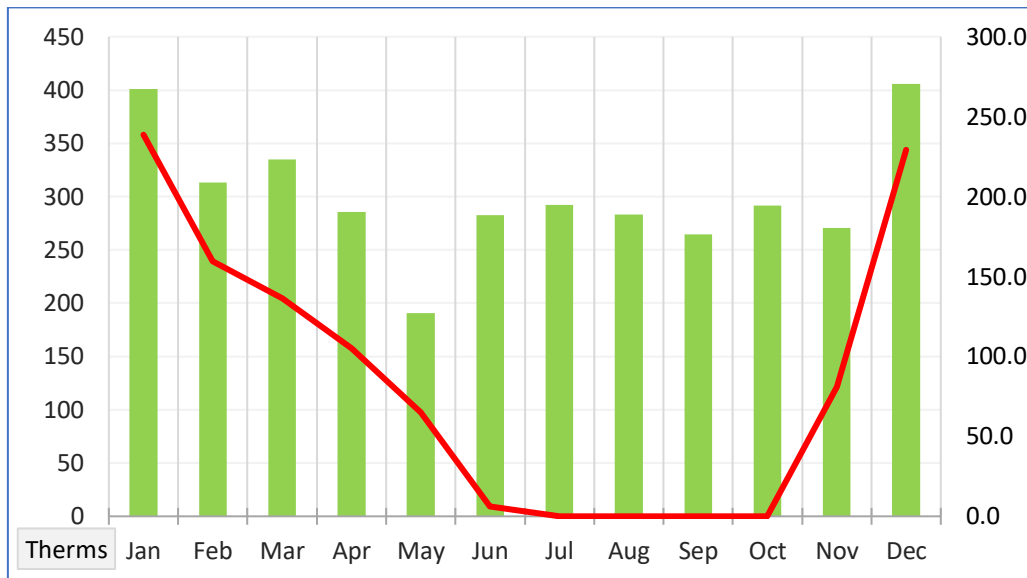


Figure 25: FSSF Natural Gas consumption versus HDD



3.2.2. Regression Analysis

In the present regression analysis, the historical figures used for the modeling are found in the below table.

Table 10: Regression Analysis Variables

Month	Occupancy	HDD
Unit	Room Nights	°C.Days
Jan-22	928	239
Feb-22	1,159	160
Mar-22	1,936	136
Apr-22	1,873	105
May-22	2,410	65
Jun-22	2,387	6
Jul-21	2,497	0
Aug-21	1,129	0
Sep-21	1,093	0
Oct-21	1,420	0
Nov-21	1,319	81
Dec-21	1,449	229
Total	19,600	1,021

Using these variables, and the energy consumption for the baseline period of each month, the coefficients of the regression model are obtained, leading to the below:

$$\text{KWh(per month)} = (-0.021) \times \text{OCC} + (0.4) \times \text{HDD} + (297.6)$$

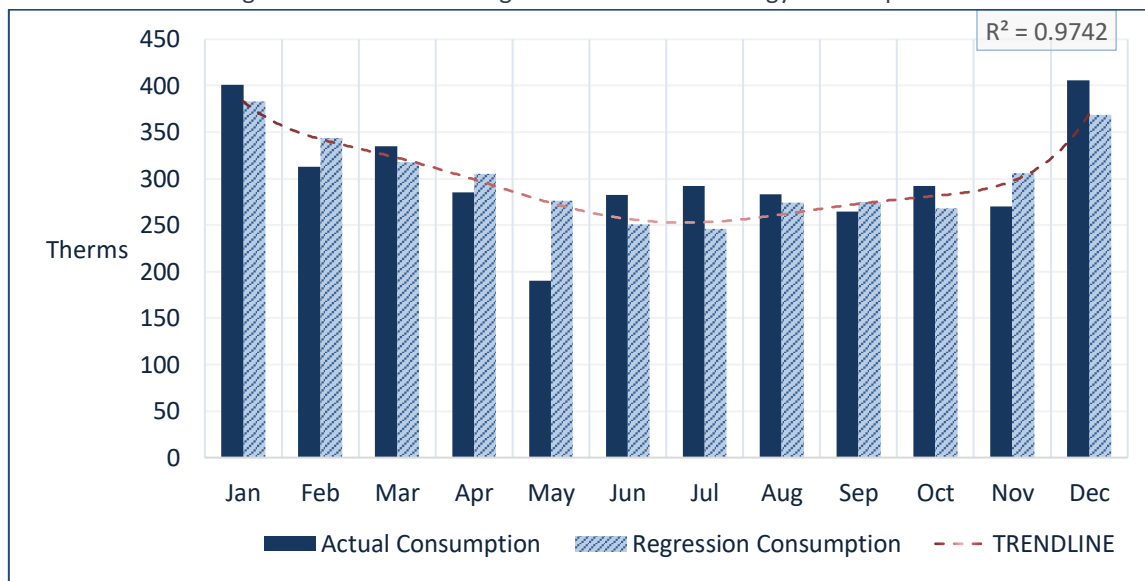
The projected consumption would be then calculated based on the above equation and the following table provides both the real baseline electricity consumption as well as the projected one if the regression equation is used.

Table 11: Natural Gas Projected Consumption, Residuals, and variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	Therms	Therms	Therms	%
Jan-22	401	383	18	4%
Feb-22	313	344	-31	-10%
Mar-22	335	318	17	5%
Apr-22	285	305	-20	-7%
May-22	191	276	-86	-45%
Jun-22	282	251	31	11%
Jul-21	292	246	46	16%
Aug-21	283	274	9	3%
Sep-21	264	275	-11	-4%
Oct-21	292	268	24	8%
Nov-21	270	306	-36	-13%
Dec-21	406	368	38	9%
Total	3,615	3,615	0	0%

In the case of FSSFE, R^2 was seen to be 0.9742, which means that the model can be considered as an acceptable model.

Figure 26: Natural Gas Regression and Actual Energy Consumption



3.3. Water

3.3.1. Water Overall Consumption and Cost

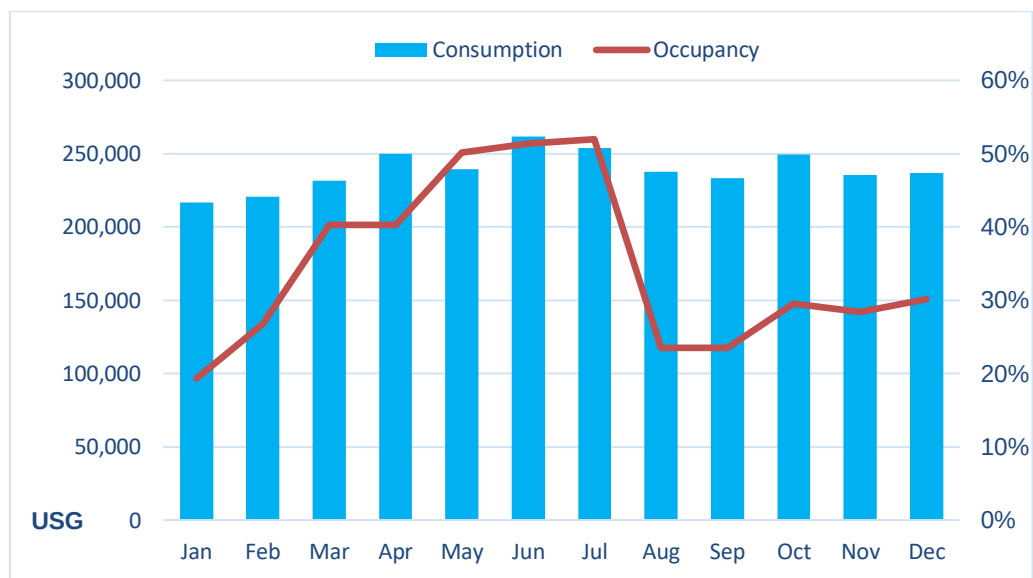
The total water consumption for the baseline year is at **2,866,962 USG** as shown in the following table.

Table 12: FSSFE Monthly Water Consumption and Cost

Month	Consumption		Rate	Deviation
Unit	Gallons	\$	\$/Gallons	-
Jan-22	216,865	\$8,675	\$0.04	-10%
Feb-22	220,682	\$8,827	\$0.04	-8%
Mar-22	231,700	\$8,874	\$0.04	-3%
Apr-22	249,705	\$10,388	\$0.04	4%
May-22	239,425	\$9,960	\$0.04	0%
Jun-22	261,689	\$10,991	\$0.04	9%
Jul-21	253,788	\$10,152	\$0.04	6%
Aug-21	237,642	\$9,506	\$0.04	-1%
Sep-21	233,451	\$9,338	\$0.04	-2%
Oct-21	249,548	\$10,007	\$0.04	4%
Nov-21	235,520	\$9,491	\$0.04	-1%
Dec-21	236,947	\$9,547	\$0.04	-1%
Total	2,866,962	\$115,755	\$0.04	-
Maximum	261,689 (Jun)	10,991 (Jun)	0.04 (Jun)	9% (Jun)
Average	238,914	\$9,646	\$0.04	-
Minimum	216,865 (Jan)	8,675 (Jan)	0.04 (Mar)	-10% (Jan)

The graph below shows the monthly water consumption variation during the baseline year.

Figure 27: FSSFE Water Consumption Monthly Variations



3.3.2. Water Consumption Breakdown

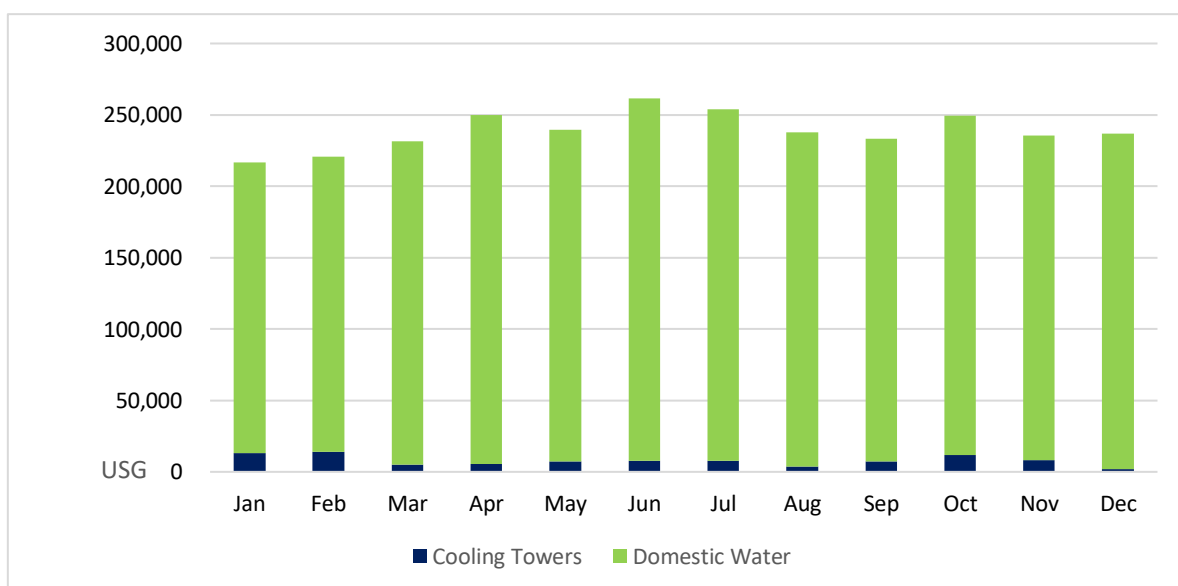
Water in FSSFE has two end uses: domestic usage cooling towers. The table below represents the total water breakdown in the hotel during the baseline period whereby it is seen that cooling towers represent only 4.6% of the total consumption , primarily due to the mild climate across the year of San Francisco. It has been notified that the average water consumption due to evaporation per cooling tower is about 3 gallons per hour.

Table 13: Water Breakdown During the Baseline Period

Month	Cooling Towers	Domestic Water	Total
Unit	Gallons	Gallons	Gallons
Jan-22	13,133	203,732	216,865
Feb-22	14,179	206,503	220,682
Mar-22	5,002	226,698	231,700
Apr-22	5,719	243,986	249,705
May-22	7,490	231,935	239,425
Jun-22	7,932	253,757	261,689
Jul-21	7,543	246,245	253,788
Aug-21	3,492	234,150	237,642
Sep-21	7,230	226,221	233,451
Oct-21	11,983	237,565	249,548
Nov-21	8,101	227,419	235,520
Dec-21	2,058	234,889	236,947
Total	93,862	2,773,100	2,866,962
Maximum	14,179 (Feb)	253,757 (Jun)	261,689 (Jun)
Average	7,822	231,092	238,914
Minimum	2,058 (Dec)	203,732 (Jan)	216,865 (Jan)

The below chart shows the monthly water consumption per end use.

Figure 28: FSSFE Water Breakdown Variations



It is important to note that the property's plumbing fixtures meet fully California Title-20 Water Efficiency standards which are seen in the below table.

Table 14: California Title 20 – Water Efficiency Standards

Plumbing Fixture	Maximum Rate
Kitchen faucets	1.8 gpm
Public lavatory faucets	0.5 gpm
Toilets	1.28 gpf
Urinal flush valves	0.125 gpf
Residential lavatory faucets	1.5 gpm
	1.2 gpm
Shower devices	2.0 gpm
	1.8 gpm

In fact, during the latest renovation, all plumbing fixtures were replaced with high water efficiency as a standard features. During the field audit, it was verified that guest rooms' faucets and showers meet the above seen standards while public toilets also included infrared operation for urinals.

Figure 29: Up: Guest rooms' aerator (Left), Shower (Right)
Down: public toilet's faucet (Left), Urinal (Right)



3.4. District Heating

3.4.1. District Heating Overall Consumption and Cost

As previously mentioned, space heating in FSSFE as well as the production of domestic hot water for the upper floors are provided through a district heating plant. Steam reaches the hotel from one side of the heat exchangers in two separate locations: in the 48th floor, providing space heating and domestic hot water to the guest rooms, and in the B1 floor, providing space heating to the lower floors.

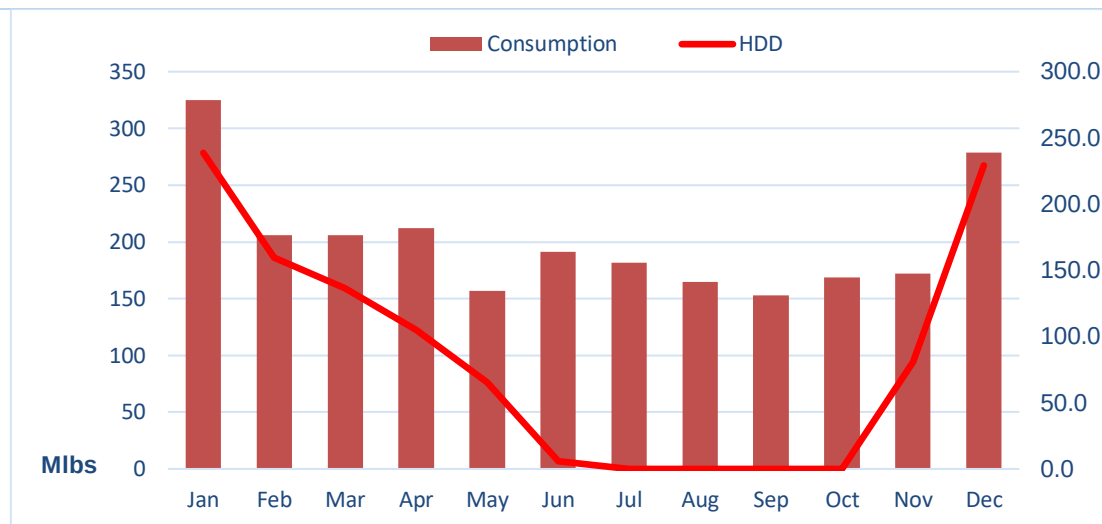
The table below depicts the monthly consumption of district heating through the baseline year. The total consumption is **2,416 Mlbs** with an average tariff of **40.988 \$/Mlb**.

Table 15: FSSFE Monthly District Heating Consumption and Cost

Month	Consumption		Rate	Energy Value	Deviation
Unit	Mlbs	\$	\$/Mlb	KWh	-
Jan-22	325	\$13,351	\$41.08	99,125	38%
Feb-22	206	\$8,679	\$42.13	62,830	2%
Mar-22	206	\$8,677	\$42.12	62,830	2%
Apr-22	212	\$8,896	\$41.96	64,660	5%
May-22	157	\$6,641	\$42.30	47,885	-28%
Jun-22	191	\$8,059	\$42.17	58,286	-5%
Jul-21	182	\$7,028	\$38.66	55,449	-11%
Aug-21	165	\$6,378	\$38.70	50,264	-22%
Sep-21	153	\$5,937	\$38.78	46,696	-32%
Oct-21	169	\$6,929	\$41.07	51,454	-19%
Nov-21	172	\$7,055	\$41.04	52,430	-17%
Dec-21	279	\$11,406	\$40.91	85,034	28%
Total	2,416	\$99,034	\$40.988	736,941	-
Maximum	325 (Jan)	13,351 (Jan)	42.30 (May)	99,125 (Jan)	38% (Jan)
Average	201	AED 8,253	AED 40.910	61,412	-
Minimum	153 (Sep)	5,937 (Sep)	38.66 (Jul)	46,696 (Sep)	-32% (Sep)

The graph below shows the district heating consumption monthly variations vs. HDD. As it can be clearly seen, there is a direct correlation between both variables, as district heating is the main source of space heating for the facility, therefore its consumption will increase during the winter when HDD are high, and will decrease during the summer when HDD are low.

Figure 30: FSSFE District Heating Monthly Variations vs. HDD



3.4.2. Regression Analysis

In the present regression analysis, the historical figures used for the modeling are found in the below table.

Table 16: Regression Analysis Variables

Month	Occupancy	HDD
Unit	Room Nights	°C.Days
Jan-22	928	239
Feb-22	1,159	160
Mar-22	1,936	136
Apr-22	1,873	105
May-22	2,410	65
Jun-22	2,387	6
Jul-21	2,497	0
Aug-21	1,129	0
Sep-21	1,093	0
Oct-21	1,420	0
Nov-21	1,319	81
Dec-21	1,449	229
Total	19,600	1,021

Using these variables, and the energy consumption for the baseline period of each month, the coefficients of the regression model are obtained, leading to the below:

$$\text{KWh(per month)} = (-0.00085) \times \text{OCC} + (0.5) \times \text{HDD} + (160.2)$$

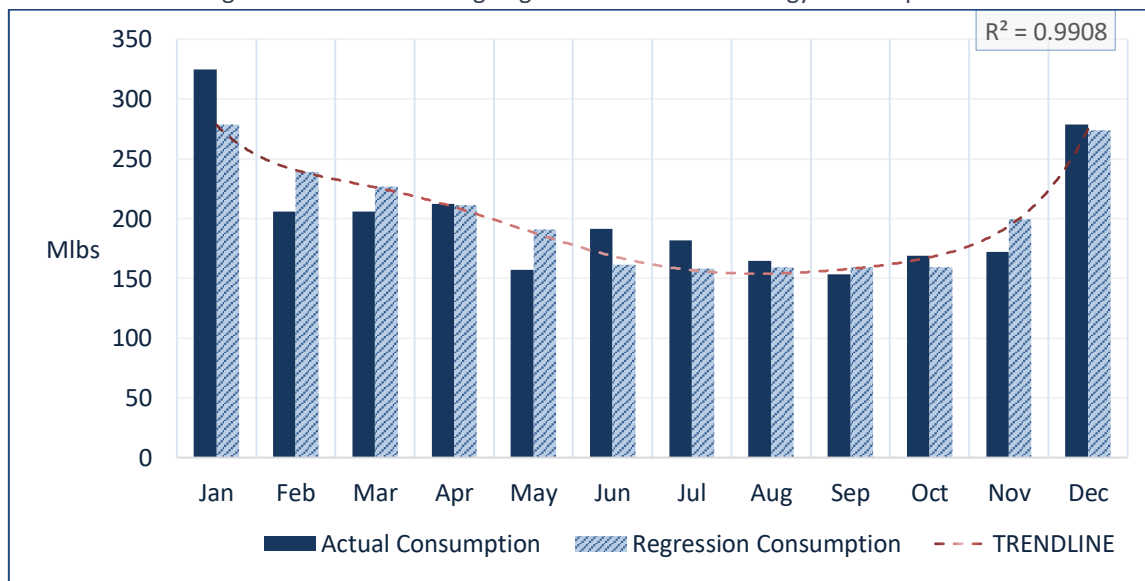
The projected consumption would be then calculated based on the above equation and the following table provides both the real baseline electricity consumption as well as the projected one if the regression equation is used.

Table 17: District Heating Projected Consumption, Residuals, and variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	MLbs	MLbs	MLbs	%
Jan-22	325	279	46	14%
Feb-22	206	239	-33	-16%
Mar-22	206	227	-21	-10%
Apr-22	212	211	1	0%
May-22	157	191	-34	-21%
Jun-22	191	161	30	16%
Jul-21	182	158	24	13%
Aug-21	165	159	6	3%
Sep-21	153	159	-6	-4%
Oct-21	169	159	10	6%
Nov-21	172	200	-28	-16%
Dec-21	279	274	5	2%
Total	2,416	2,416	0	0%

In the case of FSSFE, R^2 was seen to be 0.9908, which means that the model can be considered as an acceptable model.

Figure 31: District Heating Regression and Actual Energy Consumption



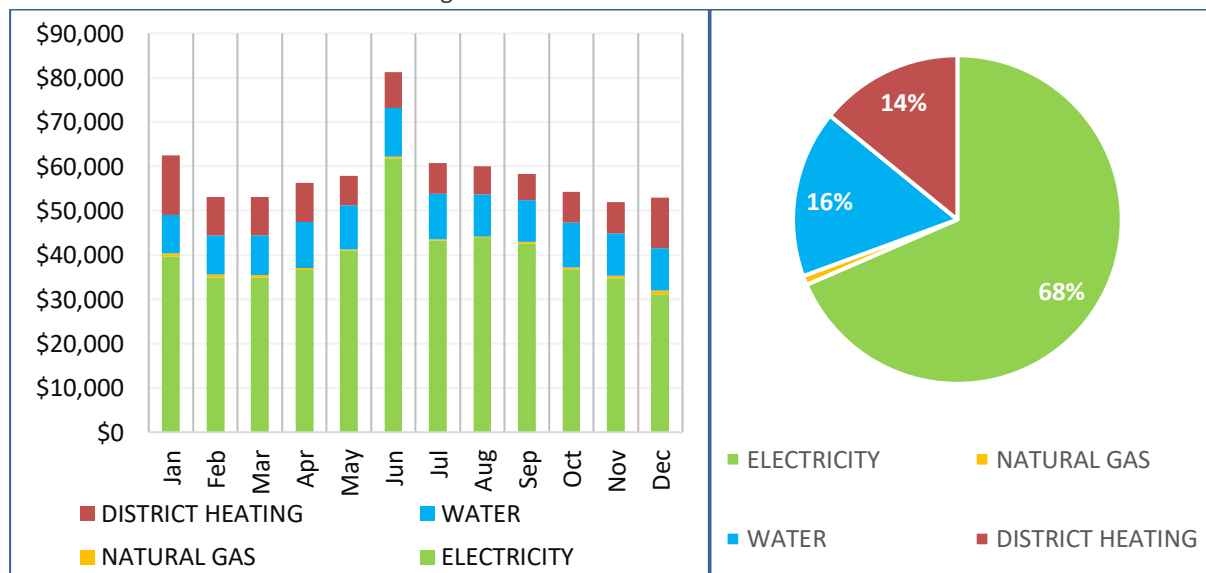
3.5. Utilities Cost Breakdown

The below table provides the monthly cost of each utility supply during for the baseline year.

Table 18: Utilities Cost Breakdown

MONTH	ELECTRICITY	NATURAL GAS	WATER	DISTRICT HEATING	TOTAL
Unit	\$	\$	\$	\$	\$
Jan-22	\$39,588	\$830	\$8,675	\$13,351	\$62,444
Feb-22	\$34,948	\$639	\$8,827	\$8,679	\$53,093
Mar-22	\$34,855	\$643	\$8,874	\$8,677	\$53,049
Apr-22	\$36,547	\$468	\$10,388	\$8,896	\$56,298
May-22	\$40,952	\$343	\$9,960	\$6,641	\$57,896
Jun-22	\$61,720	\$517	\$10,991	\$8,059	\$81,286
Jul-21	\$43,270	\$345	\$10,152	\$7,028	\$60,794
Aug-21	\$43,824	\$345	\$9,506	\$6,378	\$60,053
Sep-21	\$42,570	\$386	\$9,338	\$5,937	\$58,231
Oct-21	\$36,740	\$499	\$10,007	\$6,929	\$54,175
Nov-21	\$34,772	\$546	\$9,491	\$7,055	\$51,865
Dec-21	\$31,117	\$816	\$9,547	\$11,406	\$52,886
Total	\$480,903	\$6,377	\$115,755	\$99,034	\$702,069
Maximum	61,720 (Jun)	830 (Jan)	10,991 (Jun)	13,351 (Jan)	81,286 (Jun)
Average	\$40,075	\$531	\$9,646	\$8,253	\$58,506
Minimum	31,117 (Dec)	343 (May)	8,675 (Jan)	5,937 (Sep)	51,865 (Nov)
Percentage	68%	1%	16%	14%	100%

Figure 32: Utilities Cost Breakdown



As it can be seen, electricity represents the largest share of the facility's total utility's cost with 68%, followed by Water at 16%, district heating at 14% and finally Gas at only 1%.

3.6. Energy & Water Use Intensity, Carbon Footprint

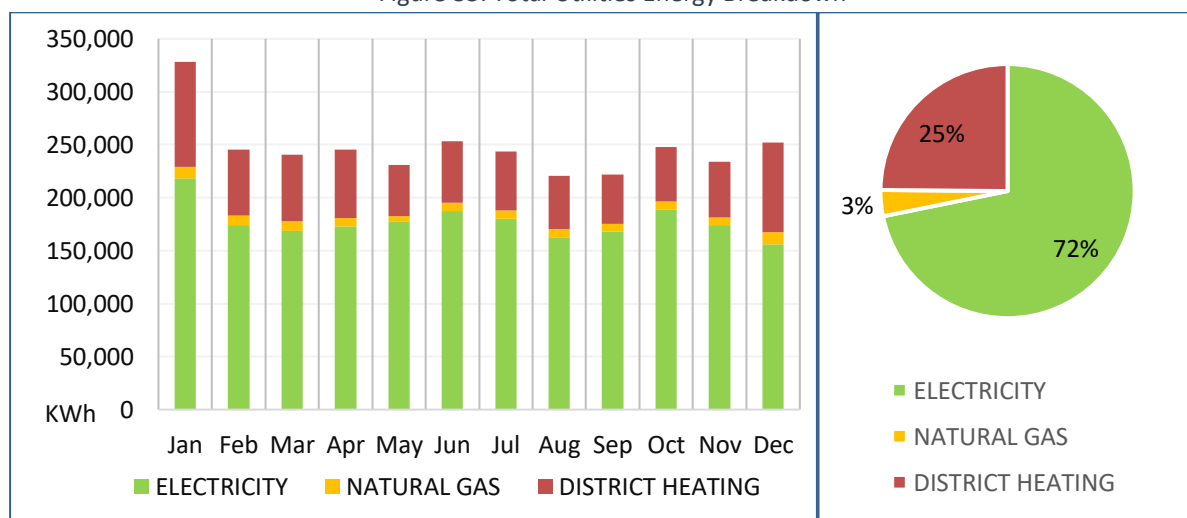
3.6.1. Energy Use Intensity (EUI)

To be able to aggregate all the utilities energy consumption, the energy consumption of all energy sources is converted to a single energy unit KWh (Kilowatt Hour). The breakdown for the energy sources consumption is shown in the following table. As can be seen, electricity accounts for 72% of the energy sources energy consumption, while District Heating accounts for 25% and Natural Gas takes the remaining 3%.

Table 19: Total Utilities Energy Breakdown

MONTH	ELECTRICITY	NATURAL GAS	DISTRICT HEATING	TOTAL
Unit	KWh	KWh	KWh	KWh
Jan-22	218,370	10,927	99,125	328,422
Feb-22	174,338	8,529	62,830	245,697
Mar-22	168,517	9,128	62,830	240,475
Apr-22	173,136	7,777	64,660	245,573
May-22	177,536	5,192	47,885	230,613
Jun-22	187,455	7,697	58,286	253,437
Jul-21	180,293	7,957	55,449	243,699
Aug-21	162,693	7,711	50,264	220,668
Sep-21	168,063	7,202	46,696	221,960
Oct-21	188,505	7,954	51,454	247,912
Nov-21	174,041	7,368	52,430	233,839
Dec-21	156,233	11,063	85,034	252,330
Total	2,129,180	98,506	736,941	2,964,626
Maximum	218,370 (Jan)	11,063 (Dec)	99,125 (Jan)	328,422 (Jan)
Average	177,432	8,209	61,412	247,052
Minimum	156,233 (Dec)	5,192 (May)	46,696 (Sep)	220,668 (Aug)
Percentage	72%	3%	25%	100%

Figure 33: Total Utilities Energy Breakdown

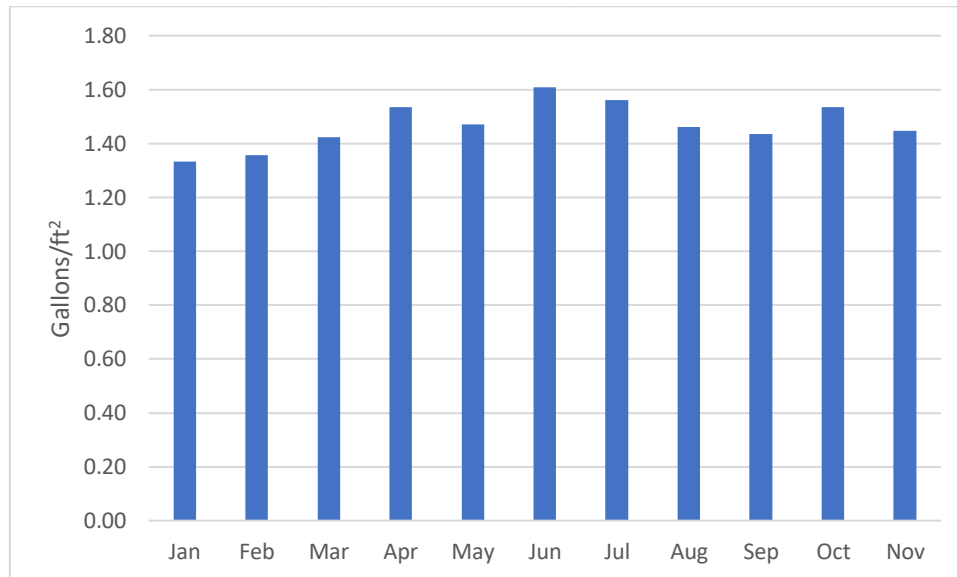


The total annual energy footprint in FSSFE is **18.22 KWh/ft²**.

3.6.2. Water Use Intensity (WUI)

The graph below represents the monthly Water Use Intensity (WUI) of FSSFE during the baseline period, which is calculated by dividing the monthly water consumption by the total built up area of the hotel in square feet. As it can be seen, the highest value of the WUI is during June with a value of 1.61 Gallons/ft², and the lowest value being in January with 1.33 Gallons/ft².

Figure 34: FSSFE Monthly WUI Variations



The total annual Water Use in FSSFE is **189.66 USG/ft²**.

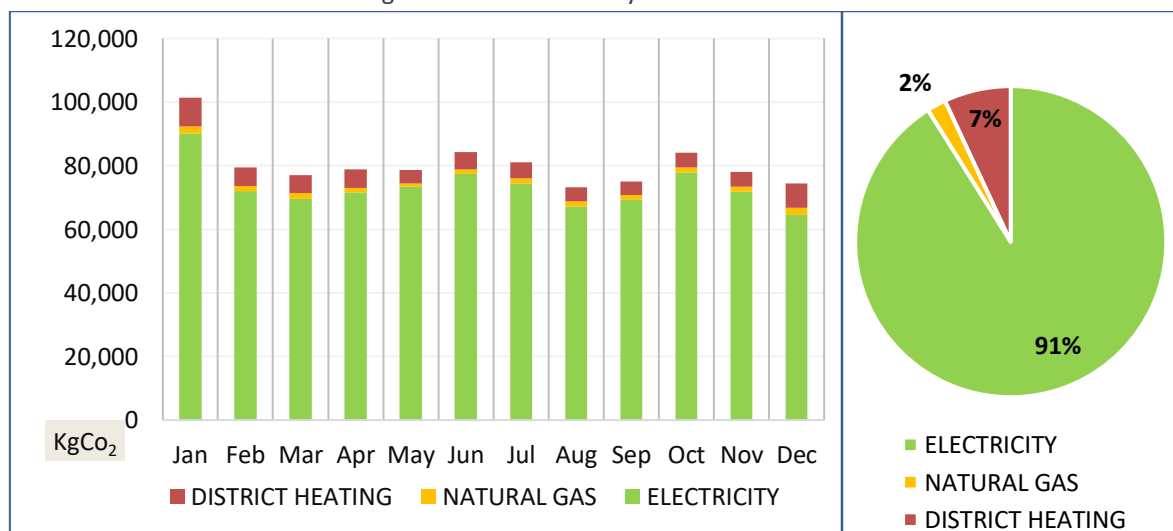
3.6.3. Carbon Emissions Footprint

The carbon emissions calculation is done by multiplying the consumption of each utility (electricity, district heating, and natural gas) by the emissions factor of each. The electricity emissions factor varies between countries and regions, with the San Francisco average of 0.6257 KgCO₂/KWh being taken in this report while applying a 33% factor off due to the purchase of Renewable Energy from the utility. Natural Gas Emission Factor is 5.3 KgCO₂/Therm and District Heating Emission Factor is 5.3 KgCO₂/Mlb. The **total annual carbon emissions** are at a total of **722,363 KgCO₂**.

Table 20: FSSFE Carbon Emissions Monthly Breakdown

MONTH	ELECTRICITY	NATURAL GAS	DISTRICT HEATING	TOTAL
Unit	KgCO ₂	KgCO ₂	KgCO ₂	KgCO ₂
Jan	90,179	2,125	9,016	101,319
Feb	71,995	1,659	5,714	79,368
Mar	69,591	1,776	5,714	77,081
Apr	71,499	1,513	5,881	78,892
May	73,316	1,010	4,355	78,681
Jun	77,412	1,497	5,301	84,210
Jul	74,454	1,548	5,043	81,045
Aug	67,186	1,500	4,572	73,257
Sep	69,403	1,401	4,247	75,051
Oct	77,845	1,547	4,680	84,072
Nov	71,872	1,433	4,769	78,074
Dec	64,518	2,152	7,734	74,404
Total	879,270	19,160	67,025	965,455
Maximum	90,179 (Jan)	2,152 (Dec)	9,016 (Jan)	101,319 (Jan)
Average	73,273	1,597	5,585	80,455
Minimum	64,518 (Dec)	1,010 (May)	4,247 (Sep)	73,257 (Aug)
Percentage	91%	2%	7%	100%

Figure 35: FSSFE Monthly Carbon Emissions



FSSFE's annual carbon footprint is at **3.11 kgCO₂/ft²**.

3.6.4. CHSB – Cornell Sustainability Hotel Benchmark – Real Consumption

The Cornell Sustainability Hotel Benchmark (CHSB) provides the most comprehensive benchmarking tool in the hospitality industry as it is based on a global database of more than 15000 properties across all regions and climate. The below table provides the key metrics of FSSFE as compared to various levels of the CHSB for 8 Key Measures. Clearly, the property has higher marks in nearly all Measures, except the intensity ones and fared relatively well in water measures. There are two key reasons, the current level of energy consumption is large compared to the number of rooms, but also some metrics have been affected by the low occupancy rate during the baseline year. In fact, increasing the occupancy rate will not lead to a prorated energy or water consumption and the property will drop its metrics in many Measures from the 'High' level to somewhere between 'Median' and 'Upper Quartile'.

Table 21: CHSB benchmark – using real consumption

MEASURE	Count	Low	Lower Quartile	Mean	Median	Upper Quartile	High	FSSFE
MEASURE 1: Hotel Carbon Footprint Per Room (kgCO ₂ e)	68	1,287	2,099	2,897	2,746	3,538	5,275	4,660.4
MEASURE 2: Hotel Carbon Footprint Per Occupied Room (kgCO ₂ e)	70	4.74	7.12	9.64	8.83	11.52	18.28	36.86
MEASURE 3: Hotel Carbon Footprint Per Square Foot (kgCO ₂ e)	80	1.57	3.35	4.30	4.02	5.21	8.24	4.44
MEASURE 4: Hotel Energy Usage Per Occupied Room (kWh)	70	22.98	33.09	46.03	42.99	55.18	90.04	151.26
MEASURE 5: Hotel Energy Usage Per Square Foot (kWh)	80	6.81	15.65	20.46	18.80	24.33	39.64	18.22
MEASURE 6: Hotel Water Usage Per Occupied Room (L)	68	58.03	333.95	415.94	406.08	473.84	936.85	553.7
MEASURE 7 : Hotel Water Usage Per Square Foot (L)	74	34	149	199	189	239	449	66.7
MEASURE 8: Hotel % Energy From Renewables (%)	80	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	22

4. Electrical Energy Load Analysis & Energy Balance

4.1. Electrical Load Categories

A crucial milestone in energy audits is the buildup of the energy balance. This process starts with acquiring the load inventory – electrical end uses, and then running a complete simulation based on the operational schedule, load factors and behavioral statistics of each item in the inventory. The micro and macro energy consumption analysis of energy systems including lighting, heating, air conditioning, ventilation and building envelope, serves the purpose of identifying where and how a facility uses energy. The breakdown of the energy consumption by end-use would be the energy balance, which will provide an important baseline for the roadmap in developing the Energy Efficiency Measures (EEMs) that would optimize the overall energy costs.

Information and Database Process

FSSFE's Load inventory data is populated into an Energy Analysis System including assets name/category, actual electrical specification (KW) and operational schedule over 12 months (hours per day, week, month, or season). The second step is performing a total simulation and energy modeling of the facility with full bills reconciliation as per the baseline values.

In the present report, electrical loads were divided into the following categories:

1- Lighting

This category corresponds to all types of existing lighting fixtures.

2- HVAC Fans

This category corresponds to all extraction and supply fans and fan coil units responsible for the required circulation of air in the facility.

3- Cooling

This category corresponds to all chillers in all buildings, as well as the cooling towers

4- Motors

This category corresponds to all chilled water pumps, water pumps (hot and domestic), circulation, pool pumps, and lifts.

5- Kitchen and Refrigeration

This category corresponds to all electricity driven Kitchen and Refrigeration equipment.

6- Others

This category corresponds to all electrical driven equipment such as IT loads, CCTV, and other miscellaneous equipment.

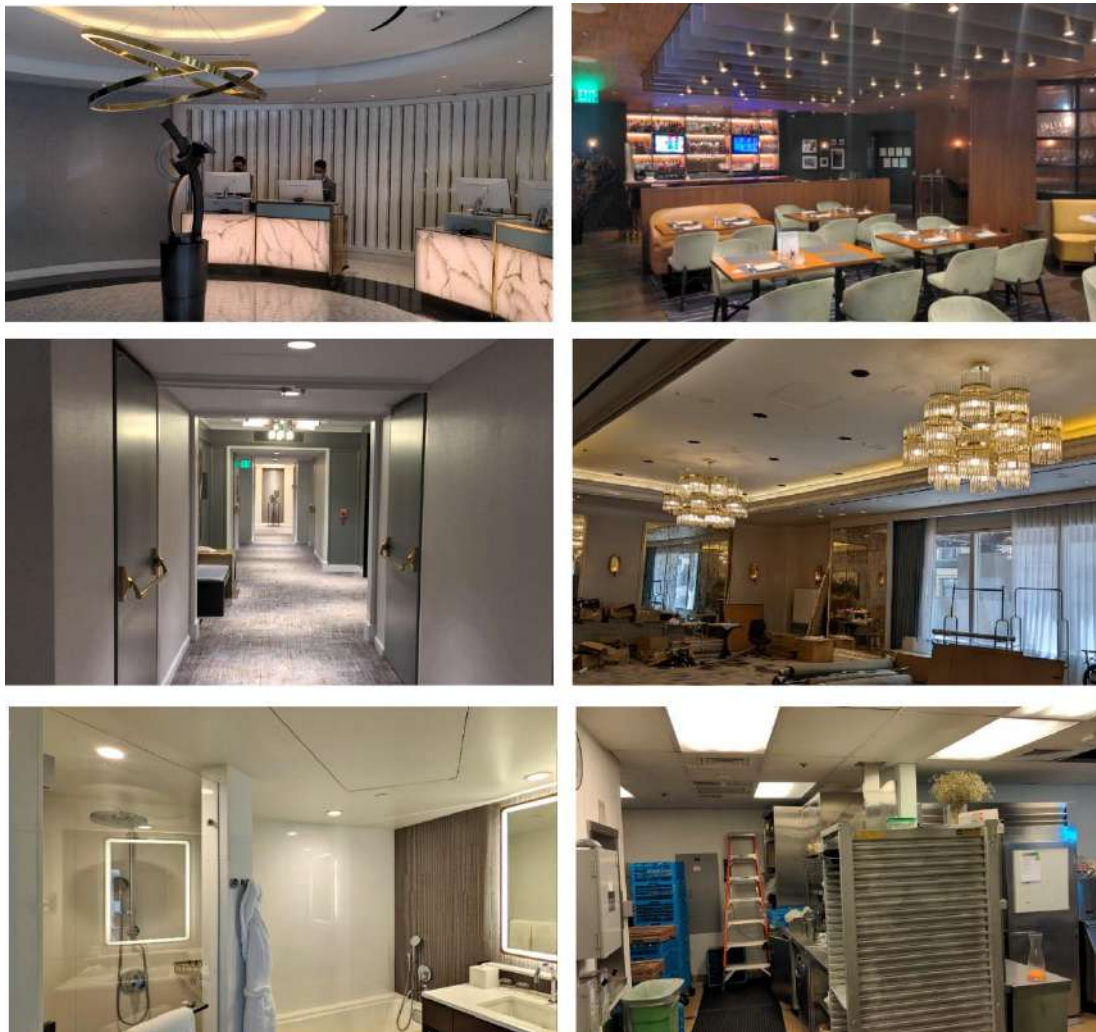
4.1.1. Lighting

Lighting in hotels represent generally a crucial aesthetic element and has high load factors as the hours of operation of the common areas and the BOH are typically close to 24/7.

In the case of FSSFE, existing lighting infrastructure is hybrid with primarily LED fixtures in the guest rooms' floors as well as the lobby and main restaurant – all installed during the last renovation, while the BOH still includes a large number of T8 fluorescent and the 2nd floor still includes halogen and Incandescent fixtures,

The below pictures show the lighting fixtures in different areas on FSSFE.

Figure 36: FSSFE Lighting

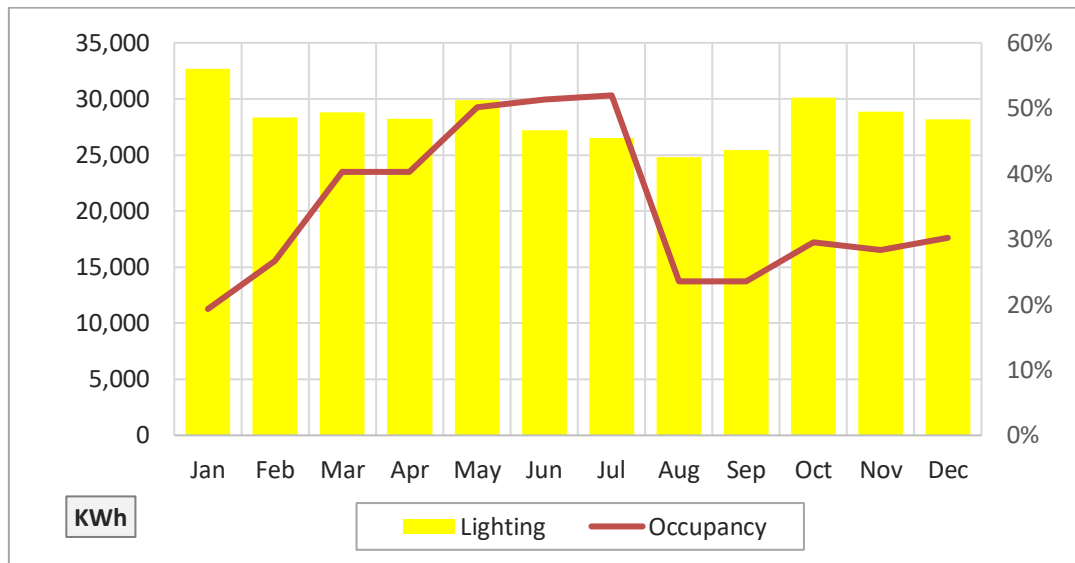


- Top: First Floor Corridor Lighting (Left); GF Restaurant Lighting (Right)
- Middle: Meeting Room Lighting (Left); Guest Room Bathroom Lighting (Right)
- Bottom: BOH Lighting

Lighting control is a mix of manual and automated ones . Typically, lobby and most part of BOH areas, just like guest corridors have their lighting ON 24/7. Guest rooms' lighting are controlled by the GRMS while offices, technical rooms and meeting rooms (unused currently) are manually controlled.

Compiling all the lighting inventory of FSSFE and simulating their operation according to the hours of usage and the bills reconciliation would yield to the following monthly energy consumption.

Figure 37: Lighting monthly energy consumption

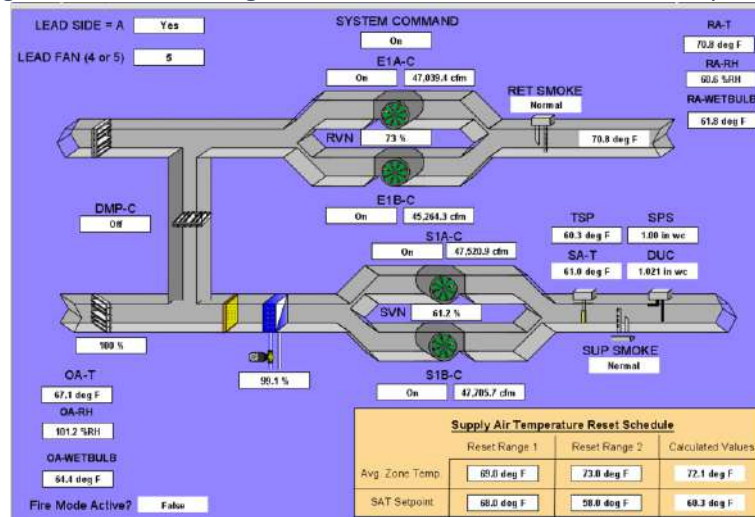


The total **Lighting** category electrical consumption for the baseline period is **338,973 KWh**.

4.1.2. HVAC Fans

The HVAC Fans category includes all the FCUs, and supply/extraction fans installed in the facility. It is important to note that FSSFE has one large Air Handling Units serving the 1st floor area which is serviced electrically from the tower's feeders – and therefore not part of the property's electricity billing; however, chilled water serving the unit is supplied from FSSFE lower floor's chilled water plant.

Figure 38: AHU#4 serving 1st floor – connected to Tower's electricity feeder



The whole HVAC system of FSSFE has a pneumatic control – since the original installations, while the BMS has very limited control on the total HVAC Fans infrastructure. On the other hand, a large number of equipment are still dating back from 1986 with their lifecycle well overdue.

a. FCUs

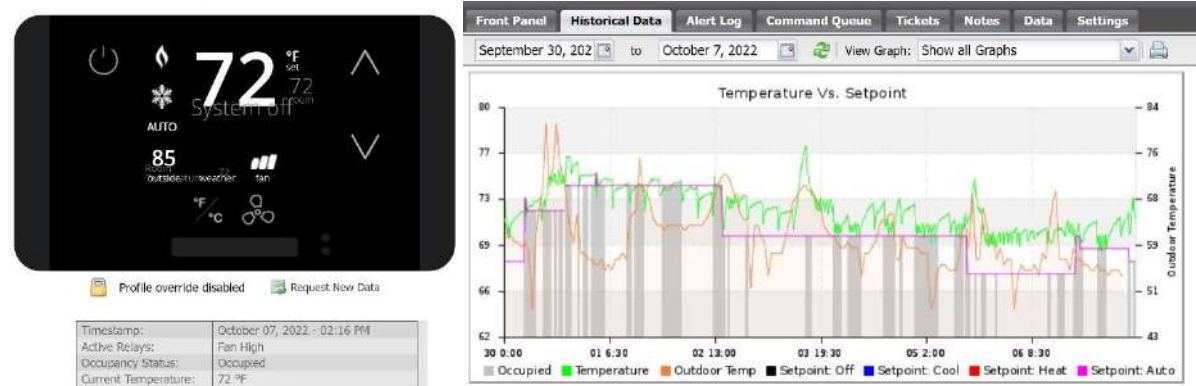
Excluding the above mentioned large AHU, the HVAC design of FSSFE is primarily revolving around FCUs. Accordingly, there is a large number (>200) serving all the guest rooms, as well as all common and BOH areas. Most of these units date back from the original construction in 1986 and have therefore more than 35 years of life. The below snapshot showcases the specifications of one typical guest rooms corridor's FCU.

Figure 39: Guest rooms' corridor FCU specifications

MARIN ORIENTAL SAN FRANCISCO		LOCATION 37TH FLOOR CORRIDOR	
SERVICE ROOM SERVICE KITCHEN		FEEDER FEU-322 VIA PNL 37 RL2-21	
FAN COIL UNIT MANUFACTURER INTERNATIONAL		SP SIZE 0.4 16	
SERIAL NUMBER		CFM 1300	
		FILTER SIZE 16 X 25 X 1	
MOTOR MANUFACTURER		MODEL NO:	
SERIAL NO:		CATALOG NO:	
DRIVE END BEARING		OPPOSITE END BEARING	
EFFICIENCY		FRAME	
FLA		TYPE	
ENCLOSURE TYPE		CODE	
VATTAGE 830	VOLTAGE 115	RPM 1	PH 1

FCUs are controlled via thermostats and manual operation (ON/OFF) without a central platform giving a visibility on the operations for the BOH and FOH units while guest rooms FCUs are controlled via the Telkonet EcoCentral GRMS system.

Figure 40: Telkonet Thermostat (Left) connected to the GRMS platform (Right)



b. Fans

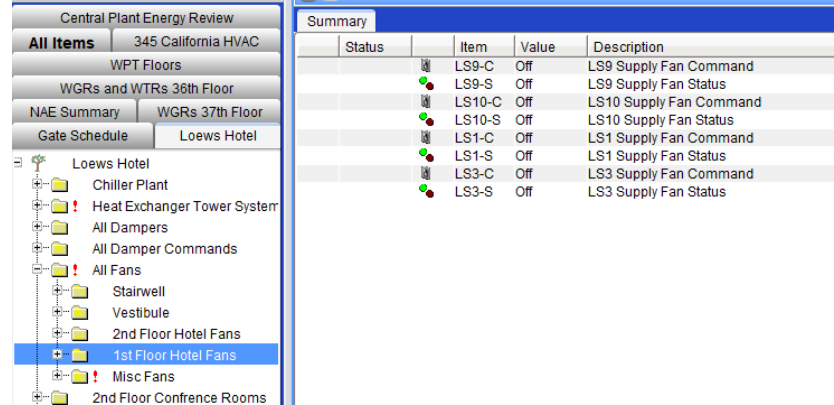
The other category of installed fans are of the Fresh Air and Exhaust Fans including the kitchen Hood . These fans vary from 0.5 HP to 15 HP in rated power and have been mostly installed also since the original construction of 1986.

Figure 41: Supply Fan SF19 (Left), Kitchen Extractor Fan EF 60 (Right)



None of the Fans has a VFD while their control (On/Off) is done for their majority from the BMS as can be seen in the below snapshot. In addition, the property implements a schedule of operation via the BMS on a number of these fans.

Figure 42: BMS Fans Control (Below); Fans' scheduling (Next Page)





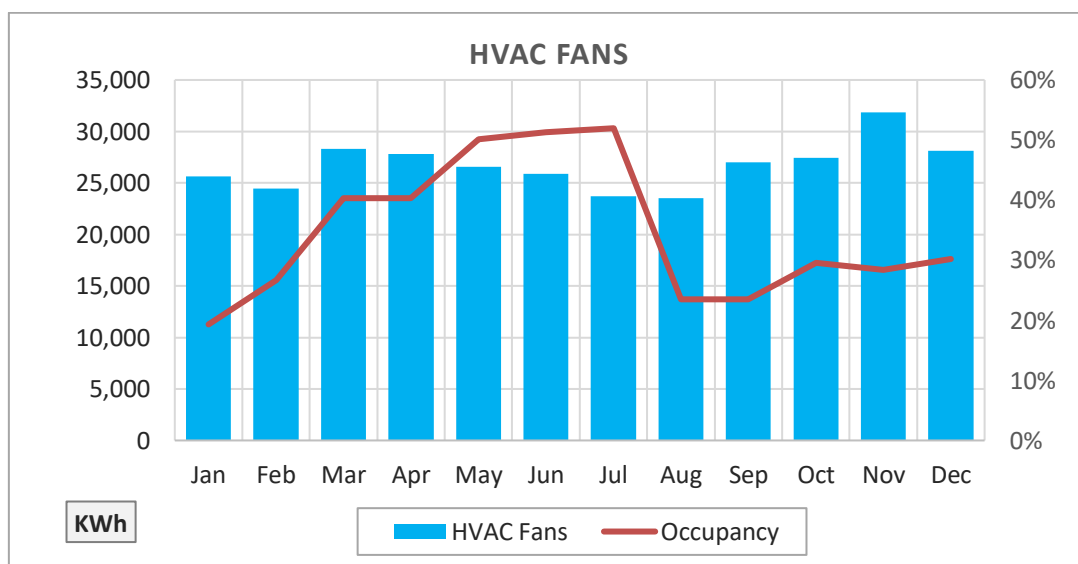
Compiling the total installed HVAC Fans' installed power and annual electricity consumption would lead to the below table broken down between FCUs and the other Fans.

Table 22: HVAC Fans Sub-Categories Loads and Consumptions

SUB-CATEGORY	LOAD		ENERGY	
Unit	KW	%	KWh	%
Fan Coil Units	21.6	24%	162,137	51%
Exhaust Fans	42.2	48%	99,598	31%
Life Safety Fans	13.1	15%	30,921	10%
Supply Fans	11.8	13%	27,794	9%
Total	89	100%	320,450	100%

Following the energy simulation and bills reconciliation, the following table summarizes the load of each sub-category and the corresponding energy consumption for the baseline year.

Figure 43: HVAC Fans monthly energy consumption



The total **HVAC Fans** category electrical consumption for the baseline period is **320,450 KWh**.

4.1.3. Cooling

The cooling energy end use category comprises the chillers, cooling towers as well as the few Split units installed in a few locations in the premises and the heat pump installed on the roof. As previously indicated, all the HVAC system in FSSFE is pneumatically controlled.

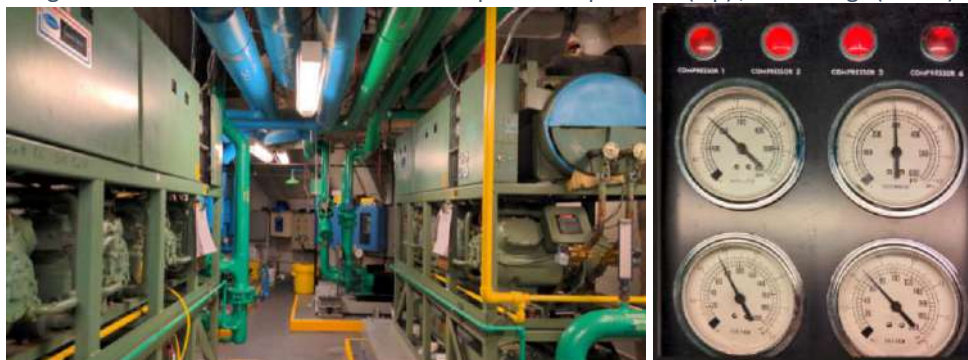
Figure 44: Analog measurements of readings in the pneumatic control system



a. Chillers

Chillers 3 and 4 serve the guest rooms and are located in the 36th floor ; they are two Carrier Water-cooled chillers of the model 30HR 160 D 600, each with a capacity of 149.5RT and 4 compressors running on the refrigerant R22. Both chillers have been installed in 1986 and one of them is in a poor technical condition while the other is relatively operating well. The original COP of both chillers was 3.7 as per the datasheet (see below ARI ratings) but a derating of up to 30% can be considered after more than 30 years of operation and the current real COP is more likely to be close to 2.8.

Figure 45: Chillers 3 and 4 with the 4 compressors operation (Up) , ARI ratings (Down)



ARI RATINGS													
UNIT 30	CAPACITY		INPUT POWER (kW)	COOLER FLOW		COOLER PRESSURE DROP		CONDENSER FLOW		CONDENSER PRESSURE DROP		EER	IPLV
	Tons	kW		Gpm	L/s	Ft of Water	kPa	Gpm	L/s	Ft of Water	kPa		
HR070	69.4	244.1	60.1	166.3	10.49	8.6	25.7	208.2	13.14	7.0	20.9	13.9	17.5
HR080	77.2	271.5	68.6	185.1	11.68	10.6	31.7	231.6	14.61	10.2	30.5	13.5	17.1
HR090	85.5	300.7	77.4	204.9	12.93	12.9	38.5	256.5	16.18	9.8	29.3	13.3	17.3
HR100	98.0	344.7	85.4	234.8	14.81	9.0	26.9	294.0	18.55	11.5	34.3	13.8	18.4
HR110	105.0	369.3	93.8	251.6	15.87	10.3	30.8	315.0	19.87	10.8	32.2	13.4	18.3
HR120	112.0	393.9	102.3	268.6	16.95	11.7	34.9	336.0	21.20	10.3	30.8	13.1	18.2
HR140	135.0	474.8	126.0	323.7	20.42	9.4	28.1	405.0	25.55	14.4	43.0	12.9	16.3
IR160	149.8	526.8	141.6	359.2	22.66	11.6	34.6	449.4	28.35	10.4	31.1	12.7	16.1

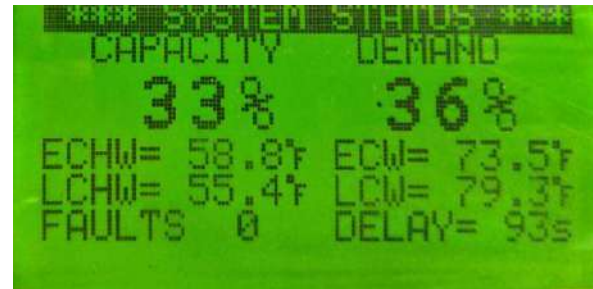
Chiller 5 is located in the 4th floor mechanical room and serves the lower floors of the property. It consists of three connected Multistack water-cooled chiller of the model

MS050XC1H2W2AA with a cooling capacity of 50 RT each and 2 scroll compressors. The total installed capacity is therefore 150RT with 6 compressors running with the R410 refrigerant . This chiller has been installed in 2012 and is in a good operating condition. Its original COP is 5.4 and a derated one close to 5 which is a large improvement compared to chillers 3 and 4.

Figure 46: Chiller 5 – Multistack with 6 compressors (Left) ; Nameplate of one unit of 2 compressors (Right)



As per the right snapshot, Supplied Chilled Water was at 55.4°F while the Return one was at 58.8°F with a ΔT of only 3.4°F (1.8°C) which is an indication of an over pumping of chilled water in the end users' loops. On the other hand Condenser water was sent to the cooling tower at 79.3°F and returning at 73.5°F



Control of the chillers is done through the BMS although there are no graphical interface but only a tabular list as seen in the following snapshots showing the controls of Chillers 3 & 4 and chiller 5.

Figure 47: BMS Chilled Water Plant Controls – Chiller 3&4 (Below); Chiller 5 (Next Page)

38th Floor	CHWP14VFD-O	100.0 %	Chilled Water Pump#14 VFD - Override
39th Floor	CHWP15VFD-O	100.0 %	Chilled Water Pump#15 VFD - Override
40th Floor	CWP16VFD-O	100.0 %	Condenser Water Pump #16 VFD - Override
41st Floor	CWP17VFD-O	100.0 %	Condenser Water Pump#17 VFD - Override
42nd Floor	CWBPVVLV-O	0.0 %	Condenser Water By-Pass Valve - Override
43rd Floor	CHWBYPVLV-O	0.0 %	Chilled Water By-Pass Valve - Override
44th Floor	CH3-C	On	Chiller 3 - Command
45th Floor	CH3CHWISOVLV-C	Open	CH3 Chilled Water Isolation Valve
46th Floor	CH3CWSOVLV-C	Open	CH3 Condenser Water Isolation Valve
47th Floor	CH4-C	Off	Chiller 4 - Command
48th Floor	CH4CHWISOVLV-C	Close	CH4 CHW Iso Valve Command
49th Floor	CH4CWSOVLV-C	Close	CH4 CW Iso Valve Command
Chiller 5	CHWP14VFD-C	On	CHWP-14 VFD Command
Heat Exchanger 1 B	CHWP15VFD-C	Off	CHWP-15 VFD Command
	CWP16VFD-C	On	CWP-16 VFD Command
	CWP17VFD-C	Off	CWP-17 VFD Command
	TOWER-START	On	
	CHW VFD Speed Command	100.0 %	CS Input (Float)
	CW VFD Speed Command	100.0 %	CS Input (Float)
	Lag Chiller Stage Up Delay	10.0 minu...	CS Input (Float)
	CHW Supply Temp Differential	2.0 deg F	CS Input (Float)
	Lead Chiller	3	CS Input (Float)
	Chiller Off Delay	10.0 minu...	CS Input (Float)
	Lag Chiller Stage Down Delay	10.0 minu...	CS Input (Float)
	Chilled Water Supply Setpoint	58.0 deg F	CS Input (Float)
	CHW Pump Off Delay	5.0 minut...	CS Input (Float)
	OSA Lockout Setpoint	35.0 deg F	CS Input (Float)
	CW Pump Off Delay	3.0 minut...	CS Input (Float)
	OSA-T Mapped	58.4 deg F	CS Input (Float)
	CHW Bypass Setpoint	40.0 deg F	CS Input (Float)
	CW Bypass Setpoint	60.0 deg F	CS Input (Float)

Status	Item	Value	Description
	CTHILVL-A	Normal	Cooling Tower High Level Alarm
	CTLOWVL-A	Normal	Tower Low Level Alarm
	CWP19-S	On	Condenser Water Pump 19 Status
	PCHWP18-S	On	Primary CHW Pump 18 Status
	CT5A-S	On	Tower 5 Status Jumped -Point not Avail.
	CT5B-S	Off	Cooling Tower 5 Status
	CH5-A	Alarm	Chiller 5 Alarm
	CWP19-C	On	Condenser Water Pump 19 Command
	CWP19-O	100.0 %	Condenser Water Output
	PCHWP18-C	On	Primary CHW Pump 18 Command
	PCHWP18-O	0.0 %	Primary Chilled Water Output
	ISO-VLV GRP C-C	On	Chiller 5 CHW Iso Valve Group C
	ISO-VLV GRP B SL-C	Off	Isolation Valve B Command
	ISO-VLV GRP A SL-C	On	Valve-A CH1& CH2 Isolation
	CT5A-C	On	Tower 5A Command
	CT5A-O	0.0 %	Tower 5A Output
	CT5B-C	On	Tower 5B Command
	CT5B-O	0.0 %	Tower 5B Output
	CH5-EN	On	Chiller 5 Enable
	CTV-O	30 %	Tower Bypass Valve Output
	SYSTEM-EN	True	Chiller 5 System Master Enable
	CHWTEMP-DIFF	14.0 deg F	Chilled Water Differential Temperature
	CHWSUPPLY-TEMP	65.8 deg F	Chilled Water Supply Temperature
	CHWRETURN-TEMP	66.3 deg F	Chilled Water Return Temperature
	CHWDT-SP	20.0 deg F	Chilled Water Diff Temperature Setpoint
	BYP-CW-SP	70.0 deg F	Bypass Valve Condenser Temp Setpoint
	Tower Fan Setpoint	70.0 deg F	Condenser Water Temp Setpoint
	PMPDOWN-TIME	300.0 seconds	Pump Shutdown Time Delay
Operator Override	CHW-DP SP	5.0 psi	Chilled Water Differential Pressure SP
	OA-T	62.0 deg F	Outside Air Temperature
	PCHWR-T	66.2 deg F	Primary CHW Return Temp
	PCHWS-T	65.8 deg F	Primary CHW Supply Temp
	CWS-T	60.9 deg F	Condenser Water Supply Temperature
	CH5CWL-T	61.2 deg F	Chiller 5 CW Leaving Temperature
	CHW-P	9.4 psi	Chilled Water System Pressure

The above two snapshots clearly shows a poor Chilled water ΔT in both cases (2.4°F for Ch3&4 and only 0.5°F for Ch5) as well as Condenser Water ΔT . On the other hand, Chilled Water and Condenser Pumps were seen to be operating at full speed (100%). It is worth noting that in the case of CH5, the ΔT setpoint on CHWS was 20°F and the readings showed a 14°F ΔT while the Supply – Return calculation showed the 0.5°F.

The other note is that the Chilled Water Temperature supply is set at a relatively high level of 58°F which is possible due to the San Francisco mild climate across the year..

b. Cooling Towers

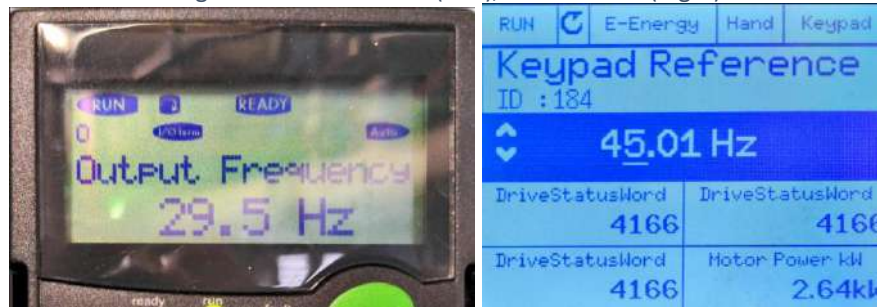
As previously mentioned, each chilled water plant has a cooling tower which serves the related water cooled chillers. Accordingly, Chillers 3 and 4 have one cooling tower with 2 motors but one VFD installed in 1986, whereas the cooling tower for chiller 5 has 2 fans (5a and 5b) both running on VFDs and installed in 2016 - during the site visit one VFD was running and the other was off.

The operation of the cooling towers is in par with the chillers and is virtually continuous across the year albeit with low speed fans due to the VFDs as can be seen in the picture in the following page.

Figure 48: Cooling Towers for Chiller 3,4 (Left) and 5 (Right)



Figure 49: VFD on CT#3 (Left), VFD on CT#5 (Right)



c. Heat Pump/Split Units

There is one Heat Pump installed on the roof of the hotel and used to cool the glass bridge across the upper floors whenever needed. The unit is a Carrier 38AD024 model installed in 1986 with a rated power of 5 RT and using R22 as a refrigerant. While it is still being in good operation, its lifecycle is well overdue and its COP is today in the best cases topping 2.

On the other hand, the property has a small number of split units servicing elevator/technical rooms as well as in the receiving areas.

Figure 50: FSSFE Heat Pump



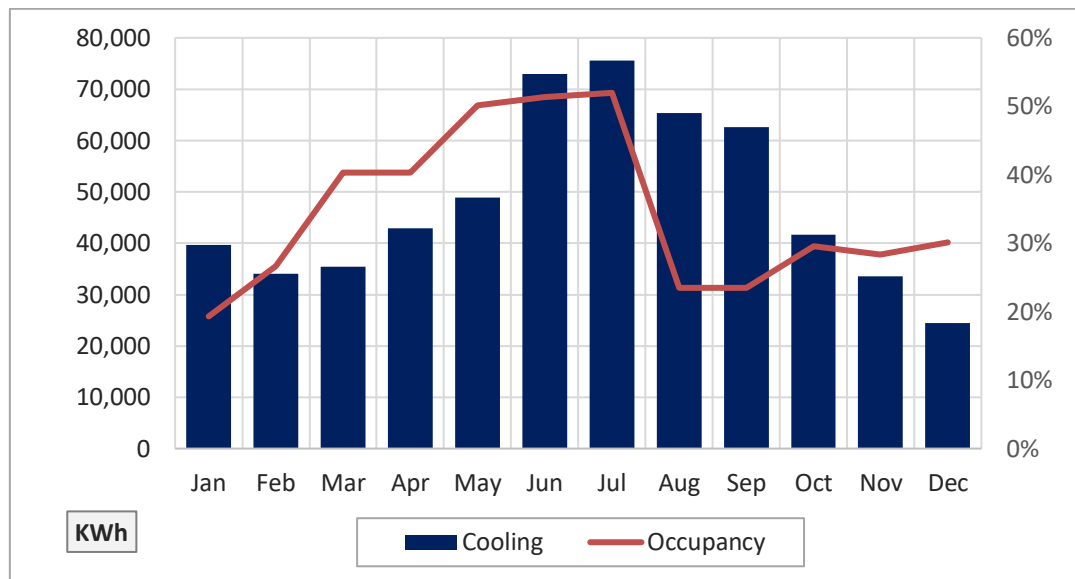
The below table provides the breakdown of the cooling end uses in terms of installed power (KW) and electricity consumed (KWh).

Figure 51: Cooling Sub-Categories Loads and Consumption

Figure 51: Cooling Sub-Categories Loads and Consumption				
SUB-CATEGORY	LOAD		ENERGY	
Unit	KW	%	KWh	%
Chillers	447.2	96%	531,503	92%
Cooling Towers	10.8	2%	13,517	2%
Split Units	4.5	1%	24,748	4%
Heat Pump	3.4	1%	7,213	1%
Total	466	100%	576,981	100%

Following the overall electricity simulation performed for all the assets and the bills reconciliation, the below chart provides the monthly electricity consumption for the cooling end user category for the baseline year which was seen to top **576,981 KWh** annually.

Figure 52: Cooling Monthly Energy Consumption



4.1.4. Motors

This end use category includes all pumps – cold water and hot water pumps, circulation pumps, primary chilled water pumps. The motors category comprises also all the service lifts and guest elevators.

The key installed and operating pumps in FSSF are here below

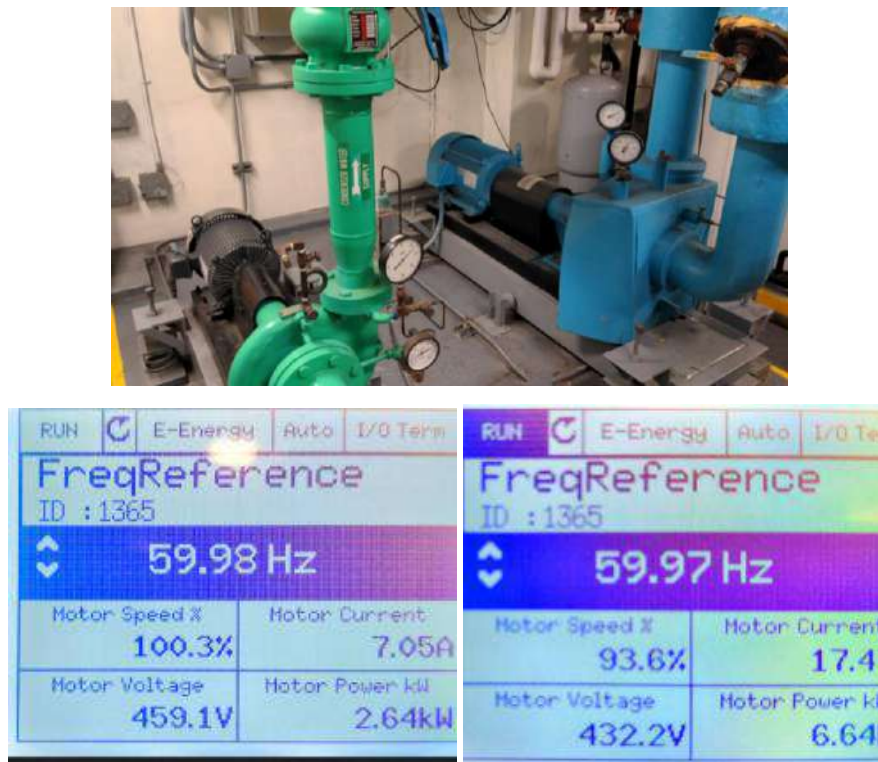
- 2x11KW primary chilled water pumps, P14 and P15, both running on VFDs for chillers 3 and 4 and with a guaranteed efficiency of 92.4%. During the field audit, the VFD was seen to be operating at full speed as can be seen in the below snapshots.
- 2x5.5KW condenser pumps, P16 and P17, both running on VFDs for chillers 3 and 4 and with a guaranteed efficiency of 92.4%. As for the Primary CWP, the Condenser Pump's VFD was seen to be running at full speed during the field audit.

Figure 53: Primary Chilled water and condenser pumps for Chillers 3 and 4 (UP), Respective VFDs operation (Down)

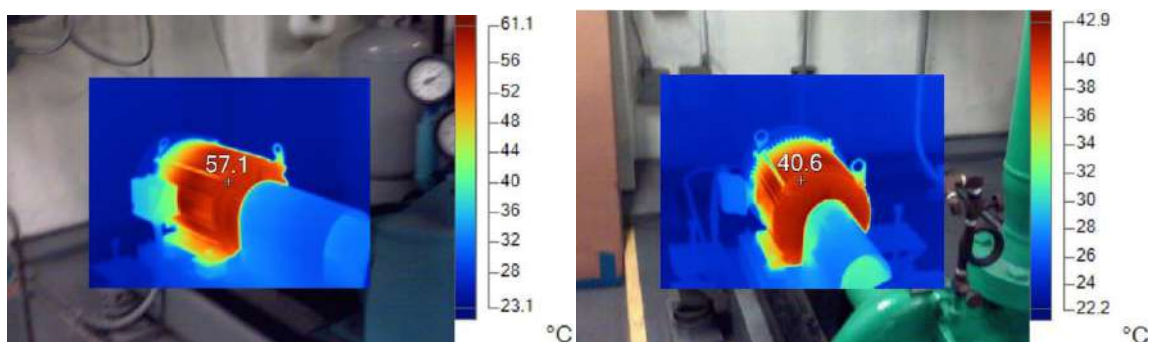


- 1x11KW Primary Chilled Water Pump, P18, and 1x5.5KW Condenser Pump, P19, running on VFDs for Chiller 5.. Both are equipped with VFDs with the running conditions seen in the snapshots in the pictures in the following page. Their guaranteed efficiencies as per their nameplate are 88% and 87% respectively.

Figure 54: Chiller 5 Primary Chilled Water and Condenser Pumps



The below thermal images show the temperature of both pumps' motors during operation. Both are within permissible ranges as per the class ranges of the pumps.

Figure 55: Condenser and Chilled Water Pumps temperature during operations (4th floor)

- 2x11KW Hot Water Circulation pumps, P7 and P8, running on VFDs. Their efficiency is 91% with a rated PF of 0.85. During the field audit, the operating pump was running at 96% speed.

Figure 56: Hot Water Circulation Pumps



- 2x2.2KW Hot Water Pumps for hot water delivery from the heat exchangers to the hotel, both running on VFDs

Figure 57: Hot Water Pumps



- Air compressors: There are two units used for the pneumatic controls, one installed in B1 and one in the 36th floor. Both have a rated capacity of 3.6KW with a rated pressure of 175Psi and installed in 2015 and 2016. They are in good operating conditions.

Figure 58: Air Compressor for the pneumatic controls



- **Guests and Service Lifts:** 4 elevators are connected to the hotel's electricity feeders (#13,#14,#17,#18) which are installed since the beginning of the operation of the tower.

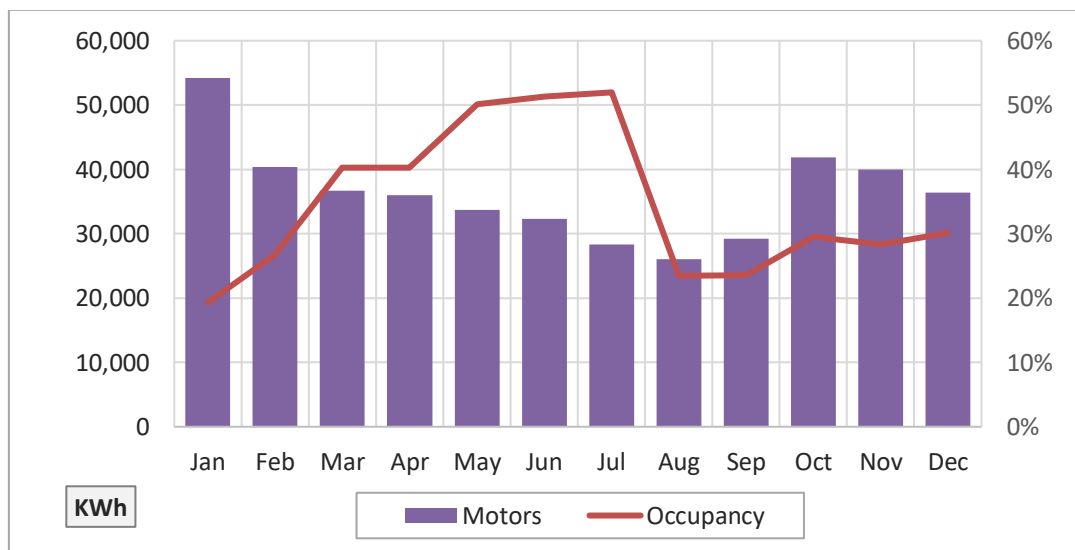
Figure 59: Guest (Left) and Service (Right) Lifts



It is to be noted that a number of pumps and motors servicing the hotel are connected to the Tower's electricity feeders such as the main cold water booster pumps .

Following the energy simulation and bills reconciliation, the total **Motors** category energy consumption for the baseline period is **435,040 KWh** with the monthly variations as per the below chart.

Figure 60: Motors Monthly Energy Consumption



4.1.5. Kitchen and Refrigeration

The kitchen end use category includes all the refrigeration and electrically driven kitchen equipment across all kitchens and restaurants of FSSFE (fridges, freezers, cold rooms, ovens, heaters, microwaves...), as well as all the ice machines installed in the guest rooms' floors. At the present time, the F&B operations in the property are limited to in room dining and to the only outlet in the 1st floor. The main kitchen is in the 2nd floor whereas the 1st floor outlet includes a pantry and there is also the in room dining kitchen accessed from the 36th floor. Overall the kitchen and refrigeration equipment don't represent both a large installed power nor electricity consumption.

Figure 61: Kitchen Equipment (Left) and Cold Rooms (Right)

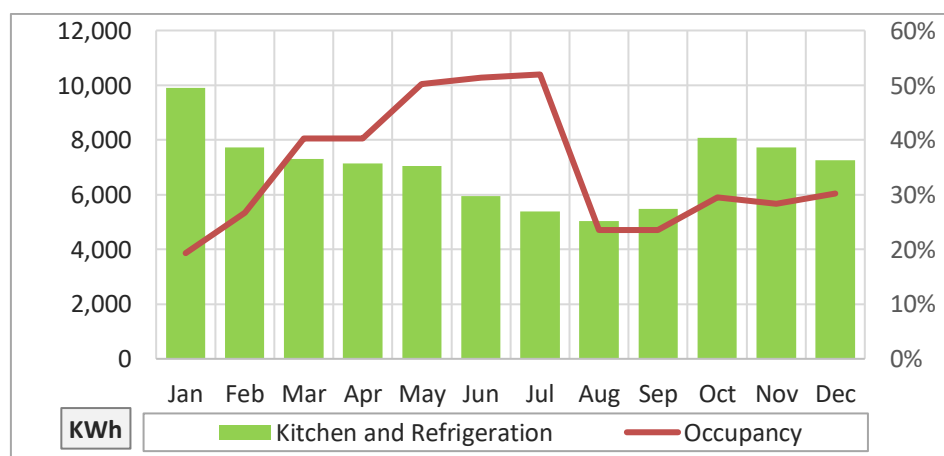


There are five Walk In freezers spread between the receiving dock to the kitchens' areas. It was noticed that doors didn't have vinyl curtains .



The total **Kitchen and Refrigeration** category electrical consumption for the baseline period is **84,041 KWh** with the monthly variations as per the below chart.

Figure 62: Kitchen and Refrigeration Energy Consumption



4.1.6. Others

This category comprises all the remaining electrical driven equipment mainly the low current equipment (loads, Telecom, CCTV) as well as electronic appliances (such as TV and minibars) loads, and other miscellaneous equipment, in addition to the laundry equipment..

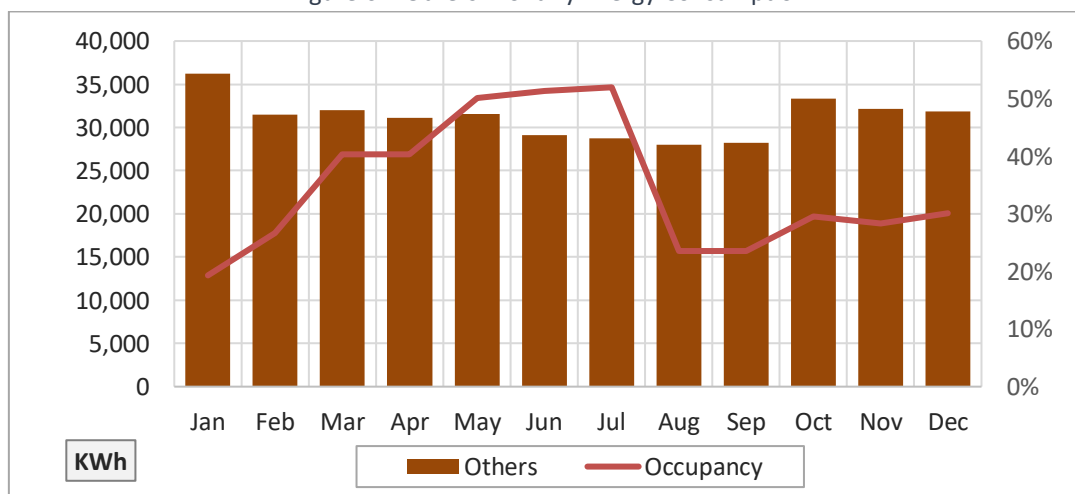
Figure 63: IT Equipment (Left) and Laundry Equipment (Right)



- Top: Ice Machine in Floors (Left); Servers (Right)
- Bottom: Washing Machines (Left); Ironer and Washing Machines (Right)

The total energy consumption of the **Others** category is **373,694 KWh** with the monthly variations as per the below chart.

Figure 64: Others Monthly Energy Consumption



4.2. Energy Balance

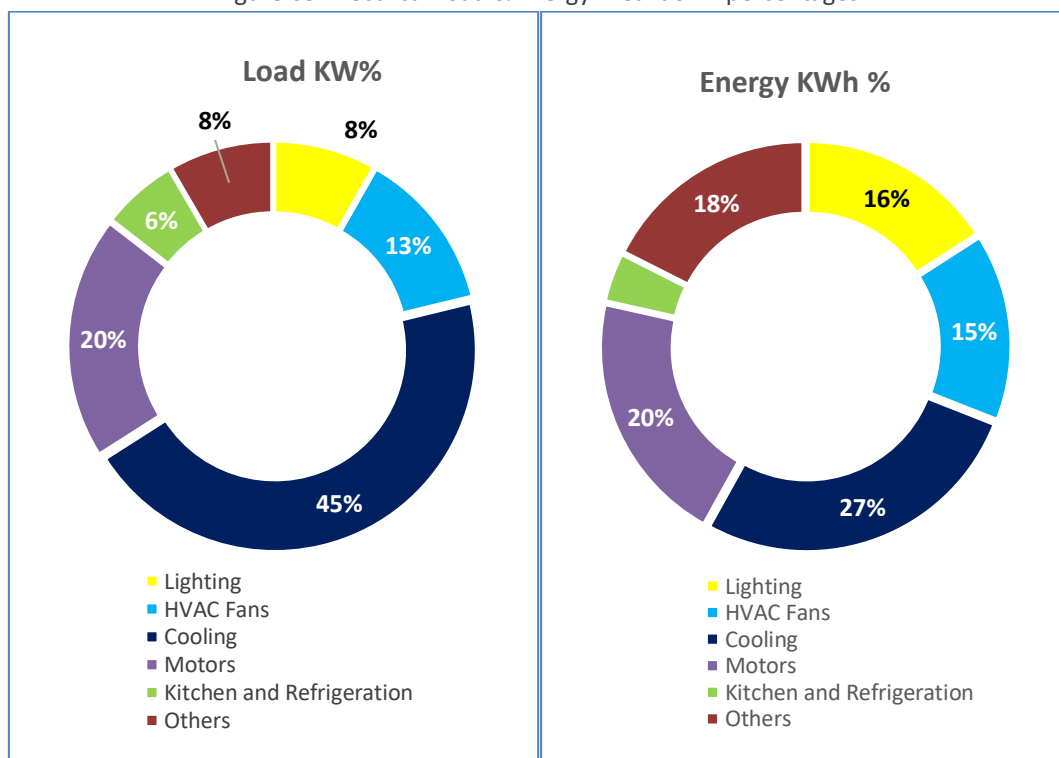
The Electrical Energy Balance provides the breakdown per chosen end-use category for both the installed electrical loads (KW) and the annual energy consumed (KWh). This is achieved following the energy simulation and bills reconciliation completion as per the load inventory, the related operations and the inspection done during the field energy audit.

The following table and figure provide the percentage shares of categories' total installed electrical power in KW and energy consumption in KWh.

Table 23: Electrical Load & Energy Consumption Breakdown by Category

Category	Load		Energy			
Unit	KW	%	KWh	€	Kg. CO ₂	%
Lighting	86	8%	338,973	76,562 €	212,096	16%
HVAC Fans	136	13%	320,450	72,378 €	200,506	15%
Cooling	467	45%	576,981	130,319 €	361,017	27%
Motors	204	20%	435,040	98,260 €	272,205	20%
Kitchen and Refrigeration	64	6%	84,041	18,982 €	52,585	4%
Others	88	8%	373,694	84,404 €	233,820	18%
Total	1,044	100%	2,129,180	480,903 €	1,332,228	100%
Maximum	Cooling - 467, 45%		Cooling - 576,981 KWh, 130,319, 27%			
Minimum	Kitchen and Refrigeration - 64, 6%		Kitchen and Refrigeration - 84,041 KWh, 18,982, 4%			

Figure 65: Electrical Load & Energy Breakdown percentages.



The Cooling category shows the highest installed load at 45% of the total, followed by the Motors at 20%. Whereas, the Cooling category also has the highest share of electricity consumption at 27% of the total.

Figure 66: Annual energy Consumption per end use

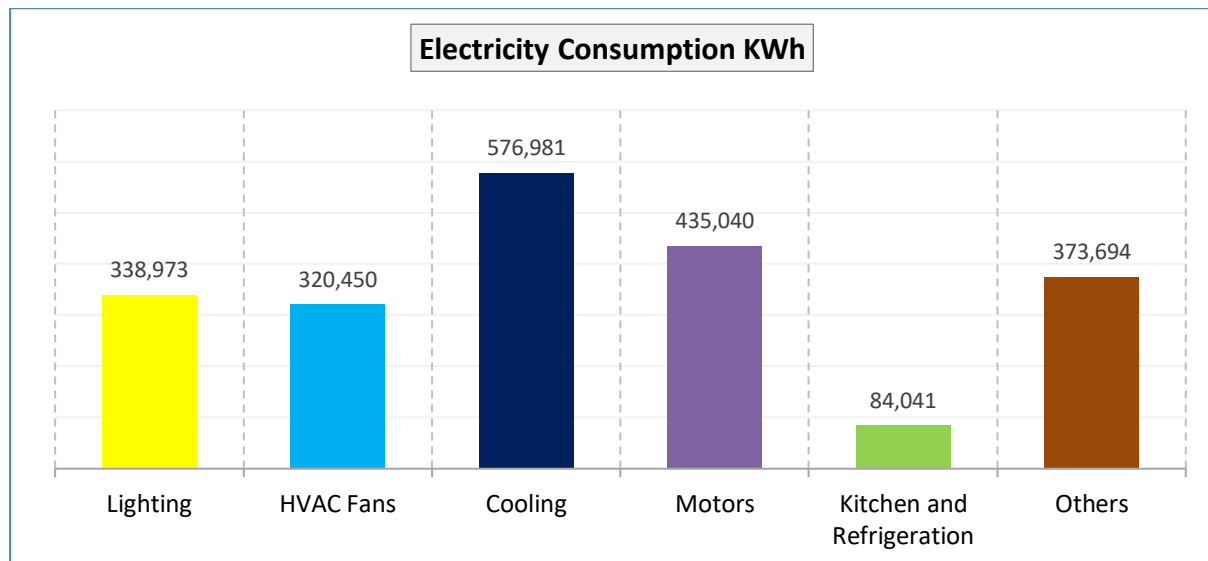
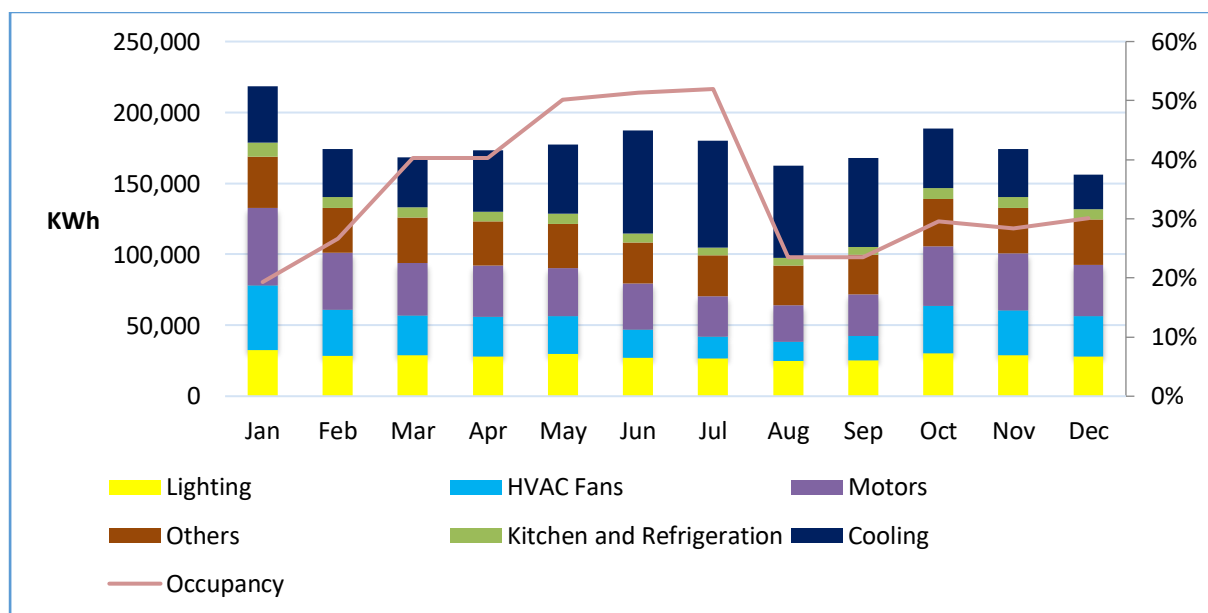


Figure 67: Monthly Electricity Consumption per end use



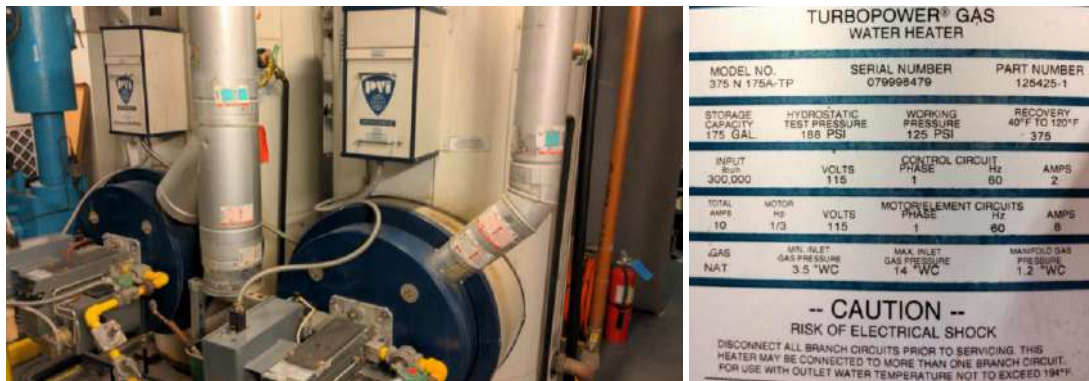
5. Thermal Energy

FSSFE has hot water boilers, as well as a district heating network connection to meet the overall thermal demands (space heating and domestic hot water) of the property .

Gas Boilers system

FSSFE has 2 identical TURBOPOWER gas boilers located in the 4th floor mechanical room producing hot water for the lower floors. Each gas boiler directly heats the 175 Gal storage which is then distributed through hot water pumps.

Figure 68: Hot Water Boilers (Left) and Nameplate (Right)

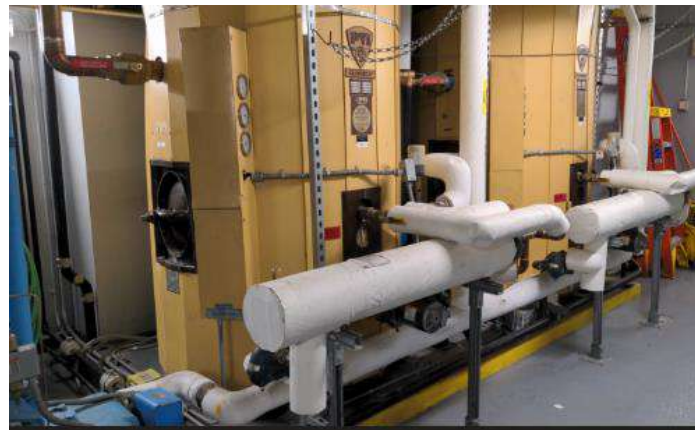


HOT WATER HEATERS / HEAT EXCHANGERS	Type / Model	Rating
WH-5	Gas / 375N175A-TP	175 Gal
WH-6	Gas / 375N175A-TP	175 Gal

District Heating Steam Systems

The hotel is connected to a District Heating network and it has 2 loops networks:

The first one produces hot water through steam in B! via HX- 7 & 8 to service the lower floors in terms of space heating whereas the second one passes via heat exchangers: HX- 3 & 5 located in the East Penthouse which serve the upper floors (guest rooms) in terms of space heating and the production of domestic hot water through HX -4 and two PVI water heaters (WH- 1 & 2) located on 36th floor with a capacity of 500 Gal each and a temperature of 120°F.

Figure 69: HX 3 & 5 in the 48th Floor (Left) and HX 7&8 in B1 (Right)Figure 70: HX 4 in the 36th floor with steam station (Up), 2xHot Water Tanks (Down)

HOT WATER HEATERS / HEAT EXCHANGERS	Type / Model	Rating
WH-1	Steam / 1000-GA-505-S	500 Gal
WH-2	Steam / 1000-GA-505-S	500 Gal

Despite being installed since the beginning of the building operation in 1986 the steam network and its key components were seen to be in a satisfactory condition generally.

During the field visit, a thorough thermal imaging was done on the hot water and steam pipes. Some parts showed weaknesses as can be seen in the below set of images covering respectively the hot water and steam pipes.

Figure 71: Hot Water Pipes Thermal Images (Left) ; Clear related images (Right)

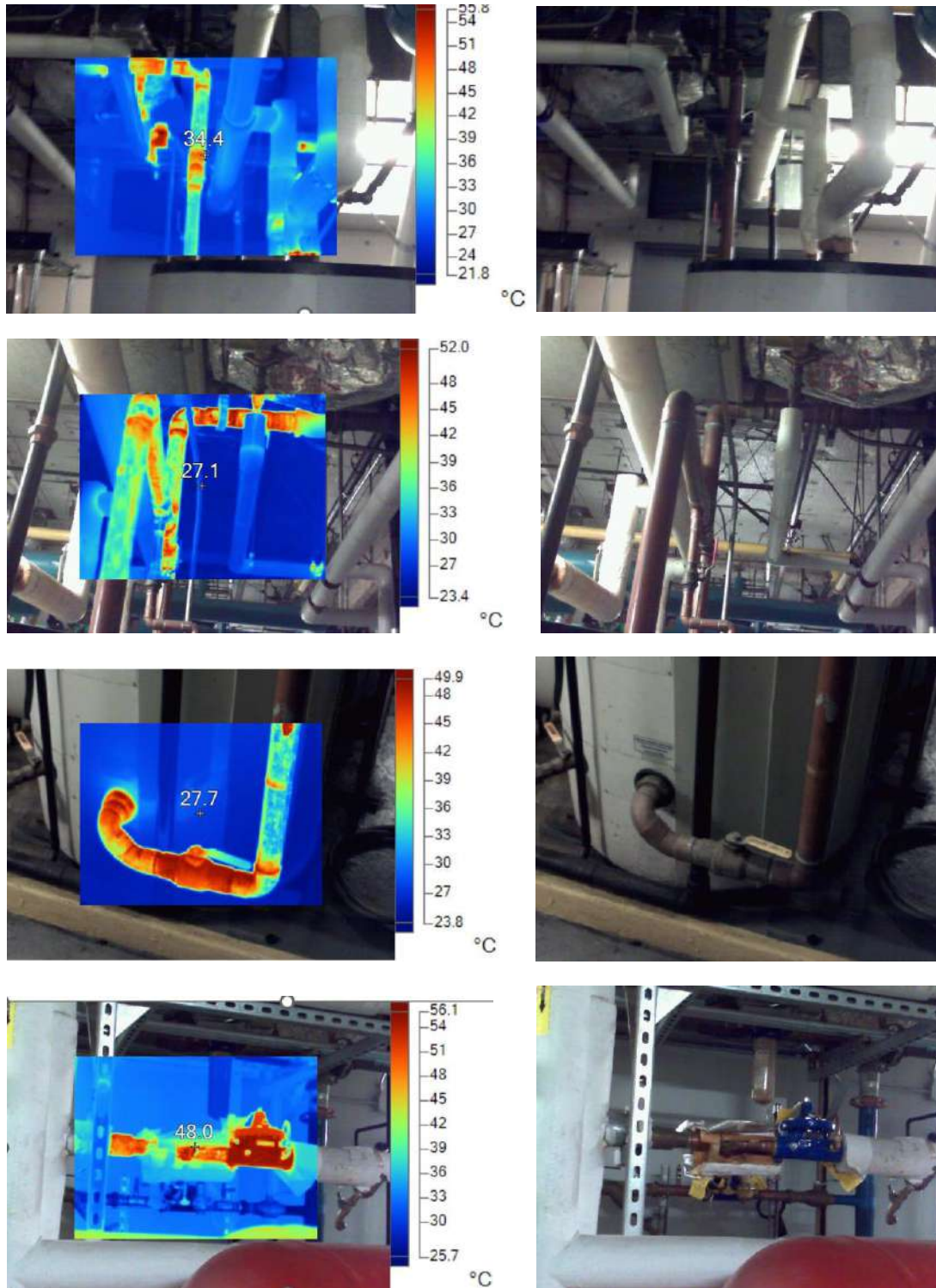
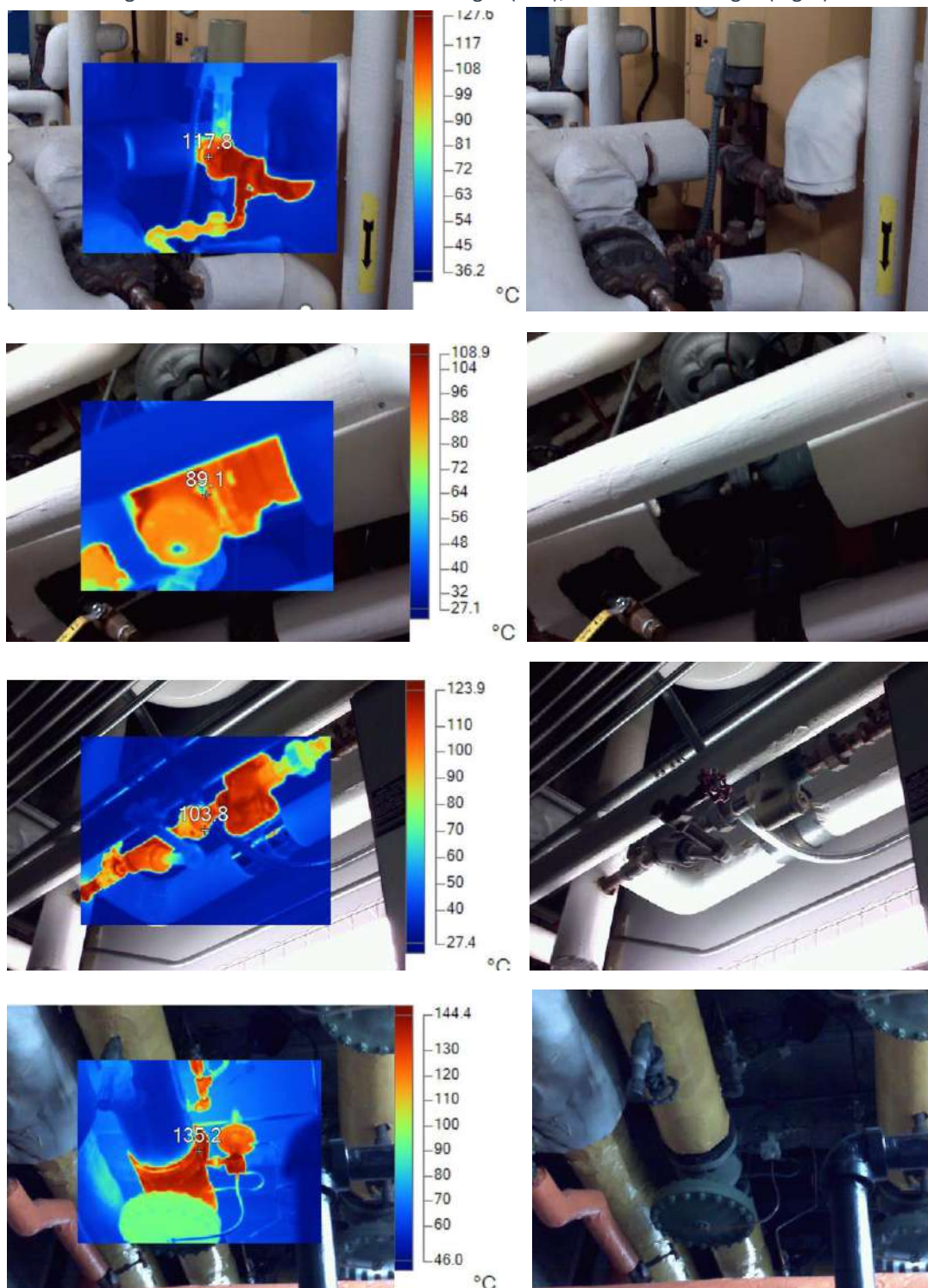


Figure 72: Steam network Thermal Images (Left); Clear related images (Right)



The table below breaks down the thermal consumption of the hotel during the baseline year from natural gas and district heating, in KWh. As it can be seen, District Heating accounts for 88% of the total thermal consumption of the hotel, with natural gas completing the last 12%.

Table 24: FSSFE Thermal Energy Breakdown

MONTH	NATURAL GAS	DISTRICT HEATING	TOTAL
Unit	KWh	KWh	KWh
Jan	10,927	99,125	110,052
Feb	8,529	62,830	71,359
Mar	9,128	62,830	71,958
Apr	7,777	64,660	72,437
May	5,192	47,885	53,077
Jun	7,697	58,286	65,982
Jul	7,957	55,449	63,406
Aug	7,711	50,264	57,975
Sep	7,202	46,696	53,897
Oct	7,954	51,454	59,407
Nov	7,368	52,430	59,798
Dec	11,063	85,034	96,097
Total	98,506	736,941	835,447
Maximum	11,063 (Dec)	99,125 (Jan)	110,052 (Jan)
Average	8,209	61,412	69,621
Minimum	5,192 (May)	46,696 (Sep)	53,077 (May)
Percentage	12%	88%	100%

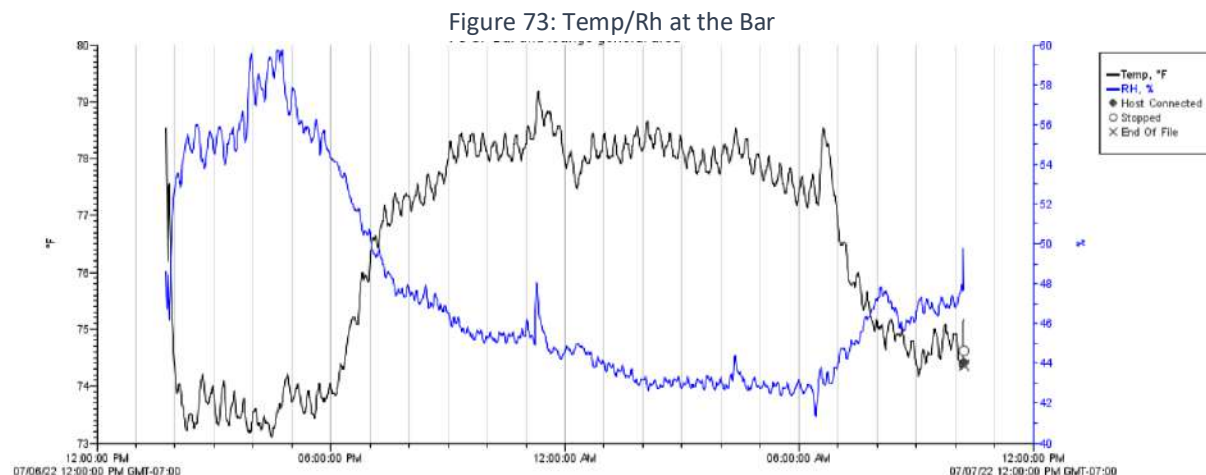
6. Indoor Air Quality data logging

Indoor air quality (IAQ) focuses on airborne contaminants including thermal comfort quality and air quality, covering therefore temperature, humidity, CO₂, CO and VOC (Volatile Organic Compounds) levels.

When it comes to standards, the main guidelines refer to standards promulgated by the American Society of Heating, Refrigerating and Air-conditioning Engineers Inc. (ASHRAE). These are found in the ASHRAE documents Ventilation for Acceptable Indoor Air Quality (ASHRAE 62-2001) and Thermal Environmental Conditions for Human Occupancy (ASHRAE 55-2004).

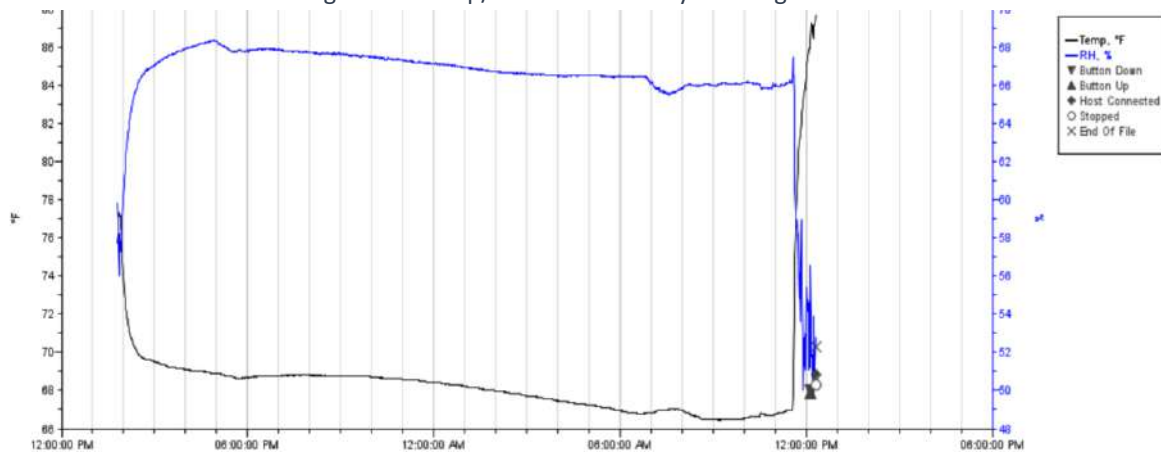
Humidity and Temperature

There is no “ideal” humidity level and temperature suitable for all building occupants. Many factors, such as personal activity and clothing may affect personal comfort. Acceptable relative humidity levels should range from 20 percent – 65 percent year-round. Levels less than 20 percent in the winter and greater than 65 percent in the summer should be considered off the standards. Elevated relative humidity can promote the growth of mold, bacteria, and dust mites, which can aggravate allergies and asthma. To achieve maximum occupant comfort, relative humidity should be maintained between 30% and 65%. The following are graphs, showing temperature and humidity plots in some spaces of FSA. These measurements were taken during the field audit of FSSF in July 2022.



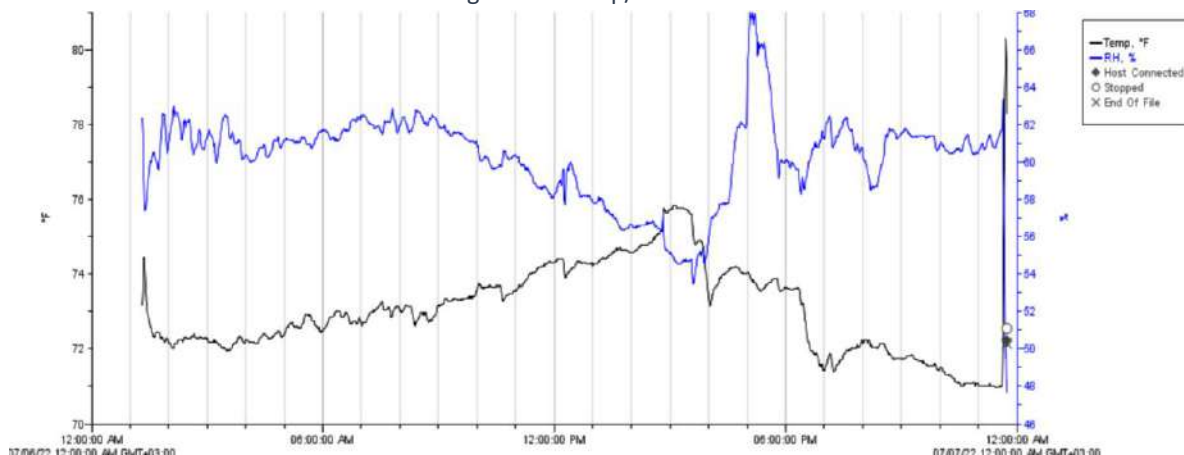
- ➔ The above chart shows that temperature varies between 73 and 79F with a clear distinction of an operation of the ventilation starting around 6h30AM and stopping at 6hPM. RH is seen to be varying between 42% and 60%.

Figure 74: Temp/Rh at the Embassy Meeting Room

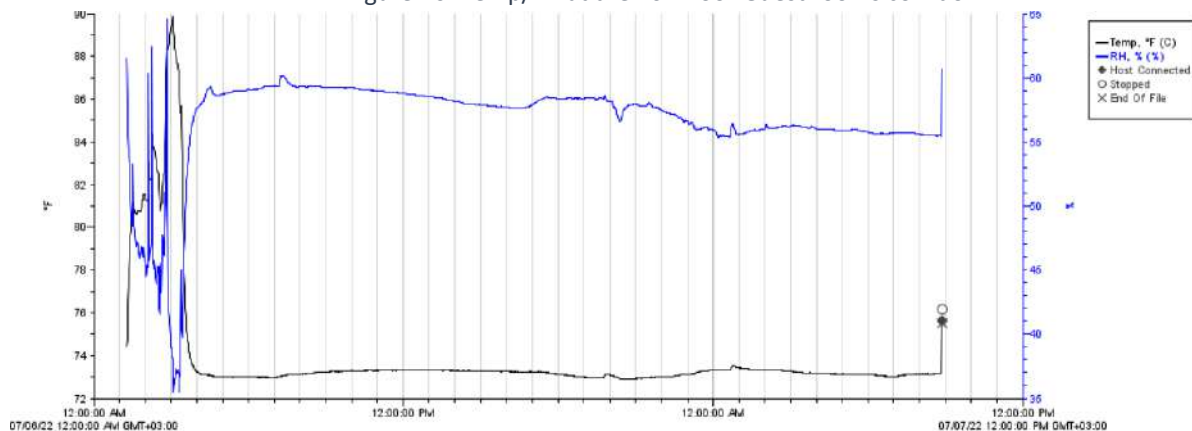


- ➔ Temperature is nearly fixed across the 24 hours of recording at an average of 68F which means that the ventilation system was running continuously. RH was also constant at about 66%

Figure 75: Temp/Rh at the Front Desk



- ➔ Temperature is seen to fluctuate in a narrow band between 72F and 74F and RH between 54% and 68% with the note that this area has large air infiltration from the main hotel's door.

Figure 76: Temp/Rh at the 46th floor Guest rooms' corridor

- ➔ Temperature and RH are seen to be fixed in guest rooms' corridor as ventilation is continuously operating .

IAQ recordings

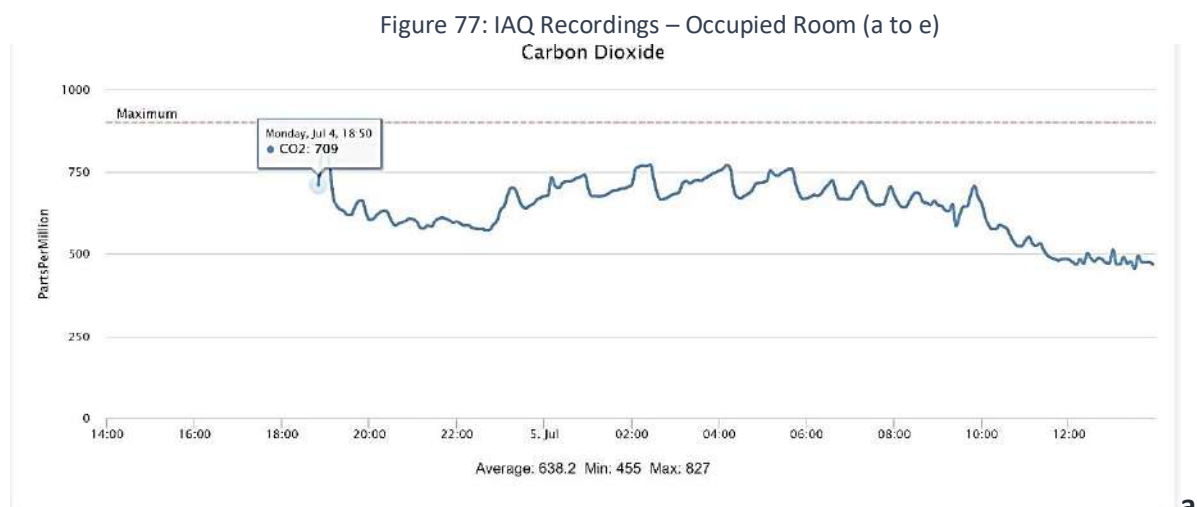
An advanced IAQ recording unit - uHoo Aura was used across the property monitoring 13 different indoor environmental parameters among them – temperature, relative humidity, carbon dioxide, various particle sizes (PM10, PM4, PM2.5, PM1), carbon monoxide, formaldehyde, and volatile organic compounds (VOCs). The IAQ unit comes with a Virus Index which is an assessment of virus survival and transmission risk (such as the coronavirus) based on air quality.

Some key IAQ parameters information are found below:

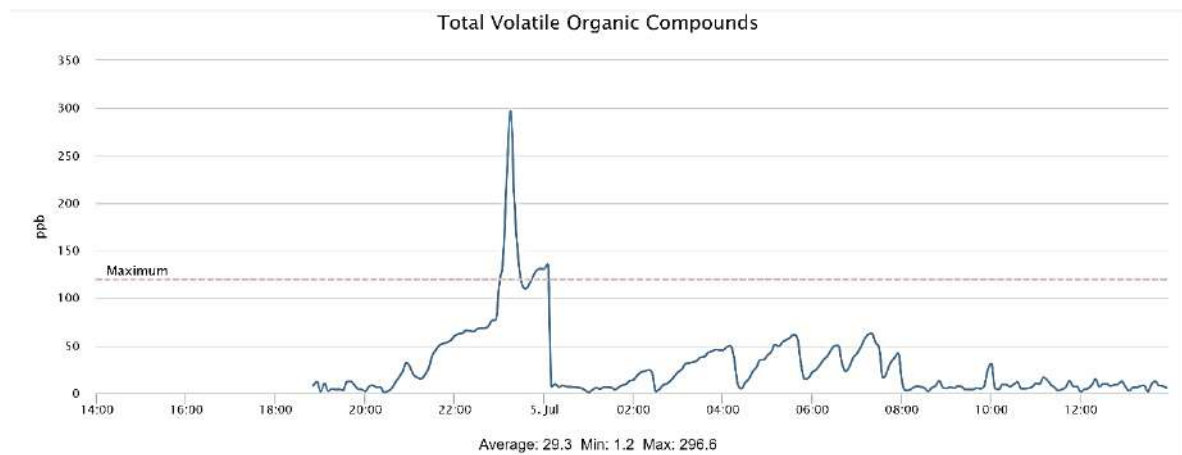
- **PM2.5:** Particulate matter is one of the most dangerous forms of pollution as the size of the particles is so small that they can get into the lungs causing numerous adverse effects. PM2.5, in particular, are particles which are 2.5 μm or less in diameter. Their threshold limit value is 25 $\mu\text{g}/\text{m}^3$, based on 24-hour data.
- **CO₂:** CO₂ is a natural compound in the air, with an average outdoor concentration of 300-400 ppm. Note that indoor levels are higher due to the confinement of indoor spaces.
- **TVOC:** The Total Volatile Organic Compound (TVOC) level is a measurement of the sum of all of the volatile organic compounds (VOC's) found in an air sample
- **Formaldehyde:** One of the most common VOCs – formaldehyde – can be emitted from numerous sources, such as furniture, incense burning, and cooking. Note that its threshold limit value is 0.1 ppm

Occupied Guest Room case:

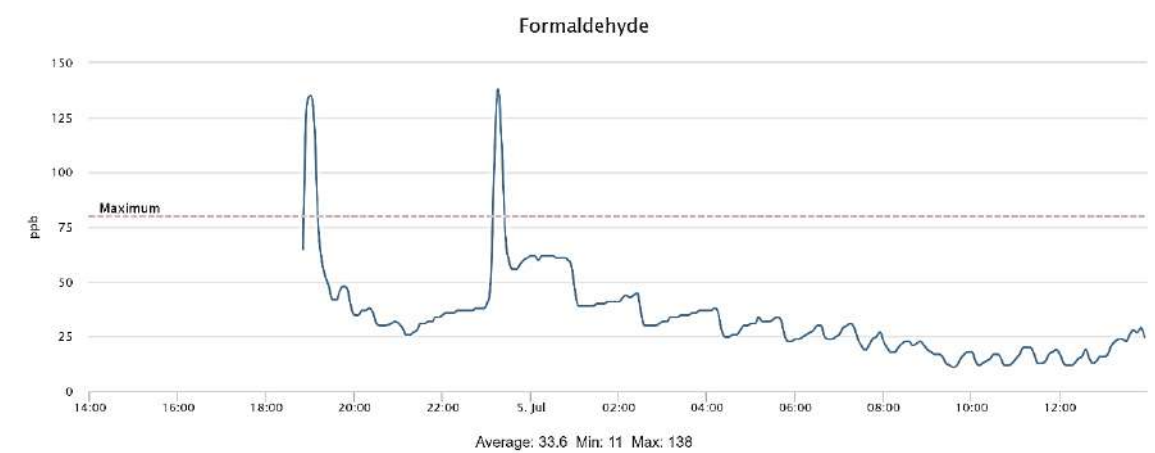
The following set of IAQ recordings were done in an occupied room whereby all parameters were seen to be within OSHA and ASHRAE St 62.1 Standards for Ventilation and Indoor Air Quality. In that respect, the below chart #a covering CO₂ showcases levels from 500 to maximum 750ppm , well within the ASHRAE standards and confirming the adequate fresh air intake in guest rooms.



The TVOC and Formaldehyde as well as PM2.5 to PM1 showed satisfactory levels. There was a sudden hike and drop in Formaldehyde as in this specific time a steamer for clothes was used.

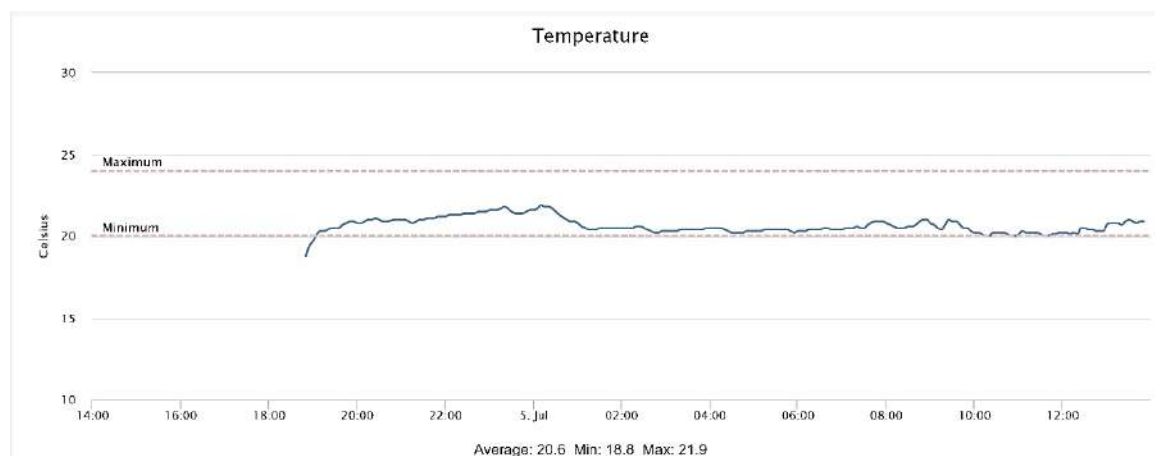


b

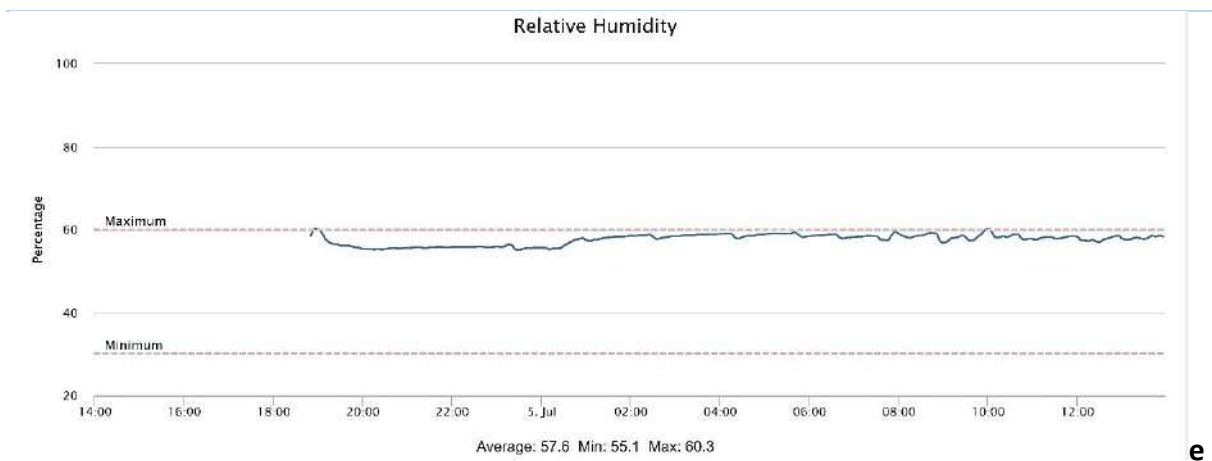


c

Temperature was seen to be varying within the close range of the thermostat setpoint of 20.5°C/69°F which is an indication of a proper operation of the thermostat sensor and the control system. On the other hand RH was seen to be satisfactory at an average of 59% (chart e)



d



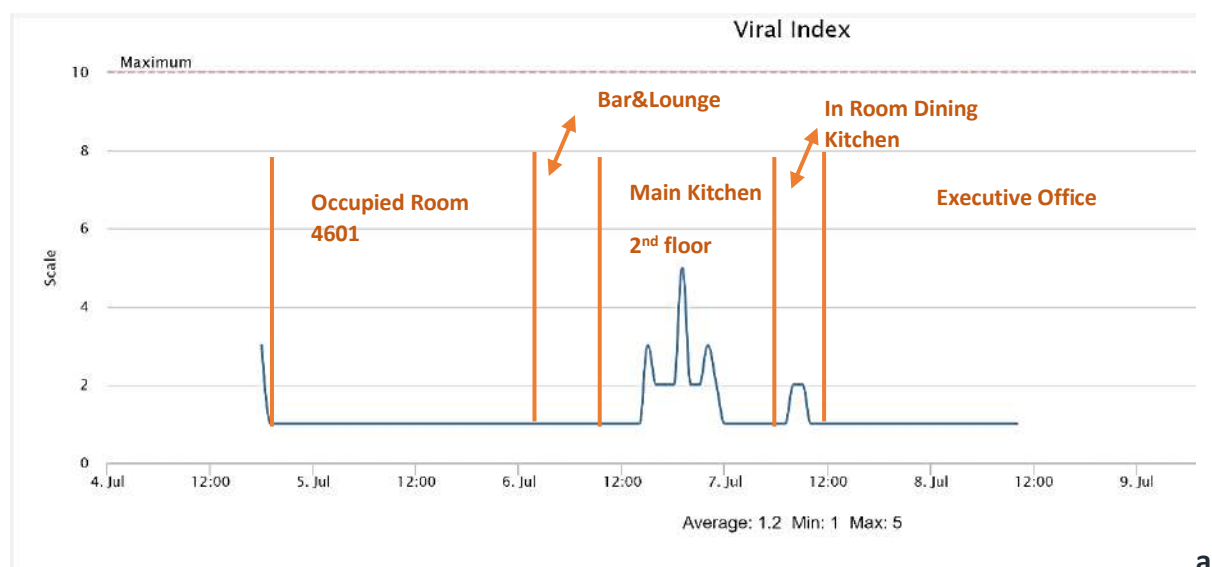
Total Recordings and locations :

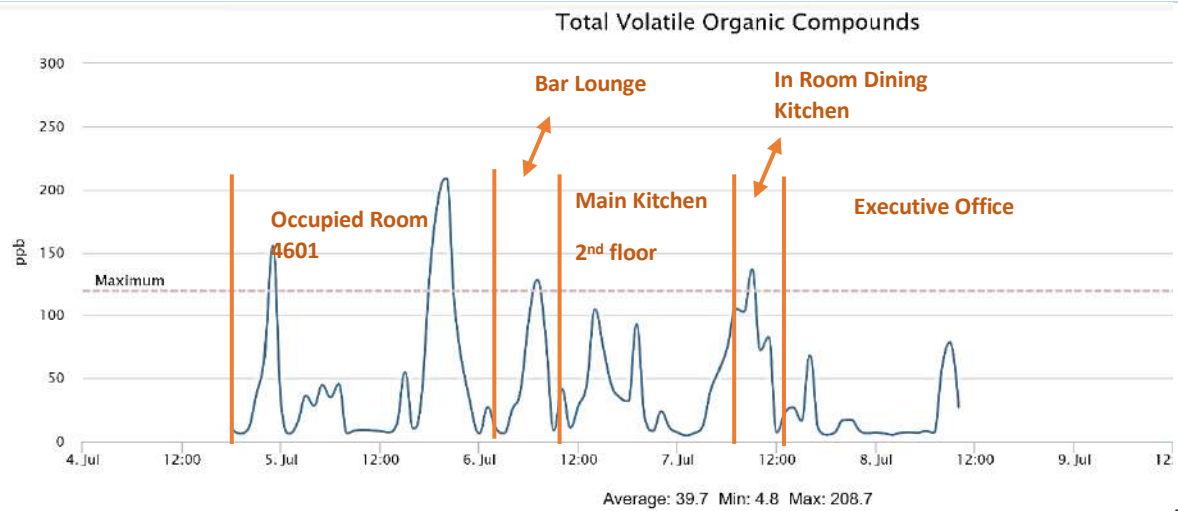
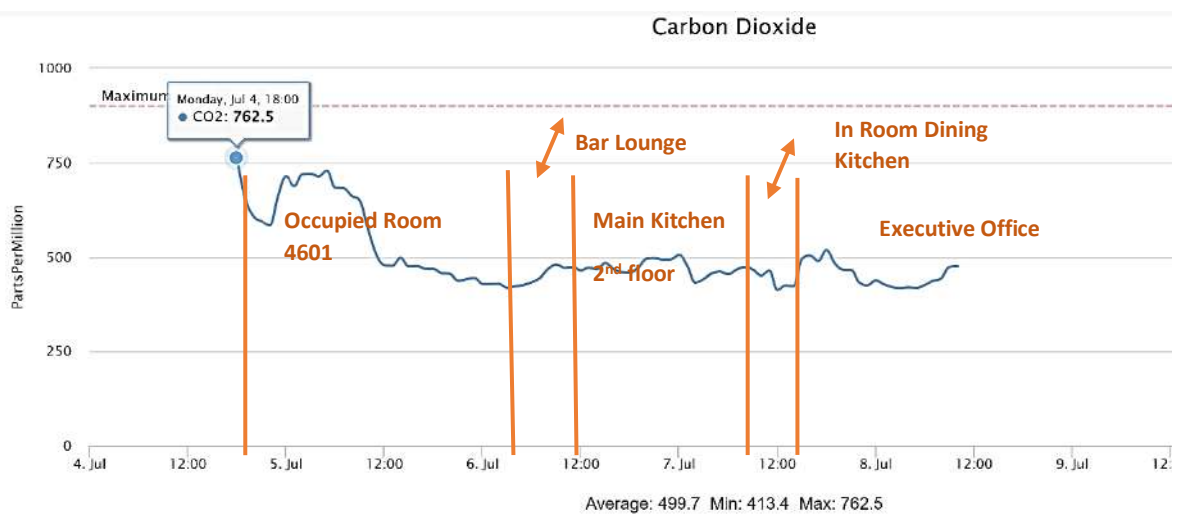
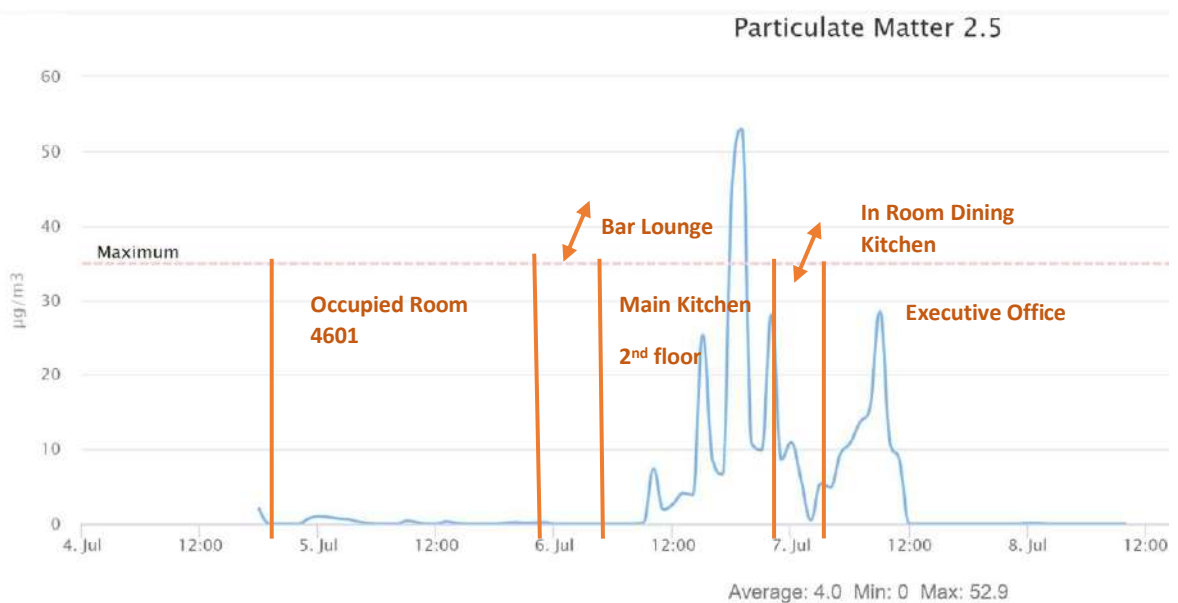
The following set of charts provide different IAQ parameters over a total of 4 days and in 5 different locations. The first one (chart a) shows the virus index across the timeline clearly indicating a very satisfactory overall IAQ.

There was some increases in TVOC , Formaldehyde and specially PM 2.5 in the Main Kitchen due to cooking.

Temperature and RH were seen to be overall within standards and are to be expected in a mild climate such as the one found in San Francisco.

Figure 78: IAQ Recordings – All locations (a to e)

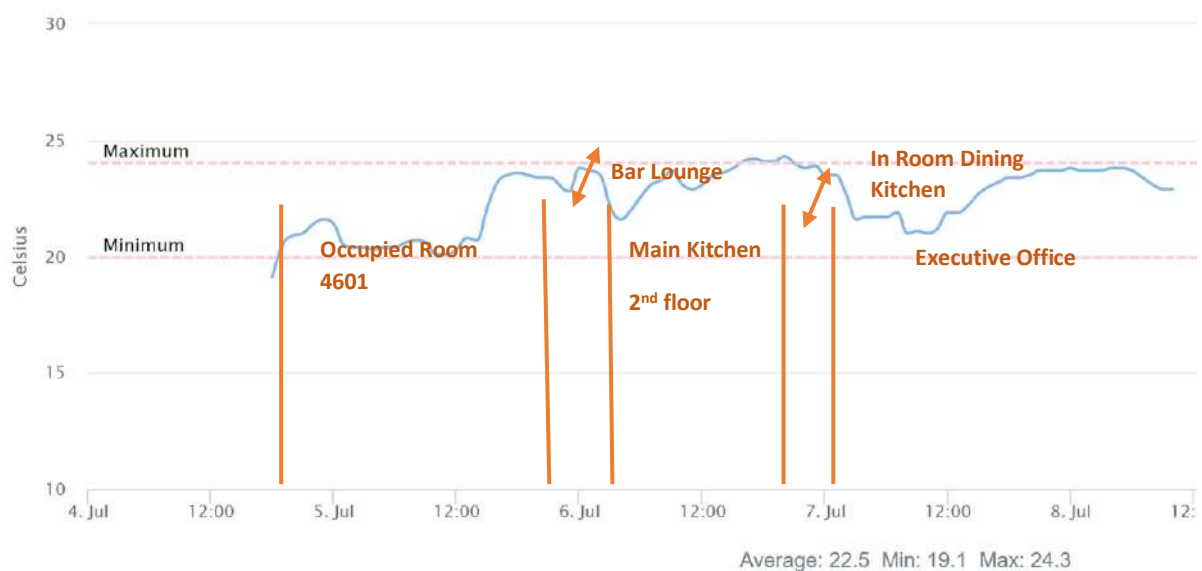


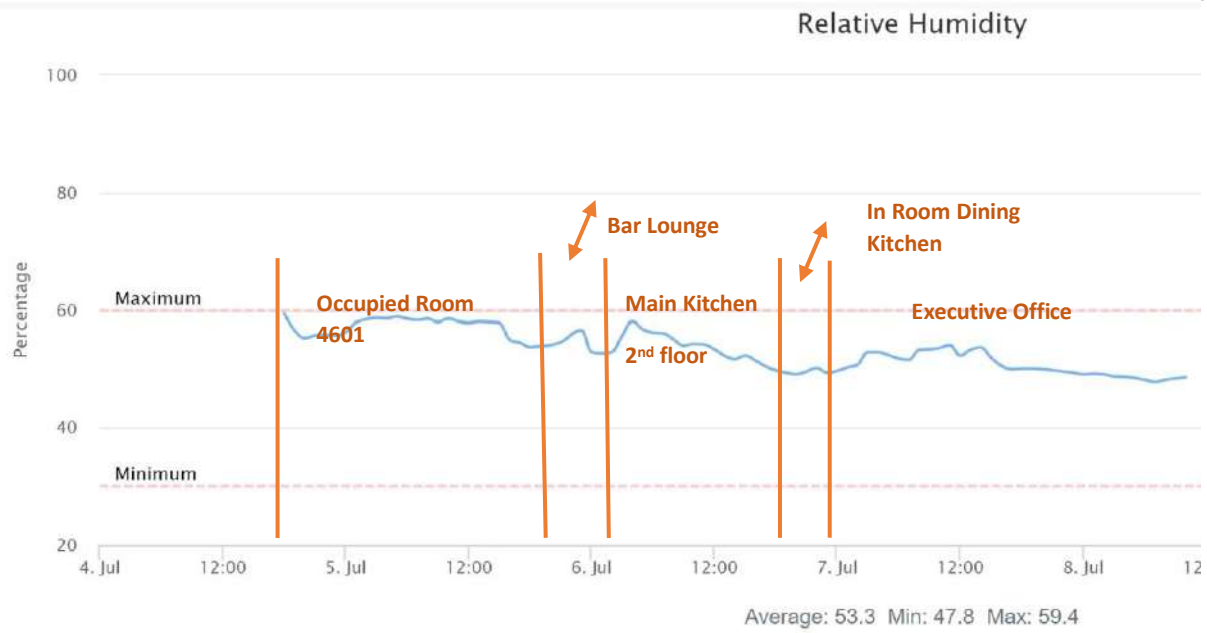
**b****c****d**

Formaldehyde



Temperature





7. Energy Efficiency Measures

The main objective of this section is to provide a roadmap that would include a number of Energy Efficiency Measure (EEM) which are all the actions that would reduce the utilities consumption. In this aim, EEG investigated a considerable number of energy efficiency solutions, which could yield to substantial energy cost and carbon footprint reductions in the event of their implementation. These opportunities include those who require very little or no CAPEX at all – Low Cost/No Cost – [LC/NC] as well as those needing capital expenditure [CAPEX]. The assessment of an EEM is its financial feasibility and technical viability. Each EEM is attributed with a color code which is a priority or urgency indicator for the EEM's implementation. The color codes adopted in this report along the proposed measures are found in the below tables:

Table 25: Energy Efficiency Measures Color Codes

Color Code	Description	Priority
	Measures with very small CAPEX and excellent payback (<1 year) – Low Cost/No Cost	A
	Measures with good payback period (<3 years)– Immediate Implementation	A
	Measures with midterm payback period (3 to 6 Years) – Strategic Implementation	B
	Measures to be done during a planned retrofit/replacement	C
	Measures with high payback period (>6 years) – Upon CAPEX Availability	D

Table 26: Energy Efficiency Measures

CODE	TYPE	TITLE
EEM - 01	CAPEX	KW Demand Management
EEM - 02	CAPEX	Lighting Retrofit
EEM - 03	LCNC	Lighting Control and Management
EEM - 04	LCNC	HVAC Control
EEM - 05	LCNC	Chilled Water: Low ΔT Mitigation
EEM - 06	LCNC	Manage Guest Rooms During Unrented Mode
EEM - 07	CAPEX	Improve Thermal Insulation
EEM - 08	CAPEX	Install Vinyl Curtains on all Cold Rooms (positive and negative temperature)
EEM - 09	CAPEX	Install VFDs on Fans
EEM - 10	CAPEX	Retrofit Legacy FCUs by ECM FCU Type
EEM - 11	CAPEX	Retrofit Electrical HW Tank in 36th floor
EEM - 12	CAPEX	Retrofit Chillers 3 and 4 serving guest rooms
EEM - 13	CAPEX	Retrofit DX Unit Servicing Sky Bridges
EEM - 14	CAPEX	BMS Upgrade and pneumatic to DDC conversion

As can be seen in the previous table, FSSFE has five key EEMs related to retrofits . In fact, the property has an ageing core MEP equipment that were not retrofitted or upgraded during the last renovation – as explained in section 3.3. Even if most are still operating satisfactorily, the advances in the last three decades as well as the need to improve energy efficiency – and remove R22 refrigerant – retrofits have been planned with clear rationale .

The other EEMs (01 to 09) fall in the Low Cost/No Cost and good payback period ones (Green and Blue colored).

In the following pages, each EEM has been developed with the rationale, technical description/recommendations and the economical analysis. At the end of each EEM sub section, there is a summary table with all the key indicators as explained here below.

EEM - XX		
XXXX		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
	0	0
0.0%	0.0%	0.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		
Annual Share of Total Utilities Cost Savings [%]		
Implementation Cost [\$]		
Simple Payback [Years]		
Net Present Value [\$]/20 years		
Internal Rate of Return [%]		

The first two lines include the EEM number and its title

The second box include the energy savings in each utility (per unit of measure) and the % savings from the same utility. Also, the CO₂ emissions reductions of the EEM is calculated

The third box include the Economical Performance with the Cost Savings , the % savings from all Utilities budget, the implementation cost, simple paybacl, IRR – Internal Rate of Return, as well as a Net Present Value calculated ove3r 20 years.

7.1. EEM-01 [CAPEX]: KW Demand Management

7.1.1. Rationale

FSSFE's electricity billing includes two main components, the electricity consumption (Kwh) and the Demand charges (based on the maximum KW drawn). The bill includes two components, one related to PG&E transmission fees and one for Clean Power Electricity production; both includes the combination of KWh and KW charges with the latter representing more than 35% of the total bill topping the \$120K annually .

These charges could be managed and reduced under a demand response and internal energy management program.

Figure 79: KW Demand charges components within PG&E Transmission section in the utility bill

06/08/2022 – 07/07/2022				
Customer Charge	30 days	@ \$31.54779		\$946.43
Demand Charge ¹				
Max Peak	396.000000	kW @ \$32.90000		13,028.40
Max Part Peak	446.400000	kW @ \$6.81000		3,039.98
Max Demand	458.400000	kW @ \$26.07000		11,950.49
Energy Charges				
Peak	43,097.700000	kWh @ \$0.19598		8,446.29
Part Peak	32,186.700000	kWh @ \$0.15878		5,110.60
Off Peak	112,170.750000	kWh @ \$0.13246		14,858.14

7.1.2. Efficiency Action and Energy Savings

Demand management strategies include the following:

- i. **Load shedding** : Once the load profile is known , a load shedding strategy will be put in place whereby unnecessary loads can be shut down
- ii. **Load Rescheduling**: Similarly, electrical loads can be rescheduled in a way to flatten the demand curve . For example, if the peak demand is seen to happen between 2h and 4hPM, the strategy would look at equipment or operations that can be rescheduled to before or after 4hPM.
- iii. **Load shift to Energy storage**: The project would consist on installing Energy Storage Systems that will be charged during off peak demand period and then will go online during the peak demand period, as can be seen in the below chart.

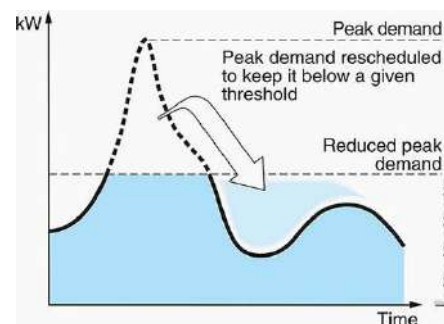
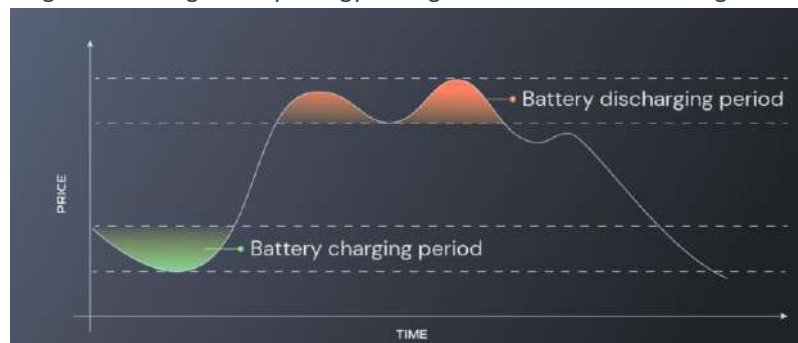


Figure 80: Using Battery Energy Storage for Peak Demand Management



The key point in Demand Management is having real time access to the property's Demand profile and use it to automate the strategies mentioned above .

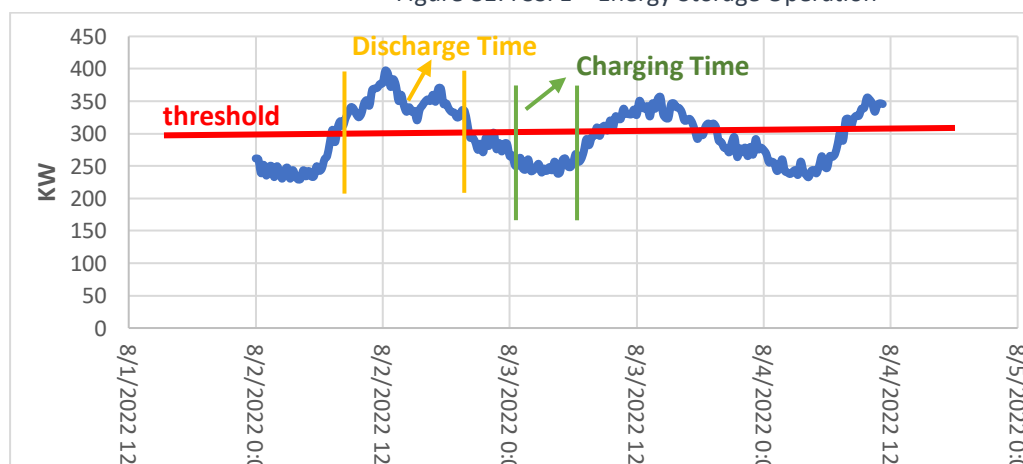
7.1.3. FSSFE Case

FSSFE can implement the three strategies previously indicated. As mentioned, the first key point would be to have a real time access to the Load Demand (KW) through either the smart meter of PG&E or by installing in house meters to understand the time frames of the demand changes. Typically the maximum demand was seen to occur between 12hPM and 4hPM.

During these times, the property can implement as a first line of actions by reviewing which loads can participate in a Load Shedding and Load Rescheduling strategies. These may include laundry's operation, some unnecessary Fans control or simply setting more aggressive settings on the chilled water plant (higher temperature setpoint and DP) which will push both chillers, the related pumps (condenser and primary) as well as cooling towers to run at a lighter load for few hours without affecting the occupants' satisfaction.

The second step would be to invest in energy storages that would shift part of the load to batteries during the few hours where peak demand is attained. This will be an automated system with a threshold control and could shift 50KW to 100KW from the grid to the batteries which would be recharged during the lowest demand time (between 12hAM and 4hAM).

Figure 81: FSSFE – Energy Storage Operation



FSSFE can start implementing the Demand Management strategies as the first two steps (Load rescheduling and Load shedding) can be planned immediately . Tracking the load profile

through the live data of PG&E portal would be necessary to understand the changes among months .

For the energy storage, the project would need detailed design to integrate it into the existing feeder with an automated response. However, the key technical aspects are as follows:

- Total energy storage should be able to provide a total net discharge of 400,000Wh over 4 hours.
- Batteries to be of Lithium Ion type.
- Design to include inverters, space location, and electrical connections.

Total savings will be from the reduction of the Max Demand on both the PG&E part (which is the major one in terms of Demand charging) and the Clean Energy Electricity purchase ones.

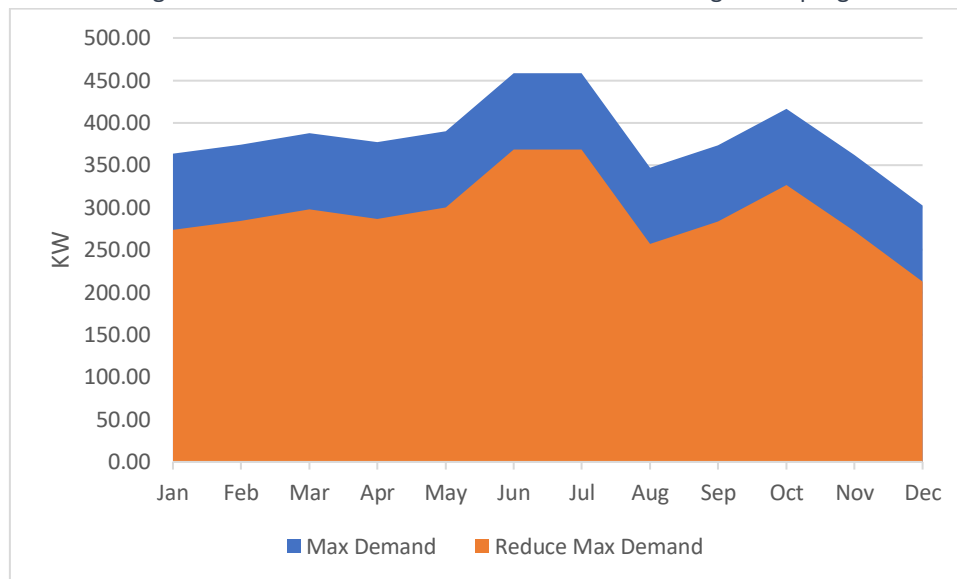
Considering only the Max Demand part of the PG&E – excluding also the Max Peak parts that are high in the summer months would provide annual savings in excess of \$26,000 . It is to be noted that a net 90KW was considered as a reduction while the energy storage would be rated to provide 100KW of load for 4 hours as mentioned.

In addition, no electricity (KWh) savings were considered although the first two steps of load shedding and load rescheduling may include energy savings on the top of the KW Demand reductions.

Table 27: Electrical Load & Energy Consumption Breakdown by Category

Month	Max Demand	New Max Demand	Tariff	Savings
-	KW	KW	\$/KW	\$
Jan	363.60	273.60	\$26.03	\$2,343
Feb	374.40	284.40	\$26.03	\$2,343
Mar	387.60	297.60	\$25.59	\$2,303
Apr	376.80	286.80	\$25.59	\$2,303
May	390.00	300.00	\$25.59	\$2,303
Jun	458.40	368.40	\$26.07	\$2,346
Jul	458.40	368.40	\$26.07	\$2,346
Aug	346.80	256.80	\$22.98	\$2,068
Sep	373.20	283.20	\$22.98	\$2,068
Oct	416.40	326.40	\$22.98	\$2,068
Nov	362.40	272.40	\$22.98	\$2,068
Dec	302.40	212.40	\$22.98	\$2,068
Total	4,610.40		\$24.66	\$26,628

Figure 82: Max Demand - Before and After the Management program



The project will involve direct actions at no CAPEX as well as the installation of the Energy Storage system and its control that would include a CAPEX.

The below table provides the key indicators of this EEM.

EEM - 01		
KW Demand Management		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
	0	0
0.0%	0.0%	0.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$26,628
Annual Share of Total Utilities Cost Savings [%]		3.8%
Implementation Cost [\$]		\$55,000
Simple Payback [Years]		2.1
Net Present Value [\$]/20 years		\$291,379
Internal Rate of Return [%]		48%

7.2. EEM – 02: [CAPEX] Lighting Retrofit ■

7.2.1. Rationale

FSSFE has upgraded all its main FOH lighting to LED lights during the last renovations including all guest rooms. However, there are large quantities of non LED lights specifically T8 fluorescent in some BOH areas and halogen/incandescent in the 2nd floor mainly.

Waiting for a failed bulb to be replaced by LED is not an effective way as lighting retrofit would provide a payback period below 18 months in most cases (and below one year for halogen replacements), specially in times of increased electricity tariffs and the impact of Demand Cost discussed in the previous EEM.

It is therefore recommended to allocate a budget to retrofit ALL non LED lights at once and optimize the related electricity consumption the earliest possible.

7.2.2. Efficiency Action and Energy Savings

At the present electricity tariffs, there is no point to wait for a lighting bulb/fixture to fail before retrofitting it with an LED equivalent. The process of retrofitting should be immediate to all non LED lighting fixtures as the ROI is very good . Even if in some areas the fit-out is not completed or initiated , as long as these areas are used , their lighting should be changed.

Some foreseen example areas are shown here below:

- i. All T8 fluorescent found mostly in and near the Main Kitchen (2nd floor) , in the 36th floor, all mechanical rooms, stairways etc...



- ii. All Halogen MR16 spotlights found in sky bridges across floors:



- iii. All Halogen MR 16/AR111/T8 Fluorescent in 2nd floor toilets/meeting rooms/corridors etc... | Chandelier's E14 candle Incandescent in 1st floor



Among the bulbs/tubes that should be retrofitted. The field audit inventoried more than 270 T8 fluorescent 36W tube as well as 100 MR16 35W, 80 E14 candle 25W/40W and 25 AR111 50W. These are the core retrofits to be done as well as T8 18W and T5 28W fluorescent tubes

All of these retrofits are very easy to be installed and will provide quick energy savings topping the 83,000 KWh annually. The below table provides the key indicators of this EEM.

EEM - 02		
Lighting Retrofit		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
78,000	0	32,699
3.7%	0.0%	3.4%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$17,617
Annual Share of Total Utilities Cost Savings [%]		2.5%
Implementation Cost [\$]		\$11,000
Simple Payback [Years]		0.6
Net Present Value [\$]/20 years		\$218,165
Internal Rate of Return [%]		160%

7.3. EEM – 03: [LCNC] Lighting Control and Management

7.3.1. Rationale

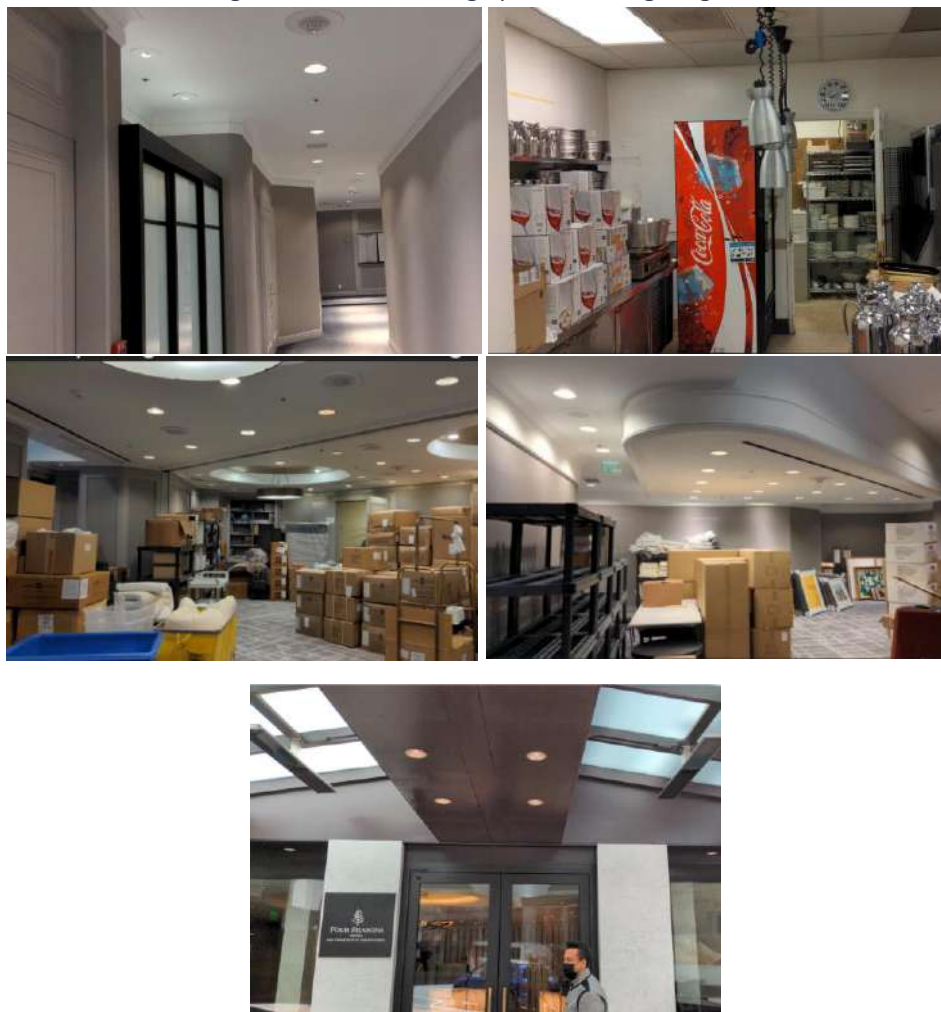
Lighting Control's concept is based on managing lighting loads across various areas of a property through human manual control or through the introduction of specific technologies. It provides a major opportunity for any facility to reduce its overall lighting energy consumption. Lighting control examples are Time control, Daylight control and Occupancy control.

Lighting control would be effective in FSSFE as there are large areas which were seen to have their lighting operating while they were vacant.

These areas include the following:

- Large areas in the 2nd floor from the unused meeting rooms to unused pantries and toilers
- Technical rooms
- Housekeeping pantries
- Stairways

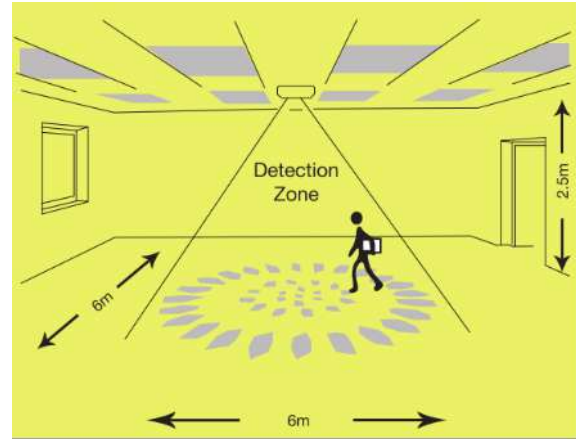
Figure 83: Areas with high potential of lighting control



7.3.2. Efficiency Action and Energy Savings

To optimize the lighting control in FSSFE, the strategy would be to install Motion Detectors or to enforce manual control in some areas. The plan to include the following:

- i. Review the possibility of daylighting control of Canopy light and skybridges
- ii. Coordinate with F&B, Housekeeping and Engineering to ensure all rooms in the 2nd floor have their lighting controlled across the days .
- iii. Review potential Motion Detectors installations in : Housekeeping pantries across the floors, technical rooms (in B1, 4th and 36th floors mainly)



The below table provides the key indicators of the EEM whereby the CAPEX represents the installation of about 35 units of Motion Detectors across the property.

EEM - 03		
Lighting Control and Management		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
23,230	0	9,739
1.1%	0.0%	1.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$5,247
Annual Share of Total Utilities Cost Savings [%]		0.7%
Implementation Cost [\$]		\$4,000
Simple Payback [Years]		0.8
Net Present Value [\$]/20 years		\$64,251
Internal Rate of Return [%]		131%

7.4. EEM-04: [LCNC] HVAC-Fans Control ■

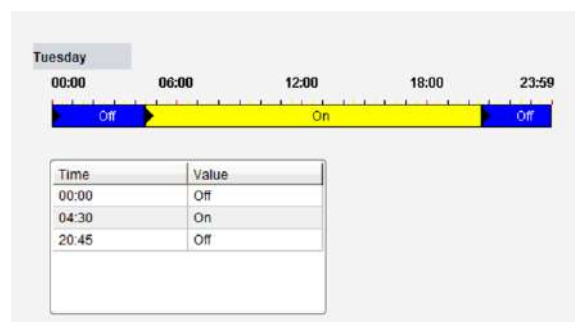
7.4.1. Rationale

Fans in FSSFE represent about 15% of the annual electricity consumption. They include mainly FCUs as well as Supply/Extractor Fans and Kitchen Hoods. At the present time, some are controlled by the GRMS (FCUs in guest rooms) while others are controlled by the BMS and its time scheduler. Further optimization can be undertaken through the application of various control strategies .

7.4.2. Aggressive Scheduling

The BMS has a scheduler module that is connected to some of the fans . However, during the field audit, the BMS analysis and the temperature recordings/data logging , there is still a good room of improvement in an aggressive scheduler setting across all the property. The climate of San Francisco allows in fact to apply aggressively such a program whereby vacant or little utilized areas (like all the 2nd floors' meeting rooms/corridors/toilets) can be operated only for a very limited number of hours per day while others can be controlled further.

Figure 84: E61 schedule of operation



The key mitigation actions are the following:

- Review the operation of each Fan (Supply and Exhaust) as well as the area it services including those serving kitchens operations. These include Fans serving penthouses.
- Review the operation of FCUs in BOH
- Implement an aggressive scheduling that should be proactively modified every day in some units, every week in others. This includes the below two scheduling possibilities:
 - I. Some units can be shut down totally night time or during vacancy such as all those serving the functions areas . During the day they can be duty cycled ON/OFF if also vacant (i.e: 1 hour ON/ 1 hour OFF) if needed. This would be better than maintaining them operational continuously.
 - II. Some units that can't be shut down fully at night time, can be duty cycled in low occupancy period such as night time or known low circulation times . Duty cycling can vary from unit to unit and can be as low as 45mn ON/15mn OFF to larger window of time.

7.4.3. Setpoint Changes

Also, setpoints are not meant to be kept constant across the days, weeks, and months, Based on the occupancy and ambient conditions, all setpoints should be readjusted to the most optimal ways taking into consideration the mild San Francisco climates and differences between days and nights.

Actions to be taken are as follows:

- Map all FCUs and their current setpoints.
- Review potential changes across different scenarios of ambient conditions and occupancy
- Ensure BMS operator change applies the strategies on a daily basis while logging all modifications – if possible or else set strategies manually.
- The same would be applied to chilled water temperature production – which was seen already at high levels but could be reset even higher under specific time of the day or climatic conditions

Applying these strategies would yield important energy savings at no CAPEX as the controls will be done strictly via the BMS and manually when it comes to changing setpoints to some of the FCUs. The below table provides the key indicators of this EEM,

EEM - 04		
HVAC Control		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
100,557		42,155
4.7%	0.0%	4.6%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$22,712
Annual Share of Total Utilities Cost Savings [%]		3.2%
Implementation Cost [\$]		
Simple Payback [Years]		-
Net Present Value [\$/20 years]		\$295,437
Internal Rate of Return [%]		-

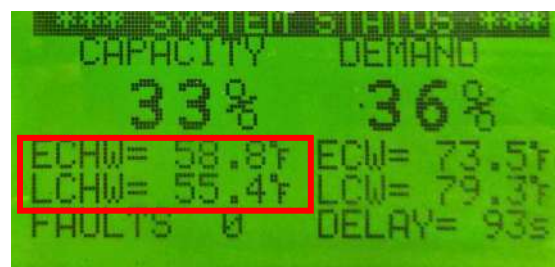
7.5. EEM – 05: [LCNC] Chilled Water: Low ΔT Mitigation

7.5.1. Rationale

During the field audit it was seen that the Delta Temperature (ΔT) of both chilled water plant was very low as can be seen in the below snapshot. Chiller 5 was barely 3.4°F while Chillers 3&4 it was 2.1°F!

Figure 85: Chiller 5 Supply/Return Chilled Water – Chiller 5 a & b, Chiller 3&4 c

	CHWTEMP-DIFF	14.0 deg F	Chilled Water Differential Temp
	CHWSUPPLY-TEMP	65.8 deg F	Chilled Water Supply Temperature
	CHWRETURN-TEMP	66.3 deg F	Chilled Water Return Temperature
	CHWDT-SP	20.0 deg F	Chilled Water Diff Temperature
	BYP-CW-SP	70.0 deg F	Bypass Valve Condenser Temp
	Tower Fan Setpoint	70.0 deg F	Condenser Water Temp Setpoint
	PMPDOWN-TIME	300.0 seconds	Pump Shutdown Time Delay
Operator Override		CHW-DP SP	5.0 psi
			Chilled Water Differential Pressure



	CH3CIR2DIS-PR	136.9 psi	CH3 Circuit 2 Discharge Pressure
	CH4CHWS-T	60.4 deg F	CH4 Chilled Water Supply - Temp
	CH4CWS-T	76.8 deg F	CH4 Condenser Water Supply - Temp
	CH4-KW	8 kW	CH4 Load
	CH4CIR1DIS-PR	63.1 psi	CH4 Circuit 1 Discharge Pressure
	CH4CIR2DIS-PR	53.2 psi	CH4 Circuit 2 Discharge Pressure
	CCWS-T	75.8 deg F	Common CWS Temperature
	CW-F	59.3 gpm	Condenser Water Flow
	CHRM-T	82.3 deg F	Chiller Room Temperature
	CCHWS-T	52.1 deg F	Common CHWS Temp
	CCHWR-T	54.5 deg F	Common CHWR Temperature
	CHW-F	341.9 gpm	Chilled Water Flow

7.5.2. Low ΔT Syndrome

In most variable-flow chilled water plants, it is assumed that Delta-T, the difference between return and supply chilled water temperature, will remain relatively constant. Because the load is directly proportional to flow rate and delta-T, if the latter is constant, it follows that flow rate must vary proportionally with the load. This delta T will affect building equipment (air handling or fan coil units) coil sizes, distribution system pumping costs, and chiller sizing and costs with associated energy costs required to produce the differential

Delta-T is very much assimilated to the cooling capacity of a chiller plant for the same chilled water flow. An increased ΔT leads to more Refrigerated Tons (RT) production for the same flow therefore less operating chillers are needed meaning less electricity consumption.

General Rule: every time the ΔT goes up by 1°C, the cooling capacity of the chiller goes up by 6.8% of the maximum cooling capacity of the chiller.

To better illustrate the impact on ΔT on the chillers' capacity, a calculation example was made on a chiller of 600TR cooling capacity to reflect the effect of increasing ΔT on the cooling capacity produced at the same chilled water flow.

For ΔT of 2°C, only 83TR are produced by the chiller plant. Thus increasing this ΔT to only 4°C will double this cooling capacity to reach 167TR always with pumping the same chilled water flow. This is logical as the related equation is :

$$Q \text{ (Btu/hr)} = 500 * GPM * \Delta T \text{ (IP Units)}$$

If we do the calculation of the calculation is based this time on a fixed ΔT -at 4°C-and a variable chilled water flow, it is seen that the greater the chilled water flow the greater cooling is transferred.

Table 28: Cooling Capacity Variation vs. Delta-T Variation (Left) ' Cooling Capacity Variation vs. Chilled Water Flow Variation (Right)

gpm	Delta-T	TR	gpm	Delta-T	TR
1000	1	42	500	4	83.3
1000	2	83	600	4	100
1000	3	125	700	4	117
1000	4	167	800	4	133
1000	5	208	900	4	150
1000	6	250	1000	4	167
1000	7	292	1400	4	233
1000	8	333	1100	4	183
1000	9	375	1200	4	200
1000	10	417	1500	4	250
1000	11	458	1800	4	300
1000	12	500	2000	4	333
1000	13	542	2500	4	417
1000	14	583	3000	4	500
1000	15	625	4000	4	667
1000	16	667			

7.5.3. Efficiency Action and Energy Savings

To mitigate the low ΔT which is reducing the efficiency of the chilled water plant and increasing the electricity consumption, two key steps are needed:

- Review the VFDs on the secondary chilled water pumps (they were seen to be running at 100% speed for Chiller 3/4 and near 100% for Chiller 5 – see below) and re-commission them to ensure their PID loop/programming is correct in a way to slow down the pumps according to the ΔP (Differential Pressure)

	CHW VFD Speed Command	100.0 %	CS Input (Float)
	CW VFD Speed Command	100.0 %	CS Input (Float)

- Proactive and aggressive setting of the ΔP . An unnecessarily high ΔP setpoint will push the pumps to operate at higher speed with more CW flow and lower ΔT

pushing chillers to run at lower efficiencies too. The importance of the Delta P is at two levels:

- It enhances the operation of the secondary pumps further by reducing the chilled water flow demand.
- It enhances the Delta Temperature (ΔT) as an over pumping will reduce Delta T and will increase the chillers' capacity/consumption.

The action to be taken is simply to change proactively the Delta P setpoint from its current levels down to the lowest possible value. This can be done on a 'trial and error' basis by going down and analyzing the overall cooling performance across the building. The current levels may be close to the final ones during peak summer seasons but from these 4 peak months levels could go gradually down to reach the lowest possible threshold in January/February.

The proposed actions are as follows:

- Set the lowest possible ΔP in peak summer months by starting to reduce the existing one by notches of 0.1 bars and analyzing the while dynamics of the chilled water system and meeting the building cooling demand. This will allow you to set the maximum setpoint of ΔP
- Repeat the above exercise in peak winter month to find the lowest possible threshold
- Have the team proactively changing the setpoint in between the above two thresholds across all months by ensuring it is always to the minimum possible. Parameters affecting the cooling demand such as the ambient temperature, humidity and occupancy should be tracked along with the above figures.

The impact of mitigating the low ΔT and applying the described actions across the seasons and conditions will be at the level of both the chiller's capacity and reducing the CHW primary pumps' speed and electricity consumption. The projected savings of this EEM along with the other Key indicators are found in the below table.

EEM - 05		
Chilled Water: Low ΔT Mitigation		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
41,195	0	17,270
1.93%	0.0%	1.9%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$9,304
Annual Share of Total Utilities Cost Savings [%]		1.3%
Implementation Cost [\$]		
Net Present Value [\$/20 years]		\$121,032
Internal Rate of Return [%]		-

7.6. EEM – 06: [LCNC] Manage Guest Rooms During Unrented Mode ■

7.6.1. Rationale

Guest rooms have been fitted with new smart thermostats and the Telkonet EcoCentral GRMS during the last renovation which certainly improved their operational efficiencies in terms of lighting and cooling demand. There are additionally two actions that could be taken for further optimization during unrented mode.

7.6.2. Efficiency Actions and Energy Savings

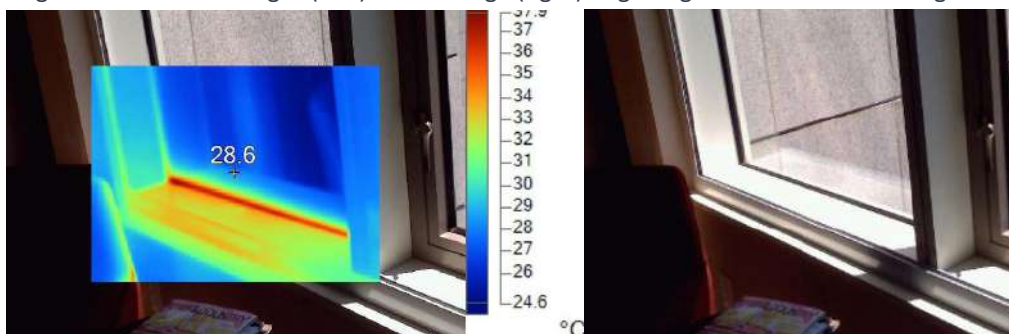
There are two key measures to be followed in all unrented guest rooms:

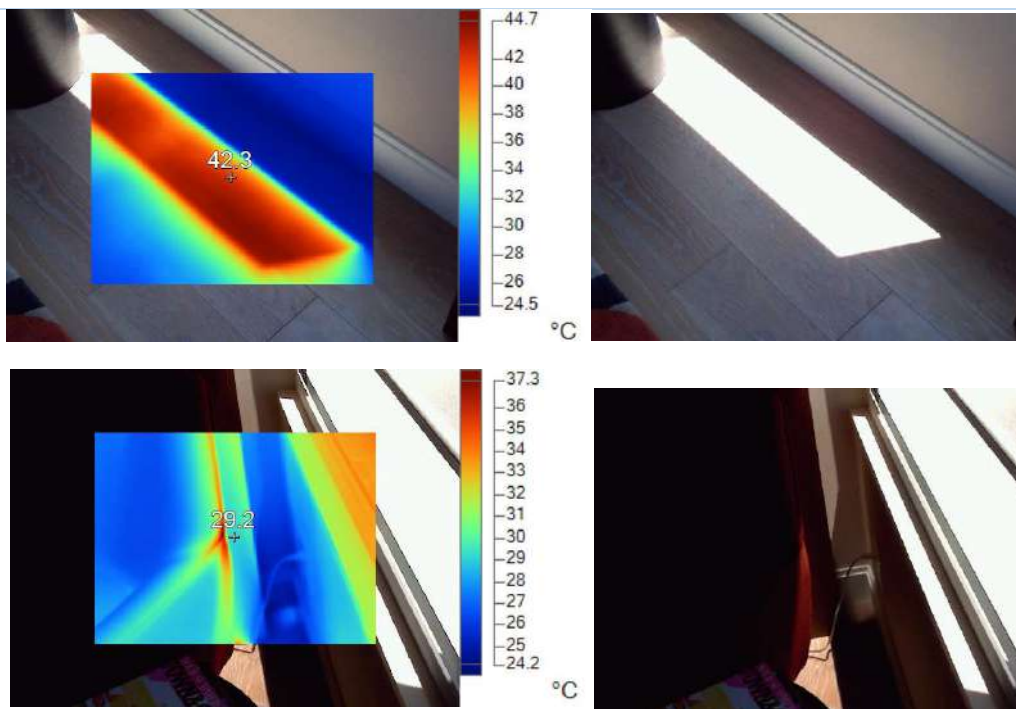
- i. Thermostats could be set back at 79°F/26°C on the cooling mode – to ensure an Auto mode doesn't push the heating mode to operate . The climate of San Francisco allows such an aggressive setting and on the other hand, there will always be the fresh air intake
- ii. It is crucial to enforce closing all curtains including blackout in all unrented rooms to minimize the solar radiations' penetration and subsequent heat gains. In fact all the building's glazing are of the single glazing type which has a window film treatment but still is a source of heat gains as can be seen in the thermal images here after.

Figure 86: Guest rooms with windows and curtains



Figure 87: Thermal images (Left) | clear image (right) of glazing and solar radiation in guest room





Applying these actions will reduce the cooling demand and the subsequent electricity consumption stemming from the chilled water plant. The below table provides the key indicators of this EEM.

EEM - 06		
Manage Guest Rooms During Unrented Mode		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	MIbs	Kg
15,400	0	6,456
0.7%	0.0%	0.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$3,478
Annual Share of Total Utilities Cost Savings [%]		0.5%
Implementation Cost [\$]		\$0
Simple Payback [Years]		-
Net Present Value [\$]/20 years		\$45,245
Internal Rate of Return [%]		-

7.7. EEM – 07: [CAPEX] Improve Hot Water/Steam network Thermal Insulation ■

7.7.1. Rationale

While flowing through the pipework, chilled/hot water/steam loses energy to the environment. These losses depends on key factors such as :

- The size, age and insulation piping system.
- The differential between the water in pipe and the ambient air temperature around the pipes.
- The time that the water/steam is in the pipe.

The overall network efficiency is an important factor in thermal energy consumption and therefore the importance of the insulation of pipes, valves and all hot surfaces.

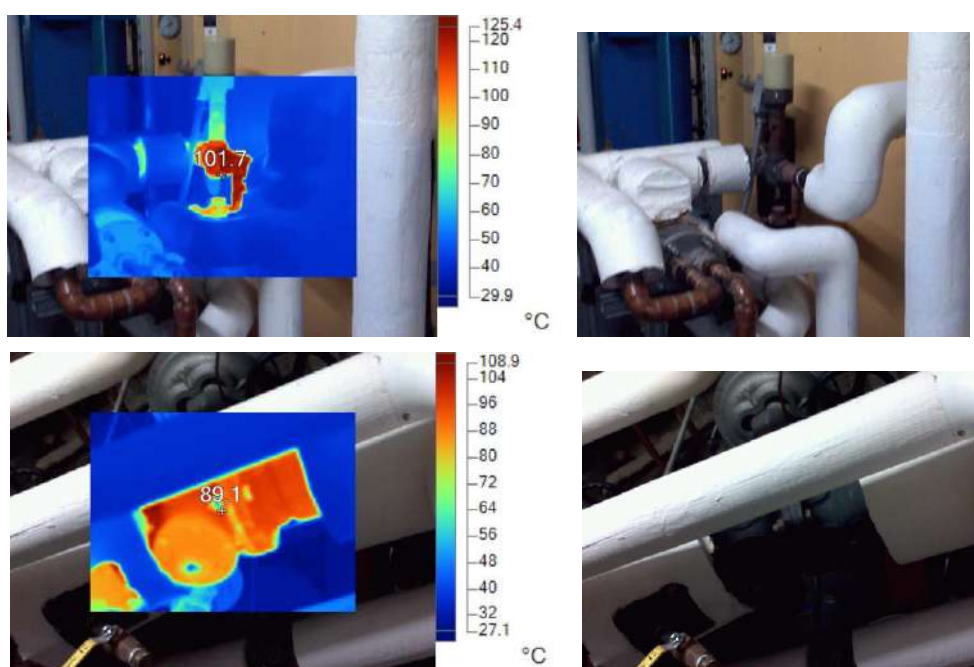
Uninsulated Hot Water/steam distribution and condensate return lines are a constant source of wasted energy. Insulation can typically reduce energy losses by 90%. Any surface over 120°F should be insulated, including boiler surfaces, steam and condensate return piping, and fittings.

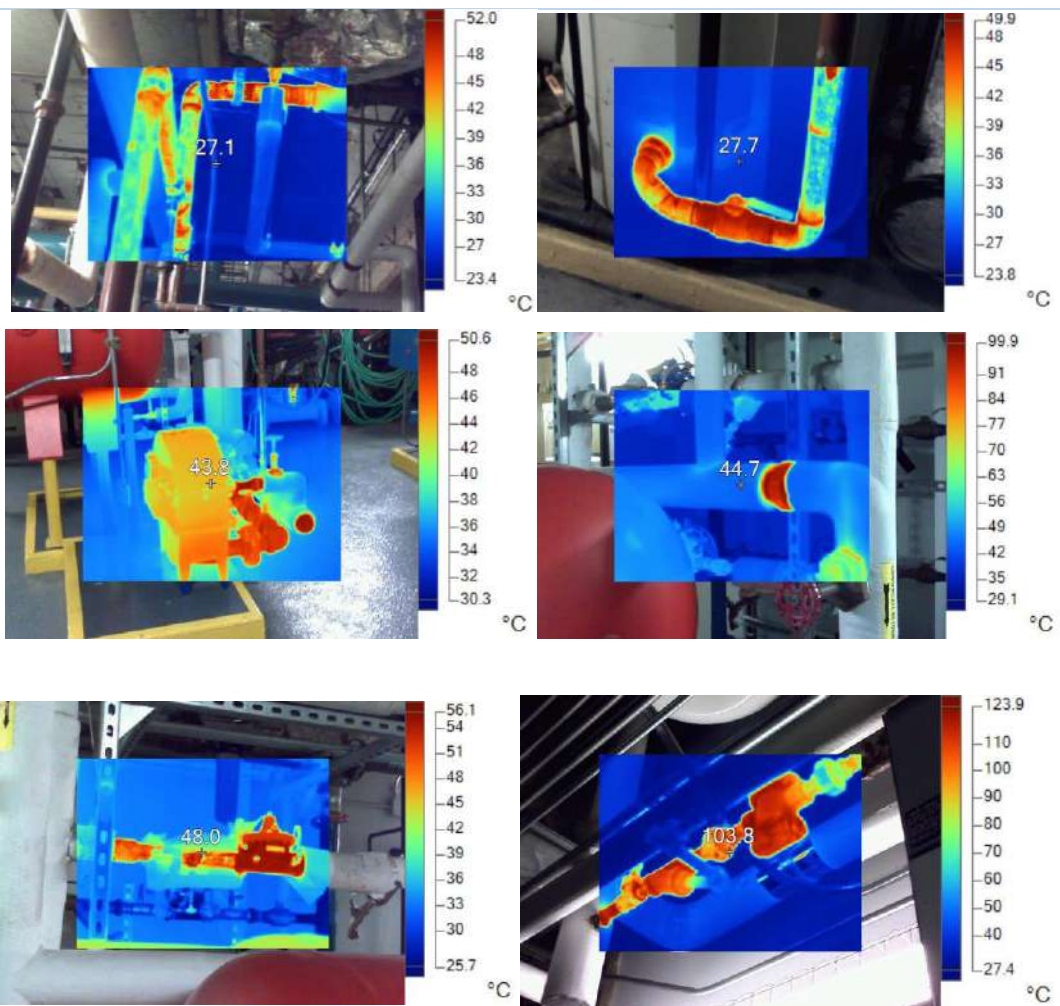
Insulation frequently becomes damaged or is removed and never replaced during steam system repair. Damaged or wet insulation should be repaired or immediately replaced to avoid compromising the insulating value.

During the field audit it was noticed that there were substantial weaknesses in the thermal insulation, mostly in the hot water and steam networks.

Below are examples of thermal images taken from the field audit clearly showcasing thermal losses.

Figure 88: FSSFE Steam and Hot Water Pipes – Thermal Losses





7.7.2. Efficiency Action and Energy Savings

it is recommended to review all hot water and steam pipes from the main intake to all visible distribution branches and connections, as well as insulating all the valves and heat exchangers. Insulation improvement can be done only at the weaknesses points as per the red areas in the thermal images. This could be a phased action done over a period of time whereby a comprehensive review through thermal imaging can be done and poorly insulated parts improved.

As mentioned, apart from the pipes insulation using fiberglass or equivalent, it is crucial to tackle valves and heat exchangers using jacket insulations as per the below examples.

Figure 89: Jacket insulation on valves and heat exchangers



As a general rule the below table provides the insulation thickness based on the fluid temperature and the pipe size.

Table 29: Insulation thickness

Fluid Operating Temperature Range (°F) and Usage	Insulation Conductivity		Normal Pipe or Tube Size (in)				
	Conductivity Btu·in./ (h·ft²·°F)	Mean Rating Temperature, °F	<1	1 to <1-1/2	1-1/2 to <4	4 to <8	≥8
			Insulation Thickness (in)				
>350 °F	0.32-0.34	250	4.5	5.0	5.0	5.0	5.0
251°F – 350°F	0.29-0.32	200	3.0	4.0	4.5	4.5	4.5
201°F – 250°F	0.27-0.30	150	2.5	2.5	2.5	3.0	3.0
141°F – 200°F	0.25-0.29	125	1.5	1.5	2.0	2.0	2.0
105°F – 140°F	0.22-0.28	100	1.0	1.0	1.5	1.5	1.5

It is not possible to calculate the potential energy and cost savings to be achieved but it can be safely assumed that should the property mitigates all the thermal weaknesses in its pipes and valves/heat exchangers, a 3% savings from the District Heating energy can be achieved . The below table provides the key indicators of this EEM.

Improve Hot Water/Steam network Thermal Insulation		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
	72	384
0.0%	3.0%	0.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$2,971
Annual Share of Total Utilities Cost Savings [%]		0.4%
Implementation Cost [\$]		\$6,000
Simple Payback [Years]		2.0
Net Present Value [\$]/20 years		\$32,647
Internal Rate of Return [%]		50%

7.8. EEM – 07: [LCNC] Install Vinyl Curtains on all Cold Rooms (positive and negative temperature) ■

7.8.1. Rationale

During the field audit, the majority of cold rooms were seen without any thermal barrier which increases the thermal losses. It is highly recommended to install vinyl curtain on all of the walk-in fridges and cold rooms to ensure all air infiltrations are reduced to the maximum. This has to cover both negative and positive temperature cold rooms.

Figure 90: Cold rooms in FSSFE



7.8.2. Efficiency Action and Energy Savings

Installation of strip curtains and Cold room door curtains are a cost-effective way to save energy in cold stores, freezer rooms and refrigerated vehicles by keeping cold air in while allowing easy access. These heavy-duty PVC strip curtains are suitable for temperatures from minus 40°C to 25°C.

Quantifying the energy savings from the above described measure is practically impossible. However, we can safely estimate that the total maximum load of the related compressors could be reduced by about 5% through the reduction of cooling demand. The below table provides the Key indicators of this EEM.



EEM - 08		
Install Vinyl Curtains on all Cold Rooms (positive and negative temperature)		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	MIbs	Kg
2,050		1,283
0.1%	0.0%	0.1%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$463
Annual Share of Total Utilities Cost Savings [%]		7.3%
Implementation Cost [\$]		\$1,000
Simple Payback [Years]		2.2
Net Present Value [\$]/20 years		\$5,023
Internal Rate of Return [%]		46%

7.9. EEM – 09: [CAPEX] Install VFDs on Fans

7.9.1. Rationale

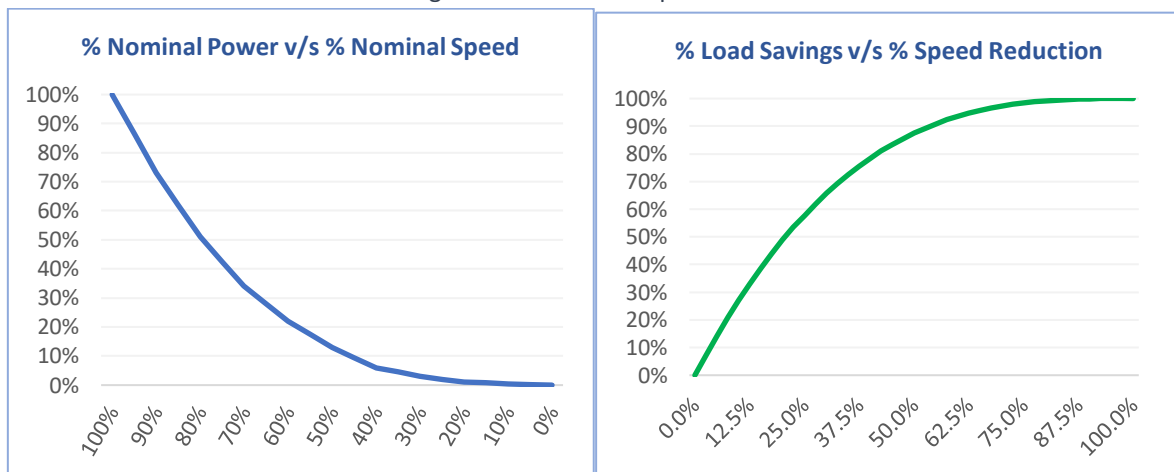
HVAC Fans represent a major electrical end use load with 15% from the total electricity consumption of FSSFE, out of which fans represent an important share.

The present EEM is to install VFDs on all Fans that still don't have this added feature, which will allow implementing the proactive speed control bringing direct savings on the Fans electricity consumption as well as the chilled water demand.

7.9.2. Efficiency Actions and Energy savings

The usage of VFD will allow savings substantial energy used (if the speed reduction is well implemented in a PID/Automatic look or proactively changed manually or via a schedule time setting). A 10% speed reduction would lead to a 27% savings on the Power used. VFDs on Fans can be implemented on a time-based operation while they could be operated similarly in on a PID Loop. The below charts showcase the relation between the Nominal Power variation and the Nominal speed as well as the Load savings (reduction in KW) and the speed reduction.

Figure 91: Power and Speed Variations



The VFD installation has multiple benefits, mainly the following:

Air Flow Control – Fan air flow can be reduced by using a VFD. A good example is when temperatures inside a facility rise, increasing the fan speed will provide a higher cooling effect. Alternatively, the fan speed will be reduced if temperature drops.

Energy Savings – If the air flow required from the fan is variable, then having the ability to reduce the fan speed reduces power consumption which results in energy savings. A VFD also reduces the power consumption when the motor is started.

No Motor Starter – When using a VFD, no motor starter is required. This can offset some of the cost of a VFD.

Extended Motor Life – Reduced power consumption and smoother start-ups may also extend motor life.

The tables below shows the list of targeted Fans in FSSFE which represent those with a power rating larger than 3KW, with their existing annual electricity consumption, the speed reduction as well as the electricity and cost savings, and finally the CAPEX .

Table 30: List of Fans and annual KWh consumption

Fan Number	Rated Power	Existing Annual Energy Consumption	Overall Average Speed Reduction	Load/Energy Reduction	Savings		CAPEX
Unit	KW	KWh	%	%	KWh	\$	\$
E-60	11.3	26,554.7	10.0%	27%	7,196	\$1,625	\$3,000
E-57	8.0	18,883.3	10.0%	27%	5,117	\$1,156	\$2,500
E-51	5.5	12,982.3	10.0%	27%	3,518	\$795	\$2,000
E-56	3.8	8,851.6	10.0%	27%	2,399	\$542	\$1,500
SF-11	3.8	8,851.6	10.0%	27%	2,399	\$542	\$1,500
SF-19	3.8	8,851.6	10.0%	27%	2,399	\$542	\$1,500
FCU-2	7.5	17,703.1	10.0%	27%	4,798	\$1,084	\$2,500
Total		102,678	-	-	27,826	\$6,285	\$14,500

You will find in the table below the key indicators of this EEM.

EEM - 09		
Install VFDs on Fans		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Therms	Kg
27,826		11,665
1.3%	0.0%	1.3%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$6,285
Annual Share of Total Utilities Cost Savings [%]		0.9%
Implementation Cost [\$]		\$14,500
Simple Payback [Years]		2.3
Net Present Value [\$]/20 years		\$67,252
Internal Rate of Return [%]		43%

7.10. EEM – 10: [CAPEX] Retrofit Legacy FCUs by EC Motors FCU Type

7.10.1. Rationale

Most of FSSFE FCUs were not replaced since the original construction and have more than 30 years for most including those in guest rooms. Their efficiency has certainly decreased and on the other hand, fans' motors and technologies have advanced in the meantime.

FCU retrofit is a strategic one that would provide benefits from a maintenance and energy efficiency perspectives. In addition, they have a far better acoustic performances compared to exiting units which means reduced noise in rooms.

The present EEM is to be done on a complete retrofit basis and based on a time planned action over three years , starting by all FCUs that are actively used everyday with high load factors.

The solution would be to retrofit all current FCUs by electronically commutated (EC) motor which are DC Motors with VFDs.

7.10.2. Efficiency Action and Energy Savings

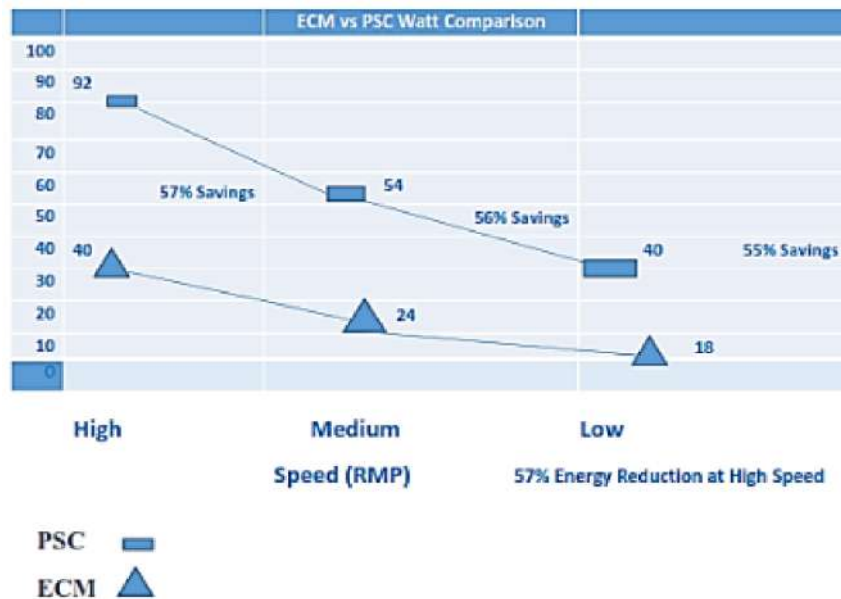
An electronically commutated (EC) motor is designed to run on an alternating current (AC) power supply, but it in fact bears a closer resemblance to a direct current (DC) motor. It is essentially a permanent magnet, brushless DC motor that incorporates on-board electronics with a 0-10v control signal which increases the energy savings.

The benefits of EC motors in hotel guest rooms include:

- Maximizing fan coil unit energy efficiency
- Bringing airflow back to manufacturer's specifications
- Lower operating costs
- Lower fan motor noise
- Drastically reduced humidity
- Improved indoor air quality and;
- Faster guest room cooling and heating.

From an energy efficiency perspective, it has been documented that in some cases EC FCUs use up to 70% less energy than traditional products and that a specific fan power (SPF) of 0.3W/l/s or lower can be achieved (compared to an SPF of 0.7W/l/s for an AC FCU).

Figure 92: ECM savings v/s PSC motors



As per the comparative efficiency chart here above, savings are the highest at full speed.

The **total electricity savings to be attained would be minimum 50% from the current total FCU's with projected savings of more than 60% due to the high initial wattage if the 1986 installed FCUs**. It is also clear that the impact of FCUs retrofits would also be on the cooling and heating demand due to their improved control efficiencies.

The below table provides the key indicators of this EEM if all FCUs in the property are retrofitted – including those in FOH and BOH.

EEM - 10		
Retrofit Legacy FCUs by ECM FCU Type		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
125,900	24	52,906
5.9%	1.0%	5.5%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$29,410
Annual Share of Total Utilities Cost Savings [%]		4.2%
Implementation Cost [\$]		\$150,000
Simple Payback [Years]		5.1
Net Present Value [\$/20 years]		\$232,561
Internal Rate of Return [%]		20%

7.11. EEM – 11: [CAPEX] Retrofit Electrical HW Tank in 36th floor ■

7.11.1. Rationale

The 36th floor includes in one of its technical room an 18KW Hot Water tank used for the in room dining kitchen. The unit has been faulty for few weeks at the time of the field audit and was due for repair .

Instead of repairing the unit, it is recommended to either connect the kitchen to the upper floors' hot water tanks or else retrofit the unit with a Heat Pump unit providing minimum 50% electricity savings which is developed in this EEM,

Figure 93: Current 18KW water heater

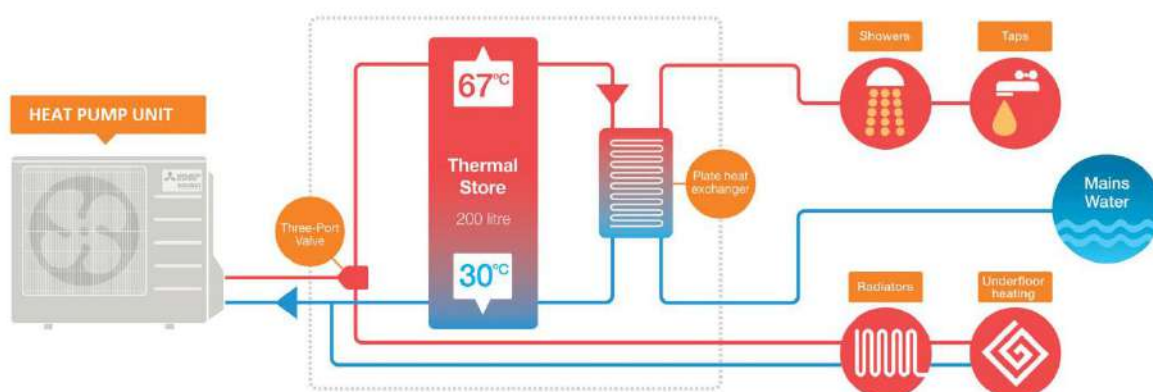


7.11.2. Efficiency Action and Energy Savings

A Heat Pump could be installed in the same floor (if not in the same room) and could have the exhausted cold air – as the unit will be running on heating mode- ducted to the corridor or to any mechanical room in the vicinity for a free cooling.

The design of the unit could be finalized according to the evaluation of the current hot water demand in the in room dining kitchen and an adequate sizing of the system. The below figure provides an example of a connection between a Heat Pump and the hot water end usages generally.

Figure 94: Example of a Heat Pump installation for water heating



The Heat Pump unit should have a minimum COP of 4 meaning a 4KW_{thermal} unit could cover the current production capacity . As indicated, because the Heat Pump is working only on heating mode, the exhausted air would have a temperature of nearly 68°F and could be used for free space cooling of vicinity areas in the technical floor.

The action plan would be as follows:

- Review if a connection from the guest rooms' hot water tanks is possible – as distances are not far away- to replace the existing electrical heating unit. If this is not possible, proceed with the below points.
- Review the hot water demand in the in room dining kitchen (average USG per day)
- According to the previous point, calculate the size of the Heat Pump needed with a 4KW unit covering fully the existing capacity.
- Allocate the space of the installation of the Heat Pump unit with the note that ideally it has to be in the same location of the current hot water unit to minimize extra pipe works.
- Check if the free cooling can be used in a nearby area
- Install the system

The below table provides the key indicators of the Heat Pump installation option. If the first alternative of connecting the in room dining to the guest rooms' hot water tanks is possible, it will certainly have a very good payback period.

EEM - 11		
Retrofit Electrical HW Tank in 36th floor		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
19,320		8,099
0.9%	0.0%	0.9%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$4,364
Annual Share of Total Utilities Cost Savings [%]		0.6%
Implementation Cost [\$]		\$10,000
Simple Payback [Years]		2.3
Net Present Value [\$/20 years]		\$46,761
Internal Rate of Return [%]		44%

7.12. EEM – 12: [CAPEX] Retrofit Chillers 3 and 4 serving guest rooms

7.12.1. Rationale

The 36th floor includes Chillers 3 and 4 that service guest rooms . They are water cooled chillers Carrier model 30HR160D600 with 4x reciprocating compressors each and a total cooling capacity of about 150RT. These units have been installed more than 30 years ago , are using the R22 refrigerant that is an ozone depleting substance, and are suffering from large maintenance costs while they have poor energy efficiency. They're a prime target for a full retrofit.

Figure 95: Current Carrier Water Cooled Chillers (CH3/CH4) serving upper floors



7.12.2. Efficiency Action and Energy Savings

The current units of Chillers 3 and 4 have a capacity of 150RT each and it has been mentioned that one of the units is in a poor condition while the second is operating fairly well.

On the other hand, the maximum cooling demand of the connected chilled water loads don't exceed the capacity of one chiller (During the field audit in July 2022, the load was barely topping the 40%) . Accordingly, the chiller retrofit should target one chiller only and leave the one with a better condition as a backup, optimizing therefore the CAPEX and ROI.

The key points to be considered for a chiller retrofit are the following:

- Total rated capacity . We recommend a capacity slightly larger than the existing ones. 180Rt instead of 150RT .
- The chiller should be modular to allow a continuity in operation during a possible failure or maintenance
- The chiller should have high part load and full load EER/COP – Minimum full load COP of 5.5
- The chiller should have VFD driven compressors
- The chiller should be of the flooded type
- Consider Heat Recovery as an option, specially that the chillers are In the same plantroom of the guest rooms' hot water tanks

Based on the above points, potential solutions may be similar to the Chiller 5 acquired 5 years ago but with the latest version or equivalent (Model Multistack VME II with multiple stacks of 2 scroll compressors). Already the difference in COP between CH3 and 4 v/s CH5 is nearly 50%, meaning an equivalent value of energy savings.

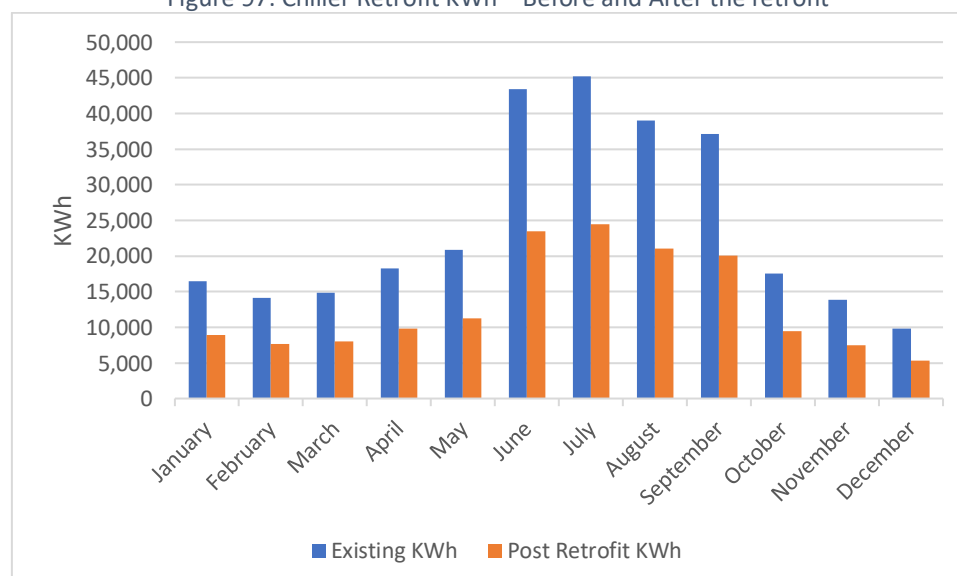
If the budget allocated is larger, the ideal solution would be to use the latest oil free magnetic bearing chillers with the Multistack MagLev units as an example. This type of chillers provide low noise, very high energy efficiency at part and full load and have a reduced maintenance cost due to the absence of oil. The COP would exceed the 6 leading to more than 60% savings from current conditions. Furthermore, a heat recovery option would provide energy savings in the production of hot water for guest rooms.

Figure 96: Water-Cooled MagLev® Modular Heat Recovery Chiller with Shell and Tube (Left) ; VME II® Scroll Modular Chiller (Right) – Source: Multistack



Retrofitting the upper floors' chillers should be a priority in the action plan due primarily to the poor current conditions of the chillers, their usage of R22, their long overdue end of lifecycle and the need of energy efficiency. The below chart provides the electricity consumption difference between the present situation (considering a real COP of 2.3 and a new chiller with a COP of 5 – knowing higher efficiency chillers could top as much as an average of 7)

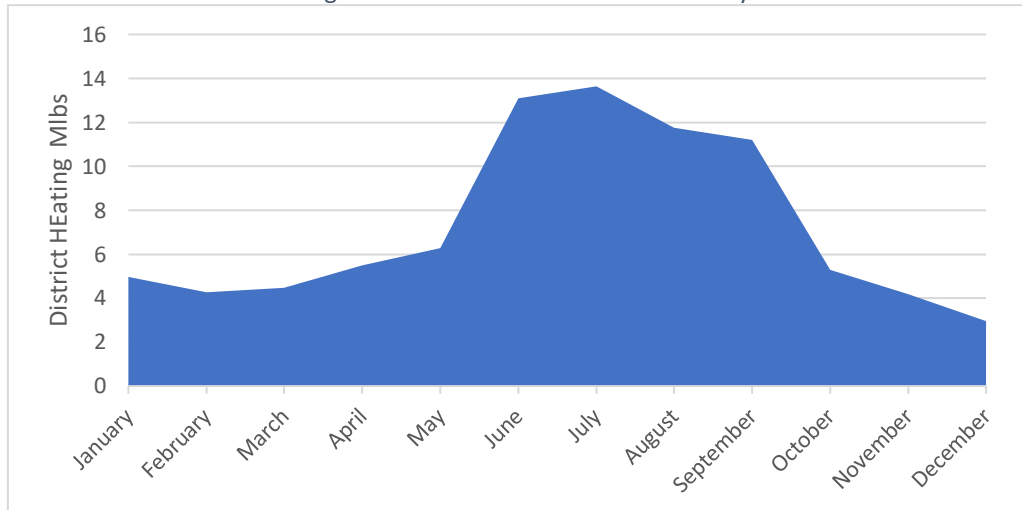
Figure 97: Chiller Retrofit KWh – Before and After the retrofit



The previous chart shows annual savings of 156,000 KWh from the chiller retrofit should we consider a new unit with a COP of 5.

In addition, if the new unit is equipped with Heat Recovery, the below chart provides the estimated savings from District Heating related to the production of hot water for the upper floors (guest rooms) .

Figure 98: Chiller Retrofit- Heat Recovery



Based on choosing the Magnetic Bearing units with heat recovery, and considering the COP improvement to be by 2.7 points (from 2.3 to 5) , the below table provides the key indicators of this EEM. Using chillers similar to the Multistack modular ones will allow an ease of installation – due to small footprint . It is to be noted that the condenser side will remain the same and the present cooling towers will be totally compatible with the new chiller. The heat recovery option would necessitate additional design for the integration of the recovered water into the existing hot water tanks. The most optimal option would be to add one buffer tank that will be heated by the recovered heat and act as the makeup water intake for the existing two hot water tanks.

Finally, while the payback is on the high side, this retrofit should be considered as a priority for the reasons mentioned previously.

EEM - 12		
Retrofit Chillers 3 and 4 serving guest rooms		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
156,863	88	68,190
7.4%	3.6%	7.5%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$39,021
Annual Share of Total Utilities Cost Savings [%]		5.6%
Implementation Cost [\$]		\$300,000
Simple Payback [Years]		7.7
Net Present Value [\$]/20 years		\$207,582
Internal Rate of Return [%]		13%

Note on the Cooling Towers

FSSFE has two Cooling Tower plants, each serving one chilled water system. The related water consumption was seen to represent only about 5% of the total consumption while the electricity consumption was also relatively low with VFD driven fans.

However, the systems can still be improved in energy efficiency if they are turned to Dry Coolers that will work perfectly well in the climate of San Francisco and will remove the hassle of water treatments and larger cooling tower maintenance.

This was not developed as an Efficiency Measure, but could be foreseen anytime the existing cooling towers have a failure and need deep maintenance. At that stage a retrofit can be planned instead of the repair

Figure 99: Example of Dry Coolers for condenser water cooling



7.13. EEM – 13: [CAPEX] Retrofit Heat Pump/DX Servicing Sky Bridges

7.13.1. Rationale

On the 48th floor , FSSFE has one Carrier DX/Heat Pump unit model 38AD024 that is used to cool the sky bridges in hot months or when needed. The unit is also considered a legacy infrastructure equipment installed more than three decades ago, using R22 refrigerant and running at low energy efficiency relative to today's standards.

Its retrofit should be planned in the next 2-3 years for both the removal of the R22 from the premises and for improved energy efficiency.

Figure 100: Current Heat Pump CW setpoint



7.13.2. Efficiency Action and Energy Savings

While the unit is still operating satisfactorily, the usage of the R22 and the fact that the unit has been installed for more than 30 years with all the advances in refrigeration efficiency that occurred afterwards are the prime reason for a planned retrofit. The new unit should be of the same type of air-cooled DX unit with the same rating of 5 RT along with high efficiency and VFD driven compressors. The minimum full load COP should be 3.3 . The below table provides the key indicators of this EEM.

EEM - 13		
Retrofit DX Unit Servicing Sky Bridges		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
3,415		1,432
0.2%	0.0%	0.2%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$771
Annual Share of Total Utilities Cost Savings [%]		0.1%
Implementation Cost [\$]		\$7,000
Simple Payback [Years]		9.1
Net Present Value [\$]/20 years		\$3,033
Internal Rate of Return [%]		11%

7.14. EEM – 13: [CAPEX] BMS Upgrade and review possible pneumatic to DDC conversion

7.14.1. Rationale

In today's building operations, a well rounded BMS is the heart of an efficient technical management. The BMS of FSSFE is a Johnson Control Metasys one that has been installed for more than two decades with some upgrades done in the past but without deep changes, It is in fact one system that caters for both the FSSFE premises as well as the ones of the 345 California Center building.

The current system has large limitations as well as virtually no Graphic User Interface which are limiting the proper control of the HVAC equipment as well as any deeper analysis.

On the other hand, as indicated in previous sections, the control system in FSSFE is a pneumatic one which is still functioning but which also carries additional maintenance and lower accuracy compared to current standards of DDC .

This EEM covers both , the upgrade of the BMS system and the long term planning of upgrading the pneumatic control to a DDC one.

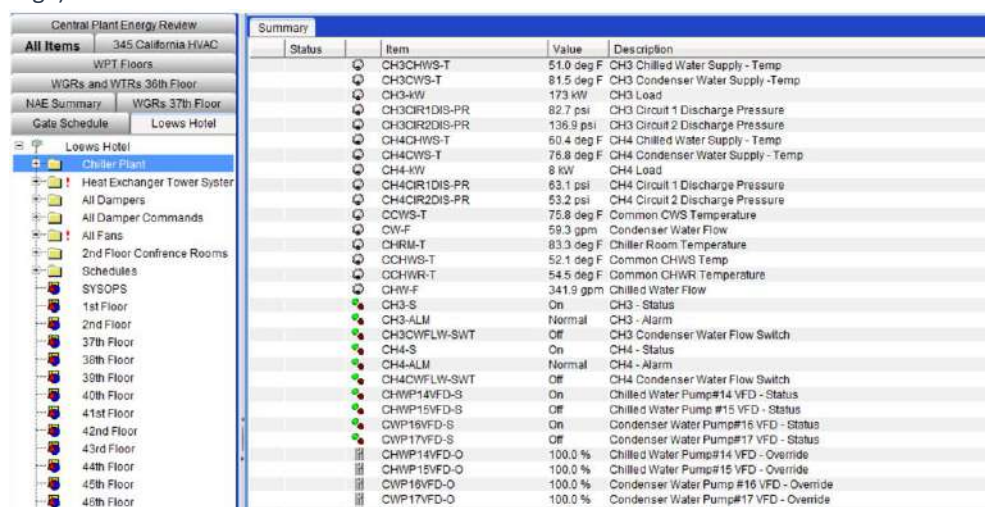
7.14.2. BMS Upgrade

A comprehensive BMS Upgrade will carry a high CAPEX and should be considered whenever the 2nd step (moving to DDC) is to be implemented .

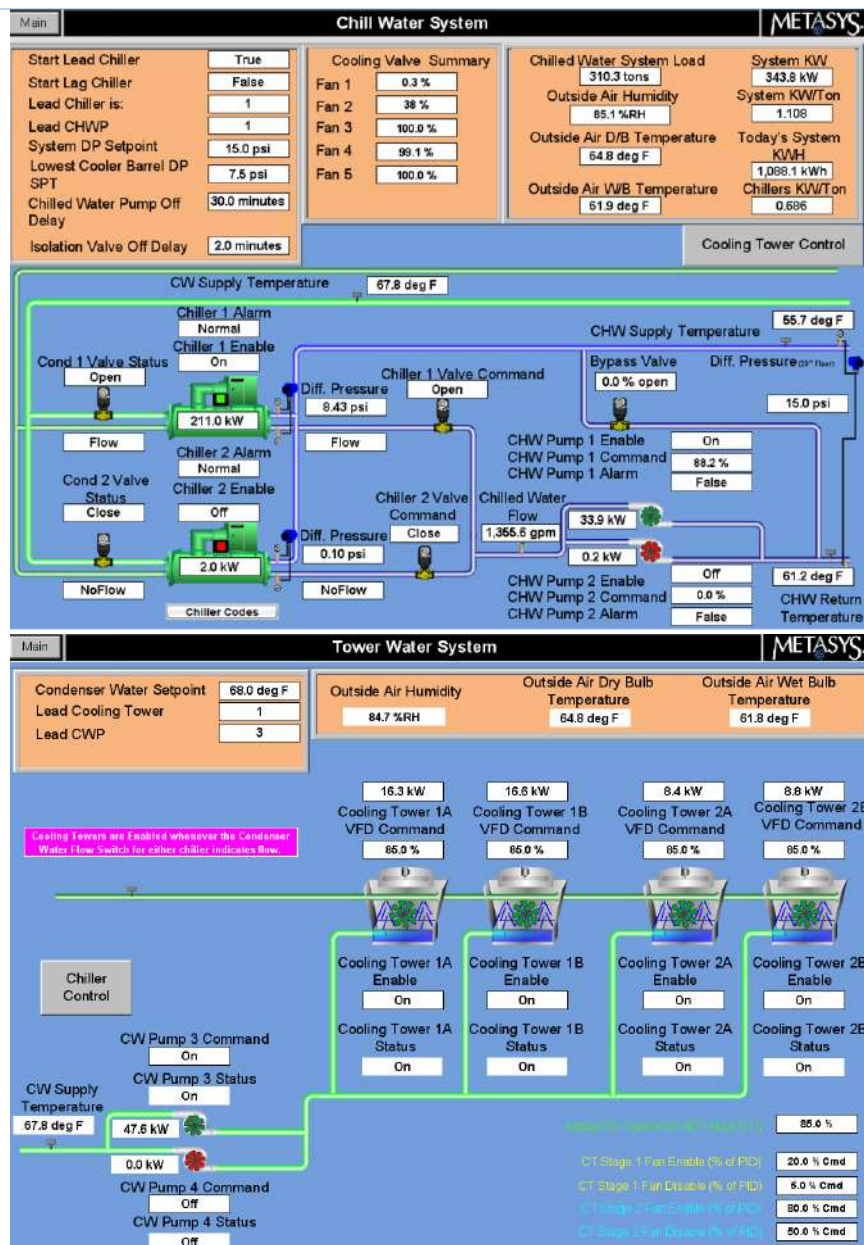
As a first step, FSSFE should work with Johnson Control to improve the current situation , and at least to bring the property in par with the BMS part related to the 345 California Centre .

The below snapshots show the difference between FSSFE BMS interface which is not graphic based and the ones used by 345 California Center on the same platform.

Figure 101: FSSFE BMS Chiller management (Below) 345 California Center building Chiller Plant BMS (Next Page)



Central Plant Energy Review		Summary			
All Items	345 California HVAC	Status	Item	Value	Description
WPT Floors			CH3CHWS-T	51.0 deg F	CH3 Chilled Water Supply - Temp
WGRs and WTRs 36th Floor			CH3CWS-T	81.5 deg F	CH3 Condenser Water Supply - Temp
NAE Summary	WGRs 37th Floor		CH3-kW	173 kW	CH3 Load
Gate Schedule	Loews Hotel		CH3OR1DIS-PR	82.7 psi	CH3 Circuit 1 Discharge Pressure
			CH3OR2DIS-PR	136.9 psi	CH3 Circuit 2 Discharge Pressure
			CH4CHWS-T	60.4 deg F	CH4 Chilled Water Supply - Temp
			CH4CWS-T	76.8 deg F	CH4 Condenser Water Supply - Temp
			CH4-kW	9 kW	CH4 Load
			CH4OR1DIS-PR	63.1 psi	CH4 Circuit 1 Discharge Pressure
			CH4OR2DIS-PR	53.2 psi	CH4 Circuit 2 Discharge Pressure
			CCWS-T	75.8 deg F	Common CWS Temperature
			CW-F	59.3 gpm	Condenser Water Flow
			CHRM-T	83.3 deg F	Chiller Room Temperature
			CCHWS-T	52.1 deg F	Common CHWS Temp
			CCHWR-T	54.5 deg F	Common CHWR Temperature
			CHW-F	341.9 gpm	Chilled Water Flow
			CH3-S	On	CH3 - Status
			CH3-ALM	Normal	CH3 - Alarm
			CH3CWLW-SWT	Off	CH3 Condenser Water Flow Switch
			CH4-S	On	CH4 - Status
			CH4-ALM	Normal	CH4 - Alarm
			CH4CWLW-SWT	Off	CH4 Condenser Water Flow Switch
			CHWP14VFD-S	On	Chilled Water Pump#14 VFD - Status
			CHWP15VFD-S	Off	Chilled Water Pump#15 VFD - Status
			CWP16VFD-S	On	Condenser Water Pump#16 VFD - Status
			CWP17VFD-S	Off	Condenser Water Pump#17 VFD - Status
			CHWP14VFD-O	100.0 %	Chilled Water Pump#14 VFD - Override
			CHWP15VFD-O	100.0 %	Chilled Water Pump#15 VFD - Override
			CWP16VFD-O	100.0 %	Condenser Water Pump#16 VFD - Override
			CWP17VFD-O	100.0 %	Condenser Water Pump#17 VFD - Override



The BMS improvement project will look at the following:

- I. To review with Johnson Control the possibilities to create full graphical interfaces for the key operations such as : Both chilled water plants (with their pumps and cooling towers), Steam intake with all connected pumps, hot water tanks, gas boilers and the connected systems as well as all supply and exhaust fans. For the larger FCUs serving public areas and treated fresh air intake, it would be crucial to check if the cooling coil valves can be connected to the system with temperature in/out
- II. Regarding the communication limitations , Johnson Control can check the possibility of a hybrid system (i.e.: analog and digital) → Covered more in the following sub section
- III. Build trends on all key operating parameters

- IV. Review the chillers sequencing operations as well as the PID feedback of all VFDs on pumps.

Generally, a BMS upgrade is considered as the central action that will allow the engineering team to enhance their visibility on the operations, implement more energy management actions and reduce their 'firefighting' and enhance the technical management of the property.

A properly implemented BMS based on the above terms would certainly provide at least 10%-15% electricity savings on all the HVAC infrastructure with various energy management control strategies implementation.

7.14.3. Convert Pneumatic Control to DDC

Pneumatic systems have been around for decades. However, increased energy and labor costs provide plenty of incentive to consider upgrading to a more efficient system. A (partial or full) conversion from a pneumatic system to a digital BMS (Building Management System) will give these long term benefits:

- **More information (actionable data) for intelligent building management.** To manage something effectively, you need data. Intelligent decisions are dependent on getting adequate information about the real-world situations inside a building. Are all your system devices and schedules functioning properly at all times? How do you really know?
Pneumatic systems, unfortunately, provide little or no data to a building owner. That dearth of data means underperforming subsystems continue wasting energy, impending breakdowns aren't noticed until it is too late, and maintenance is reduced to chasing fires instead of proactive prevention.
Digital BMS systems, on the other hand, are all about data. They automatically adjust parameters and correct conditions that would require tedious manual labor on pneumatic systems. Digital systems also provide real-time and trended data for reports and graphs that allow for easy analysis and intelligent decision-making.
- **Lower energy costs through greater operating efficiency.** Energy savings flow from data. Digital's precise measurement and control of setpoints as well as numerous automated energy savings techniques squeeze maximum efficiency out of energy dollars. Digital features that are difficult or impossible for pneumatics to provide include Demand Control Ventilation (DCV) , Intelligent predictive scheduling and operations...
- **Lower maintenance costs through less need for calibration and service of equipment.** In fact, pneumatic sensors and thermostats need calibration periodically to maintain reasonable accuracy (but typically infrequently performed in practice). Air compressors and air dryers need careful maintenance to avoid contaminating the lines with oil, and replacement costs for those items are high. On the other hand, after installation, digital systems usually provide set-and forget, self-maintaining, accurate control for long periods of time.

The general benefits of the conversion from pneumatic to DDC is also best summarized in the following comment from Lawrence Berkley Lab in California:

“Solid-state sensors and controllers used in DDC systems have considerable energy-efficiency advantages over conventional pneumatic systems. Substantial advantages are realized in calibration and maintenance, but the critical value lies in the accuracy and reliability of the DDC systems. These features can yield operational energy savings of 15% and greater when compared to the conventional pneumatic system. The inherently precise positioning of valves and dampers with EMCS control loops and blocks are responsible for these energy savings. For instance, to realize the energy-savings opportunity offered by VAV fume hoods, an 8:1 turndown ratio of the exhaust airflow is required. Pneumatic airflow systems typically lose accuracy at 25% of their span, resulting in the capability of only a 4:1 turndown ratio. Solid-state DDC systems provide the degree of precise airflow measurement and control that enables the operation of these VAV systems.”

The conversion to DDC can take two paths:

- **Full digitization and a new BMS** : This represent the ideal solution as it completely retrofits the pneumatic system to a DDC one , and will provide the greatest savings in the long run as well as the full benefits listed before from the precision monitoring and controlling of the infrastructure. However, this will carry a high CAPEX as the project will have to install new conduits, wires, valves and rip out the pneumatic infrastructure.
- **Hybrid system**: A less disruptive and possibly more cost-effective option is to replace only the parts of the particular system that make the most sense and to leave the parts that can still perform adequately in place. This results in a pneumatic and digital hybrid; for example applying new digital controls to old air handlers and roof top units is usually quite feasible and cost-effective . In this case, the key point is how do analog air pressures and binary electrical pulses talk. In fact, for a pneumatic system to talk to a digital system, a P-E (pressure to voltage) or P-I (pressure to current) transducer is used. Some, for example, can take a typical 3–15 psi signal from pneumatic devices (such as thermostats, transmit- ters, and receiver-controllers) and convert it to an analog electronic signal (e.g., 2–10 VDC or 4–20 mA). Thus, a standard pressure signal can translate into a standard analog electronic signal that serves as an input to a digital BMS, allowing the BMS to monitor the pneumatic operation.

7.14.4. Action Plan and Savings

The recommended actions to tackle these two projects are as follows:

- i. Review with Johnson Control (JC) the existing infrastructure and the BMS software user interface to check what are the possibilities of improvement with the current equipment and with little investment . The points mentioned in section 7.14.2 should be followed.
- ii. In parallel , discuss the possibilities to convert the pneumatic control to DDC with JC and specifically the two cases of full digitization or a hybrid solution. Each one

- to have its own CAPEX developed . The visibility of the HVAC system in full is the crucial point and this includes getting access to actuators and cooling/heating coils control and status on the top of the standard parameters
- iii. Take into consideration the difference in the maintenance costs and spare parts availabilities between both system with the projection for the next 10 years.
 - iv. Based on the previous points, FSSFE will decide which way to go for a 3 years implementation.

A full DDC system will allow an improvement in the overall HVAC control and the implementation of energy management strategies while ensuring the control is precise (due to the new valves and actuators with their digital control compared to existing pneumatic one) . A 10% savings on the chilled water and district heating used for space heating is to be expected by this move. However, the long term benefits are much more than the pure energy savings as they will upgrade the asset for the next two decade.

The below table provides an estimated CAPEX and savings for a full conversion and upgrade of the BMS. These savings could be fine tuned whenever the review is performed and the coverage decision is taken.

EEM - 14		
BMS Upgrade and pneumatic to DDC conversion		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	Mlbs	Kg
125,000	40	53,512
5.9%	1.7%	5.9%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$29,872
Annual Share of Total Utilities Cost Savings [%]		4.3%
Implementation Cost [\$]		\$350,000
Simple Payback [Years]		11.7
Net Present Value [\$]/20 years		\$38,578
Internal Rate of Return [%]		9%

8. EEMs Summary Figures & Implementation Notes

8.1. EEMs Summary – CAPEX, Cost Savings & Payback Period

The following table provides the summary of the Energy Efficiency Measures (EEMs) with key indicators including CAPEX, % Savings from Total Utilities Cost and Payback period.

As can be seen, if all EEMs are implemented, the total annual cost savings would be **\$198,144** at a CAPEX of **\$908,500** and a payback of **4.6 Years**.

Table 31: EEMs summary – CAPEX, Cost savings and Payback Period

CODE	EEM Name	ANNUAL SAVINGS		COST	PAYBACK
-	-	\$	KGs CO ₂	\$	Years
EEM - 01 [CAPEX]	KW Demand Management	\$26,628 (3.8%)	0	\$55,000	2.1
EEM - 02 [CAPEX]	Lighting Retrofit	\$17,617 (2.5%)	32,699	\$11,000	0.6
EEM - 03 [LCNC]	Lighting Control and Management	\$5,247 (0.7%)	9,739	\$4,000	0.8
EEM - 04 [LCNC]	HVAC Control	\$22,712 (3.2%)	42,155	\$0	-
EEM - 05 [LCNC]	Chilled Water: Low ΔT Mitigation	\$9,304 (1.3%)	17,270	\$0	-
EEM - 06 [LCNC]	Manage Guest Rooms During Unrented Mode	\$3,478 (0.5%)	6,456	\$0	-
EEM - 07 [CAPEX]	Improve Hot Water/Steam network Thermal Insulation	\$2,971 (0.4%)	384	\$6,000	2.0
EEM - 08 [CAPEX]	Install Vinyl Curtains on all Cold Rooms (positive and negative temperature)	\$463 (0.1%)	859	\$1,000	2.2
EEM - 09 [CAPEX]	Install VFDs on Fans	\$6,285 (0.9%)	11,665	\$14,500	2.3
EEM - 10 [CAPEX]	Retrofit Legacy FCUs by ECM FCU Type	\$29,410 (4.2%)	52,906	\$150,000	5.1
EEM - 11 [CAPEX]	Retrofit Electrical HW Tank in 36th floor	\$4,364 (0.6%)	8,099	\$10,000	2.3
EEM - 12 [CAPEX]	Retrofit Chillers 3 and 4 serving guest rooms	\$39,021 (5.6%)	68,190	\$300,000	7.7
EEM - 13 [CAPEX]	Retrofit DX Unit Servicing Sky Bridges	\$771 (0.1%)	1,432	\$7,000	9.1
EEM - 14 [CAPEX]	BMS Upgrade and pneumatic to DDC conversion	\$29,872 (4.3%)	53,512	\$350,000	11.7
	TOTAL	\$198,144 (28.2%)	305,366	\$908,500	4.6

8.2. EEMs Summary – Electricity (KWh) and District Heating (Mlbs) Savings

The below table provides the annual savings of every EEM and for each utility. Electricity savings are seen to be **718.755 KWh** per year representing **34%** of the actual electricity consumption and about **334 Mlbs** of district heating reduction representing **9%** of the total.

Table 32: EEM Summary – Electricity (KWh) and District Heating (Mlbs) Savings

CODE	EEM Name	ELECTRICITY	DISTRICT HEATING
-	-	KWh	Mlbs
EEM - 01	KW Demand Management	0 (0.0%)	0 (0.0%)
EEM - 02	Lighting Retrofit	78,000 (3.7%)	0 (0.0%)
EEM - 03	Lighting Control and Management	23,230 (1.1%)	0 (0.0%)
EEM - 04	HVAC Control	100,557 (4.7%)	0 (0.0%)
EEM - 05	Chilled Water: Low ΔT Mitigation	41,195 (1.9%)	0 (0.0%)
EEM - 06	Manage Guest Rooms During Unrented Mode	15,400 (0.7%)	0 (0.0%)
EEM - 07	Improve Hot Water/Steam network Thermal Insulation	0 (0.0%)	72 (3.0%)
EEM - 08	Install Vinyl Curtains on all Cold Rooms (positive and negative temperature)	2,050 (0.1%)	0 (0.0%)
EEM - 09	Install VFDs on Fans	27,826 (1.3%)	0 (0.0%)
EEM - 10	Retrofit Legacy FCUs by ECM FCU Type	125,900 (5.9%)	24 (1.0%)
EEM - 11	Retrofit Electrical HW Tank in 36th floor	19,320 (0.9%)	0 (0.0%)
EEM - 12	Retrofit Chillers 3 and 4 serving guest rooms	156,863 (7.4%)	88 (3.6%)
EEM - 13	Retrofit DX Unit Servicing Sky Bridges	3,415 (0.2%)	0 (0.0%)
EEM - 14	BMS Upgrade and pneumatic to DDC conversion	125,000 (5.9%)	40 (1.7%)
	TOTAL	718,755 34%	224 9%

8.3. Implementation Notes and Carbon Reductions

The implementation of all EEMs would depend on the CAPEX availabilities as well as the technical difficulties and the potential disruption to the operation.

EEMs 01 to EEM-09 could be however implemented without both a large CAPEX and any disruption, which is the reason they are considered as the base of the EEMs implementation.

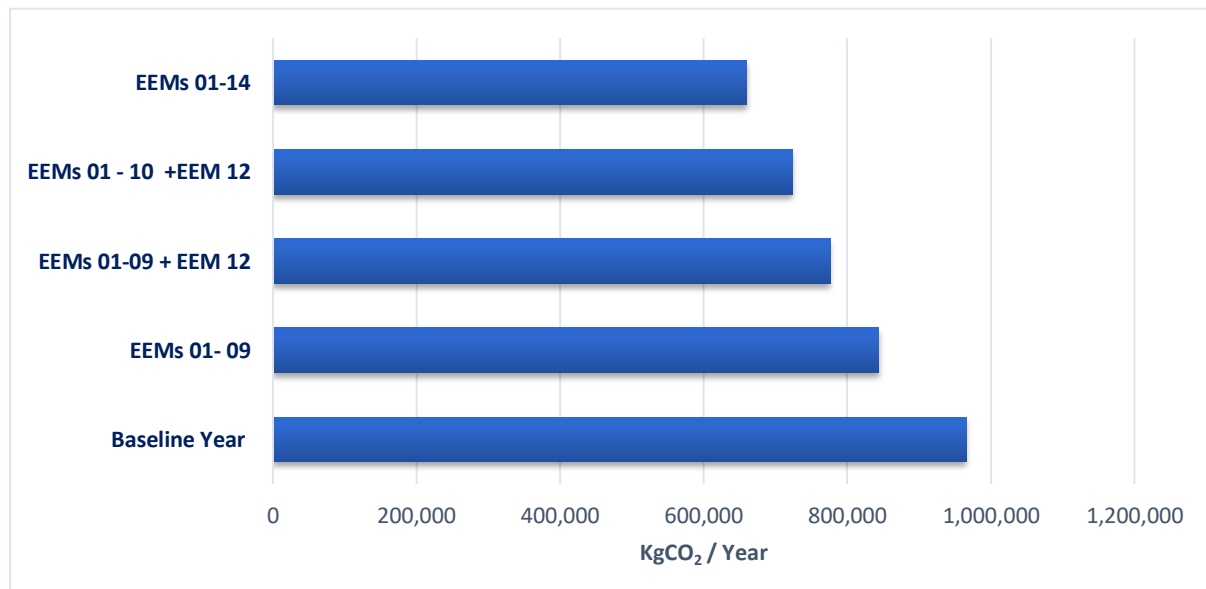
From that point, the remaining EEMs are those related to the retrofits and whereby the first priority should be given to EEM-12/retrofitting the chillers 3 and 4. This is a crucial measure for the three reasons : The units are using R22, one of the units is in poor operating condition , and the substantial energy efficiency opportunity .

The second in line would be EEM-10 as the current FCUs are still legacy ones installed for more than three decades – except those replaced for failure . This measure provides substantial energy efficiency, lower noise in guest rooms and enhanced control.

EEM-14 is a strategic one whereby upgrading the BMS and moving from a pneumatic control to a DDC one will take the property's control system to current standards and ready for the next two decades with enhanced control, energy savings and building analytics . While this measure certainly needs deep design, preparation and planning with disruptions to some operations, there is a first step that could be tackled immediately as explained in the related section – which is enhancing the current BMS platform with Graphical User Interfaces and additional trending.

Implementing the EEMs would lead to a substantial reduction in Carbon Emissions. The below chart provides the CO₂ emissions (KgCO₂/Year) from the baseline towards the full implementation with different scenarios aligned to the recommended priorities.

Figure 102: Annual Carbon Emissions under different EEMs implementation scenarios



END OF REPORT